

C8000V IPsec Testbed

Test evidence — every command run on every device

Run started	07 September 2026, 22:22 UTC
Duration	25.0 minutes
Result	215 of 215 tests passed
Suites	23
Device commands recorded	1586
Topology	hub-and-spoke: R1 hub, R2 and R3 spokes, one Linux host per router
Encryption	IKEv2 + static VTI, ESP-AES-256 / SHA256, one tunnel per spoke
Routing	eBGP inside the tunnels; AS 65001 hub, AS 65002 / 65003 spokes

Suite results

Suite	Tests	Passed	Failed	Commands	Time
01 Platform	5	5	0	10	1s
02 Configuration	5	5	0	14	0s
03 Wan	6	6	0	19	1s
04 Dmvpn	13	13	0	37	25s
05 Capture State	12	12	0	79	16s
06 Wire Encryption	3	3	0	27	34s
07 Ipsec Negative	3	3	0	63	127s
08 Management Api	6	6	0	7	4s
09 Bgp	26	26	0	624	308s
10 Host To Host	10	10	0	47	80s
11 Throughput	3	3	0	10	82s
12 Nat	10	10	0	51	79s
13 Nat Pool	13	13	0	49	92s
14 Ntp	12	12	0	39	3s
15 Syslog	10	10	0	21	47s
16 Snmpv3	13	13	0	87	16s

17 Snmp Polling	14	14	0	74	17s
18 Mtu	8	8	0	17	21s
19 Oob	9	9	0	68	31s
20 Tacacs	12	12	0	35	25s
21 Vty Acl	9	9	0	22	1s
22 Banner	6	6	0	6	21s
23 Destructive	7	7	0	180	291s

Each test below lists the commands it issued, in order, with the device that ran them and the output it returned. Router commands are marked in teal, Linux host commands in amber. Long output is trimmed to the first 14 lines; complete captures are in results/.

01 Platform

1.01 PASS Both Routers Report Catalyst 8000V Software (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **Cisco IOS[-]XE Software, Version 1d+1.1d+** (checked 2×)
- output contains **C8000V** (checked 2×)

How it was checked

R1 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 by mcpre
```

```
Cisco IOS-XE software, Copyright (c) 2005-2026 by cisco Systems, Inc.
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with ABSOLUTELY NO WARRANTY. You can redistribute and/or modify such
GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines
```

R2 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 by mcpre
```

```
Cisco IOS-XE software, Copyright (c) 2005-2026 by cisco Systems, Inc.
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licensed under the GNU General Public License ("GPL") Version 2.0. The
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with ABSOLUTELY NO WARRANTY. You can redistribute and/or modify such
GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines
```

Teardown (suite) Close All Routers

1.02 PASS Hostnames Match The Day0 Bootstrap Configs (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **\${expected}** (checked 2×)

How it was checked

R1 show running-config | include ^hostname

```
hostname c8000v-r1
```

R2 show running-config | include ^hostname

```
hostname c8000v-r2
```

Teardown (suite) Close All Routers

1.03 PASS Management Interfaces Are Up With DHCP Addresses (0.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **GigabitEthernet1s+10\0.0\2\1d+1s+YES\1s+DHCP\1s+up\1s+up** (checked 2×)

How it was checked

R1 `show ip interface brief GigabitEthernet1`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up

R2 `show ip interface brief GigabitEthernet1`

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up

Teardown (suite) Close All Routers

1.04 **PASS** Crypto Feature License Is Active (0.1s, 2 commands)

Without a feature license the crypto CLI does not exist at all, so every IPsec test downstream depends on this.

Setup (suite) Open All Routers

Acceptance criteria

- output matches **License Level:ls*S+** (checked 2×)

How it was checked

R1 `show version | include License Level`

```
License Level: network-premier
Addon License Level: dna-premier
```

R2 `show version | include License Level`

```
License Level: network-premier
Addon License Level: dna-premier
```

Teardown (suite) Close All Routers

1.05 **PASS** Management Plane Services Are Enabled (0.3s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **netconf-yang** (checked 2×)
- output contains **restconf** (checked 2×)

How it was checked

R1 `show running-config | include ^netconf-yang|^restconf`

```
netconf-yang
restconf
```

R2 `show running-config | include ^netconf-yang|^restconf`

```
netconf-yang
restconf
```

Teardown (suite) Close All Routers

02 Configuration

2.01 PASS Create A Loopback Interface (0.1s, 6 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain % **Invalid input**
- output does not contain % **Incomplete command**
- output does not contain % **Ambiguous command**

How it was checked

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Loopback100

#{output} =

R1 description ROBOT-MANAGED

#{output} =

R1 ip address 192.0.2.1 255.255.255.0

#{output} =

R1 no shutdown

#{output} =

R1 end

#{output} =

Teardown (this test) Run Keyword If Test Failed (Log, #{TEST_NAME} failed -- see the console on port 5001)

2.02 PASS Loopback Appears In The Interface Table (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **#{LOOPBACK}#{s}#{LOOPBACK_IP}#{s}+YES#{s}+manual#{s}+up#{s}+up**

How it was checked

R1 show ip interface brief Loopback100

Interface	IP-Address	OK?	Method	Status	Protocol
Loopback100	192.0.2.1	YES	manual	up	up

Teardown (this test) Run Keyword If Test Failed (Log, #{TEST_NAME} failed -- see the console on port 5001)

2.03 PASS Loopback Is Reachable From The Device Itself (0.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

R1 ping 192.0.2.1 repeat 2

Type escape sequence to abort.

Sending 2, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:

!!

Success rate is 100 percent (2/2), round-trip min/avg/max = 1/1/2 ms

R1 ping 192.0.2.1 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

Teardown (this test) Run Keyword If Test Failed (Log, #{TEST_NAME} failed -- see the console on port 5001)

2.04 PASS Loopback Is Present In The Running Config (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **description** ROBOT-MANAGED
- output contains **ip address** \${LOOPBACK_IP} \${LOOPBACK_MASK}

How it was checked

R1 show running-config interface Loopback100

Building configuration...

```
Current configuration : 92 bytes
!
interface Loopback100
  description ROBOT-MANAGED
  ip address 192.0.2.1 255.255.255.0
end
```

Teardown (this test) Run Keyword If Test Failed (Log, \${TEST_NAME} failed -- see the console on port 5001)

2.05 **PASS** Removing The Loopback Cleans It Up (0.1s, 4 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain \${LOOPBACK_IP}

How it was checked

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 no interface Loopback100

\${output} =

R1 end

\${output} =

R1 show ip interface brief Loopback100

```
^
% Invalid input detected at '^' marker.
```

Teardown (this test) Run Keyword If Test Failed (Log, \${TEST_NAME} failed -- see the console on port 5001)

03 Wan

3.01 PASS Every Router Is On The Shared WAN Segment (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **GigabitEthernet2** is **\${WAN_IP}** is **YES** is **IS** is **up** is **up** (checked 3×)

How it was checked

```
R1 show ip interface brief | include GigabitEthernet2
GigabitEthernet2    100.64.0.1      YES NVRAM  up          up
R2 show ip interface brief | include GigabitEthernet2
GigabitEthernet2    100.64.0.2      YES NVRAM  up          up
R3 show ip interface brief | include GigabitEthernet2
GigabitEthernet2    100.64.0.3      YES NVRAM  up          up
```

Teardown (suite) Close All Connections

3.02 PASS Every Router Can Reach Every Other Router On It (0.2s, 6 commands)

Including spoke to spoke, which is the precondition for a direct tunnel and the property the VTI lab deliberately did not have. Tested at the underlay, before any tunnel exists.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 6×)

How it was checked

```
R2 ping 100.64.0.2 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R3 ping 100.64.0.3 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1 ping 100.64.0.1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R3 ping 100.64.0.3 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1 ping 100.64.0.1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2 ping 100.64.0.2 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Teardown (suite) Close All Connections

3.03 PASS The Spokes Reach Each Other Directly, Not Through The Hub (0.0s, 1 commands)

One hop at the underlay. If this became two the spokes would still have connectivity, and DMVPN would still form tunnels -- but every "direct" spoke-to-spoke path would quietly be transiting the hub anyway.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${R3_WAN_IP}`
- output does not contain `${R1_WAN_IP}`
the spokes reach each other via the hub at the underlay:\n\${trace}

How it was checked

R2 `traceroute 100.64.0.3 probe 1 timeout 1 numeric`

```
Type escape sequence to abort.
Tracing the route to 100.64.0.3
VRF info: (vrf in name/id, vrf out name/id)
 1 100.64.0.3 2 msec
```

Teardown (suite) Close All Connections

3.04 **PASS** Each Router Learns The Others On The Segment (0.1s, 3 commands)

A shared segment, so every router should have resolved every other -- unlike a point-to-point link with exactly one neighbour.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${WAN_IP}${peer}` (checked 9×)

How it was checked

R1 `show ip arp GigabitEthernet2`

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.64.0.1	-	5254.00c8.0102	ARPA	GigabitEthernet2
Internet	100.64.0.2	47	5254.00c8.0202	ARPA	GigabitEthernet2
Internet	100.64.0.3	37	5254.00c8.0302	ARPA	GigabitEthernet2

R2 `show ip arp GigabitEthernet2`

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.64.0.2	-	5254.00c8.0202	ARPA	GigabitEthernet2
Internet	100.64.0.1	47	5254.00c8.0102	ARPA	GigabitEthernet2
Internet	100.64.0.3	37	5254.00c8.0302	ARPA	GigabitEthernet2

R3 `show ip arp GigabitEthernet2`

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.64.0.3	-	5254.00c8.0302	ARPA	GigabitEthernet2
Internet	100.64.0.1	47	5254.00c8.0102	ARPA	GigabitEthernet2
Internet	100.64.0.2	37	5254.00c8.0202	ARPA	GigabitEthernet2

Teardown (suite) Close All Connections

3.05 **PASS** The Underlay Carries No Site Prefixes (0.1s, 3 commands)

The WAN is a transport, not a routing domain. A site prefix appearing here would mean traffic could reach a LAN without ever entering the tunnel -- unencrypted, and invisible to every test that watches the overlay.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain `${LAN_NET}${peer}` (checked 9×)
would put site traffic on the underlay instead of the tunnel
- output contains `${WAN_NET}` (checked 3×)

How it was checked

R1 `show ip route | include GigabitEthernet2`

```
C    100.64.0.0/24 is directly connected, GigabitEthernet2
L    100.64.0.1/32 is directly connected, GigabitEthernet2
```

R2 `show ip route | include GigabitEthernet2`

```
C    100.64.0.0/24 is directly connected, GigabitEthernet2
L    100.64.0.2/32 is directly connected, GigabitEthernet2
```

R3 `show ip route | include GigabitEthernet2`

```
C    100.64.0.0/24 is directly connected, GigabitEthernet2
L    100.64.0.3/32 is directly connected, GigabitEthernet2
```

Teardown (suite) Close All Connections

3.06 PASS No Routing Protocol Runs On The WAN Segment (0.1s, 3 commands)

BGP peers over the tunnel, so the underlay should carry no adjacency at all. One here would be a second, unprotected path for the control plane.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain `${WAN_NET.rsplitt(':', 2)[0]}`. (checked 3×)

How it was checked

R1 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.2	4	65002	46	48	184	0	0	00:06:05	3
172.20.0.3	4	65003	45	48	184	0	0	00:06:04	3

R2 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	48	46	190	0	0	00:06:05	6

R3 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	48	45	196	0	0	00:06:04	6

Teardown (suite) Close All Connections

04 Dmvpn

4.01 PASS Every Router Runs One Multipoint Tunnel With No Destination (0.2s, 4 commands)

The structural difference from a static VTI. A configured destination would make this point-to-point again and there would be nothing for NHRP to resolve.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **tunnel mode gre multipoint** (checked 3×)
- output does not contain **tunnel destination** (checked 3×)
- output contains **tunnel source GigabitEthernet2** (checked 3×)
- output contains **ip address \${DMVPN_IP}\${r}** (checked 3×)
- output contains **Number of lines which match regexp = 1**
the hub has more than one tunnel interface, which is the VTI shape

How it was checked

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R1 show ip interface brief | count Tunnel

Number of lines which match regexp = 1

Teardown (suite) Restore Nhrp, Close All Connections

4.02 **PASS** The Tunnel Is Protected By IPsec On Every Router (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **tunnel protection ipsec profile** (checked 3×)

How it was checked

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.03 **PASS** NHRP Shares A Network ID And Authentication Everywhere (0.1s, 3 commands)

Both must match across the cloud; a mismatch in either leaves registrations silently unanswered.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ip nhrp network-id \${NHRP_NETWORK_ID}** (checked 3×)
- output contains **ip nhrp authentication \${NHRP_AUTH}** (checked 3×)

How it was checked

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
```

```

interface Tunnel0
description DMVPN hub
ip address 172.20.0.1 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp redirect
ip tcp adjust-mss 1360
... 6 more lines

```

R2 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.2 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

R3 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.3 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

Teardown (suite) Restore Nhrp, Close All Connections

4.04 PASS The Hub Is The Next Hop Server And The Spokes Point At It (0.1s, 4 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains `ip nhrp nhs ${R1_DMVPN_IP} nbma ${R1_WAN_IP}` (checked 2×)
- output does not contain `ip nhrp nhs`
the hub points at a next hop server, so it is behaving as a spoke
- output matches `${WAN_IP}[${r}]\s+${DMVPN_IP}[${r}]\s+UP` (checked 2×)
- `${dynamic}` has exactly 2 entries
the hub does not show two dynamically registered peers

How it was checked

R2 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.2 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

R3 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

R1 show dmvpn

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

```
# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 2 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.05 PASS This Is Phase 3, Not Phase 1 Or 2 (0.1s, 3 commands)

Redirect on the hub and shortcut on the spokes are what make the overlay reorganise itself. Without them the tests below about direct paths cannot pass, and the design is a different one.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ip nhrp redirect**
the hub sends no NHRP redirects
- output contains **ip nhrp nhs** (checked 2x)

How it was checked

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
```

```

ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp redirect
ip tcp adjust-mss 1360
... 6 more lines

```

R2 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.2 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

R3 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.3 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

Teardown (suite) Restore Nhrp, Close All Connections

4.06 PASS Both Spokes Register With The Hub (0.0s, 1 commands)

Registration is dynamic: the hub is configured with no spoke addresses at all and learns them as they arrive.

Setup (suite) Open All Routers

Acceptance criteria

- output matches `${WAN_IP}${r}}\s+${DMVPN_IP}${r}}\s+UP` (checked 2x)
 - `${dynamic}` has exactly 2 entries
- the hub does not show two dynamically registered peers

How it was checked

R1 show dmvpn

```

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
         N - NATed, L - Local, X - No Socket
         T1 - Route Installed, T2 - Nexthop-override, B - BGP
         C - CTS Capable, I2 - Temporary
         # Ent --> Number of NHRP entries with same NBMA peer
         NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
         UpDn Time --> Up or Down Time for a Tunnel
=====

```

```

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

```

```

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
... 2 more lines

```

Teardown (suite) Restore Nhrp, Close All Connections

4.07 PASS The Hub Maps Each Spoke's Tunnel Address To Its NBMA Address (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches `(?s)${DMVPN_IP}${r}]/32.*?${WAN_IP}${r}]` (checked 2×)
the hub has no NBMA mapping for `${r}`
- output contains **registered**

How it was checked

R1 show ip nhrp

```
172.20.0.2/32 via 172.20.0.2
  Tunnel0 created 00:01:16, expire 00:03:43
  Type: dynamic, Flags: registered nhop bfd
  NBMA address: 100.64.0.2
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:01:16, expire 00:03:43
  Type: dynamic, Flags: registered nhop bfd
  NBMA address: 100.64.0.3
```

Teardown (suite) Restore Nhrp, Close All Connections

4.08 PASS Every DMVPN Peering Is Encrypted (0.1s, 2 commands)

One IPsec profile protects the multipoint tunnel, so each peering formed over it -- including ones the configuration never named -- is protected by construction.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${R1_WAN_IP}` (checked 2×)
- output matches `${R1_WAN_IP}\s+Tu0\s+.*UA` (checked 2×)

How it was checked

R2 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
100.64.0.3 Tu0      100.64.0.3    100.64.0.3           00:01:12    UA
100.64.0.1 Tu0      100.64.0.1    100.64.0.1           00:01:17    UA
```

R3 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
100.64.0.2 Tu0      100.64.0.2    100.64.0.2           00:01:12    UA
100.64.0.1 Tu0      100.64.0.1    100.64.0.1           00:01:17    UA
```

Teardown (suite) Restore Nhrp, Close All Connections

4.09 PASS A Spoke Learns The Far Spoke As The BGP Next Hop (0.1s, 2 commands)

The routing half of Phase 3, and the reason "next-hop-unchanged" is configured on the hub. If the hub rewrote the next hop to itself, every spoke-to-spoke prefix would resolve to the hub, NHRP would never be asked to find the far spoke, and no shortcut could form.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${R3_DMVPN_IP}`
R2 does not see R3's tunnel address as the next hop for R3's prefix
- output contains `${R2_DMVPN_IP}`
R3 does not see R2's tunnel address as the next hop for R2's prefix

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1

```
BGP routing table entry for 10.20.3.1/32, version 188
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
```

```

65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:16:54 UTC

```

R3 show bgp ipv4 unicast 10.20.2.1

```

BGP routing table entry for 10.20.2.1/32, version 194
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:16:25 UTC

```

Teardown (suite) Restore Nhrp, Close All Connections

4.10 PASS Spoke To Spoke Traffic Works Before Any Shortcut Exists (12.1s, 4 commands)

The fallback path. With the NHRP cache cleared there is no direct entry, so the first packets go through the hub -- which is what makes the shortcut an optimisation rather than a prerequisite for connectivity.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain `${WAN_IP}${peer}`
- output contains **Success rate is**
- `${loss}[0] > 0` holds

How it was checked

```

R2 clear ip nhrp
${output} =
R3 clear ip nhrp
${output} =
R2 show ip nhrp 172.20.0.3
${output} =
R2 ping 10.20.3.1 source Loopback1 repeat 30
Type escape sequence to abort.
Sending 30, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Success rate is 96 percent (29/30), round-trip min/avg/max = 1/1/2 ms

```

Teardown (suite) Restore Nhrp, Close All Connections

4.11 PASS A Shortcut Forms On Demand (12.2s, 6 commands)

The defining behaviour. Cleared first so the entry cannot be left over from an earlier test: it has to be created by this traffic, here.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain `${WAN_IP}${peer}`
- output contains `${WAN_IP}${peer}` (checked 2x)

How it was checked

```

R2 clear ip nhrp
${output} =
R3 clear ip nhrp
${output} =
R2 show ip nhrp 172.20.0.3
${output} =
R2 ping 10.20.3.1 source Loopback1 repeat 20
Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

```


Success rate is 95 percent (19/20), round-trip min/avg/max = 1/1/2 ms

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:58
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3
```

R3 show ip nhrp 172.20.0.2

```
172.20.0.2/32 via 172.20.0.2
  Tunnel0 created 00:00:02, expire 00:04:57
  Type: dynamic, Flags: router implicit nhop bfd
  NBMA address: 100.64.0.2
```

Teardown (suite) Restore Nhrp, Close All Connections

4.12 PASS Traffic Then Leaves The Hub Out Of The Path (0.1s, 2 commands)

The point of the shortcut, and the strongest statement this suite makes: with the direct path installed, a trace from one spoke to the other reaches it in a single hop instead of transiting the hub.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${WAN_IP}[${peer}]`
- output contains `${R3_DMVPN_IP}`
- output does not contain `${R1_DMVPN_IP}`
the trace still passes through the hub, so no shortcut is in use:\n\${trace}
- `${hops}[-1]` equals 1
the far spoke is `${hops}[-1]` hops away, expected 1

How it was checked

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:57
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3
```

R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric

```
Type escape sequence to abort.
Tracing the route to 10.20.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.20.0.3 2 msec
```

Teardown (suite) Restore Nhrp, Close All Connections

4.13 PASS The Shortcut Gets Its Own IPsec Session (0.1s, 2 commands)

A direct path between two routers that were never configured with each other's addresses, protected all the same, because the profile is bound to the tunnel rather than to a peer.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${WAN_IP}[${peer}]`
- output contains `${R3_WAN_IP}`
R2 has no IPsec session with R3's NBMA address:\n\${sess}
- `${count}` has exactly 2 entries
R2 holds `${count}` tunnel sessions, expected one to the hub and one to R3

How it was checked

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:57
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3
```

R2 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
      K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
```

100.64.0.3	Tu0	100.64.0.3	00:00:02 UA
100.64.0.1	Tu0	100.64.0.1	00:01:41 UA

Teardown (suite) Restore Nhrp, Close All Connections

05 Capture State

5.01 **PASS** Capture Running Config From Every Router (12.9s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **hostname** (checked 3×)
- output matches **(?m)^endl*\$** (checked 3×)

How it was checked

R1 show running-config

Building configuration...

```
Current configuration : 21292 bytes
!
! Last configuration change at 02:22:29 UTC Tue Sep 8 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 624 more lines
```

R2 show running-config

Building configuration...

```
Current configuration : 19824 bytes
!
! Last configuration change at 02:21:14 UTC Tue Sep 8 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 579 more lines
```

R3 show running-config

Building configuration...

```
Current configuration : 19832 bytes
!
! Last configuration change at 02:21:14 UTC Tue Sep 8 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 579 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.02 **PASS** Capture Routing Table From Every Router (0.2s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **Gateway of last resort** (checked 3×)
- output contains **Codes:** (checked 3×)

How it was checked

R1 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
 ... 24 more lines

R2 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
 ... 24 more lines

R3 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
 ... 24 more lines

Teardown (suite) Close Nms, Close All Connections

5.03 PASS Capture BGP Table And Neighbours From Every Router (0.2s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show bgp ipv4 unicast

BGP table version is 184, local router ID is 172.16.1.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
> 10.20.1.1/32	0.0.0.0	0		32768	i
> 10.20.2.1/32	172.20.0.2	0		0	65002 i
> 10.20.3.1/32	172.20.0.3	0		0	65003 i
> 10.30.1.0/24	0.0.0.0	0		32768	i
> 10.30.2.0/24	172.20.0.2	0		0	65002 i

... 4 more lines

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001
 BGP table version is 184, main routing table version 184
 9 network entries using 2232 bytes of memory

```

9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 84/75 prefixes, 96/87 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:48:10.065 ago)

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

```

R2 show bgp ipv4 unicast

```

BGP table version is 190, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	172.20.0.1	0		0	65001 i
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.20.3.1/32	172.20.0.3			0	65001 65003 i
*>	10.30.1.0/24	172.20.0.1	0		0	65001 i
*>	10.30.2.0/24	0.0.0.0	0		32768	i

... 4 more lines

R2 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 190, main routing table version 190
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 90/81 prefixes, 99/90 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (00:48:09.146 ago)

```

```

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines

```

R3 show bgp ipv4 unicast

```

BGP table version is 196, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	172.20.0.1	0		0	65001 i
*>	10.20.2.1/32	172.20.0.2			0	65001 65002 i
*>	10.20.3.1/32	0.0.0.0	0		32768	i
*>	10.30.1.0/24	172.20.0.1	0		0	65001 i
*>	10.30.2.0/24	172.20.0.2			0	65001 65002 i

... 4 more lines

R3 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 196, main routing table version 196
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 99/90 prefixes, 102/93 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (00:48:09.233 ago)

```

```

Neighbor      V          AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines

```

Teardown (suite) Close Nms, Close All Connections

5.04 PASS Capture Crypto State From Every Router (0.4s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 763, #pkts encrypt: 763, #pkts digest: 763
  #pkts decaps: 754, #pkts decrypt: 754, #pkts verify: 754
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvrf/ivrf	Status
1	100.64.0.1/500	100.64.0.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1073, Session-id: 73				
Local spi: 6F52660E0B4ECD39 Remote spi: CA8260705A3473F7				

Tunnel-id	Local	Remote	fvrf/ivrf	Status
2	100.64.0.1/500	100.64.0.3/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1074, Session-id: 74				

... 3 more lines

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 767, #pkts encrypt: 767, #pkts digest: 767
  #pkts decaps: 761, #pkts decrypt: 761, #pkts verify: 761
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

R2 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvrf/ivrf	Status
1	100.64.0.2/500	100.64.0.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1070, Session-id: 68				
Local spi: CA8260705A3473F7 Remote spi: 6F52660E0B4ECD39				

IPv6 Crypto IKEv2 SA

R3 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.3

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 757, #pkts encrypt: 757, #pkts digest: 757
  #pkts decaps: 764, #pkts decrypt: 764, #pkts verify: 764
  #pkts compressed: 0, #pkts decompressed: 0
```

```
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines
```

R3 show crypto ikev2 sa

IPv4 Crypto IKEv2 SA

```
Tunnel-id Local Remote fvr/ivrf Status
1 100.64.0.3/500 100.64.0.1/500 none/none READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/128 sec
CE id: 1068, Session-id: 66
Local spi: 68C218A7F9527196 Remote spi: AEF5C6DE6B24D6A3
```

IPv6 Crypto IKEv2 SA

Teardown (suite) Close Nms, Close All Connections

5.05 PASS Capture NAT State From Every Router (0.3s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.11	192.168.10.11	---	---
---	10.30.1.10	192.168.10.10	---	---
---	10.30.1.13	192.168.10.13	---	---
---	10.30.1.12	192.168.10.12	---	---
---	10.30.1.15	192.168.10.15	---	---
---	10.30.1.14	192.168.10.14	---	---
---	10.30.1.17	192.168.10.17	---	---
---	10.30.1.16	192.168.10.16	---	---
---	10.30.1.19	192.168.10.19	---	---
---	10.30.1.18	192.168.10.18	---	---

Total number of translations: 10

R1 show ip nat statistics

```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
Tunnel0
Inside interfaces:
GigabitEthernet3
Hits: 0 Misses: 0
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
pool SITE-POOL: id 2, netmask 255.255.255.0
start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.11	192.168.20.11	---	---
---	10.30.2.10	192.168.20.10	---	---
---	10.30.2.13	192.168.20.13	---	---
---	10.30.2.12	192.168.20.12	---	---
---	10.30.2.15	192.168.20.15	---	---
---	10.30.2.14	192.168.20.14	---	---
---	10.30.2.17	192.168.20.17	---	---
---	10.30.2.16	192.168.20.16	---	---
---	10.30.2.19	192.168.20.19	---	---
---	10.30.2.18	192.168.20.18	---	---

Total number of translations: 10

R2 show ip nat statistics

```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
Tunnel0
Inside interfaces:
GigabitEthernet3
Hits: 0 Misses: 0
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
```

```
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
pool SITE-POOL: id 2, netmask 255.255.255.0
start 10.30.2.100 end 10.30.2.129
... 9 more lines
```

R3 show ip nat translations

```
Pro Inside global      Inside local      Outside local      Outside global
--- 10.30.3.11         192.168.30.11    ---              ---
--- 10.30.3.10         192.168.30.10    ---              ---
--- 10.30.3.13         192.168.30.13    ---              ---
--- 10.30.3.12         192.168.30.12    ---              ---
--- 10.30.3.15         192.168.30.15    ---              ---
--- 10.30.3.14         192.168.30.14    ---              ---
--- 10.30.3.17         192.168.30.17    ---              ---
--- 10.30.3.16         192.168.30.16    ---              ---
--- 10.30.3.19         192.168.30.19    ---              ---
--- 10.30.3.18         192.168.30.18    ---              ---
Total number of translations: 10
```

R3 show ip nat statistics

```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
Tunnel0
Inside interfaces:
GigabitEthernet3
Hits: 0 Misses: 0
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
pool SITE-POOL: id 2, netmask 255.255.255.0
start 10.30.3.100 end 10.30.3.129
... 9 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.06 PASS Capture NTP State From Every Router (0.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 292200 (1/100 of seconds), resolution is 4000
reference time is EE49EF2A.E45A1F20 (02:19:22.892 UTC Tue Sep 8 2026)
clock offset is 5.4060 msec, root delay is 50.87 msec
root dispersion is 22.30 msec, peer dispersion is 2.98 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007803 s/s
system poll interval is 128, last update was 240 sec ago.
```

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    37    128  377  0.908  5.406  2.982
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 292300 (1/100 of seconds), resolution is 4000
reference time is EE49EDE9.04189380 (02:14:01.016 UTC Tue Sep 8 2026)
clock offset is 8.5701 msec, root delay is 50.86 msec
root dispersion is 28.85 msec, peer dispersion is 4.47 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007655 s/s
system poll interval is 128, last update was 561 sec ago.
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    96    128  377  0.070  8.570  4.478
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
```



```
ntp uptime is 292200 (1/100 of seconds), resolution is 4000
reference time is EE49EDE7.4A3D7170 (02:13:59.290 UTC Tue Sep 8 2026)
clock offset is 5.9554 msec, root delay is 50.87 msec
root dispersion is 32.13 msec, peer dispersion is 3.60 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007218 s/s
system poll interval is 128, last update was 564 sec ago.
```

R3 show ntp associations

```
address          ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  192.207.55.254  2    92   128   377  0.921  5.955  3.605
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms, Close All Connections

5.07 PASS Capture Logging State From Every Router (0.2s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show logging | include Trap logging|Logging to|Syslog logging

Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

Trap logging: level informational, 930 message lines logged

Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,

000065: *Sep 8 01:34:40.638 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 0 CLI Request Triggered

000085: *Sep 8 01:34:42.278 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) started - CLI initiated

000104: *Sep 8 01:34:54.812 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) started - reconnection

000161: *Sep 8 01:35:48.352 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) stopped - CLI initiated

000163: *Sep 8 01:35:48.590 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) restored CLI initiated

R2 show logging | include Trap logging|Logging to|Syslog logging

Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

Trap logging: level informational, 977 message lines logged

Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,

000160: *Sep 8 01:35:48.345 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) stopped - CLI initiated

000162: *Sep 8 01:35:48.541 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) restored CLI initiated

R3 show logging | include Trap logging|Logging to|Syslog logging

Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

Trap logging: level informational, 850 message lines logged

Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,

000069: *Sep 8 01:34:40.581 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 0 CLI Request Triggered

000090: *Sep 8 01:34:42.411 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) started - CLI initiated

000104: *Sep 8 01:34:54.961 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) started - reconnection

000159: *Sep 8 01:35:48.371 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) stopped - CLI initiated

000161: *Sep 8 01:35:48.558 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514 (MGMT) restored CLI initiated

Teardown (suite) Close Nms, Close All Connections

5.08 PASS Capture SNMP State From Every Router (0.2s, 9 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show snmp user

User name: labmon

Engine ID: 800000000300525400C80101

storage-type: nonvolatile active

Authentication Protocol: SHA

Privacy Protocol: AES128
Group-name: LABGRP

R1 show snmp host

Notification host: 192.168.99.10 udp-port: 1162 VRFName: MGMT type: trap
user: labmon security model: v3 priv

Notification host: 192.168.99.10 udp-port: 162 VRFName: MGMT type: trap
user: labmon security model: v3 priv

R1 show snmp engineID

Local SNMP engineID: 800000090300525400C80101
Remote Engine ID IP-addr Port

R2 show snmp user

User name: labmon
Engine ID: 800000090300525400C80201
storage-type: nonvolatile active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

R2 show snmp host

Notification host: 192.168.99.10 udp-port: 1162 VRFName: MGMT type: trap
user: labmon security model: v3 priv

Notification host: 192.168.99.10 udp-port: 162 VRFName: MGMT type: trap
user: labmon security model: v3 priv

R2 show snmp engineID

Local SNMP engineID: 800000090300525400C80201
Remote Engine ID IP-addr Port

R3 show snmp user

User name: labmon
Engine ID: 800000090300525400C80301
storage-type: nonvolatile active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

R3 show snmp host

Notification host: 192.168.99.10 udp-port: 1162 VRFName: MGMT type: trap
user: labmon security model: v3 priv

Notification host: 192.168.99.10 udp-port: 162 VRFName: MGMT type: trap
user: labmon security model: v3 priv

R3 show snmp engineID

Local SNMP engineID: 800000090300525400C80301
Remote Engine ID IP-addr Port

Teardown (suite) Close Nms, Close All Connections

5.09 PASS Capture The NMS View Of Every Router (0.5s, 16 commands)

The same system group, but seen over SNMPv3 from the NMS rather than read off the console. Captured next to the CLI output so the two can be compared in the evidence PDF.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS ip -br addr; ip route

```
lo                UNKNOWN      127.0.0.1/8 ::1/128
ens3              UP            10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4              UP            192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5              UP            192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS ip -br addr; ip route

```
lo                UNKNOWN      127.0.0.1/8 ::1/128
ens3              UP            10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4              UP            192.168.10.20/24 fe80::5054:ff:febb:102/64
```

```

ens5          UP          192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10

```

```

NMS snmpget --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2

```

```

NMS snmpget --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.1

```

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293250) 0:48:52.50
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.1

```

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293250) 0:48:52.50
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.2.2.1.2

```

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.2.2.1.2

```

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.1

```

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293289) 0:48:52.89
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78

```

```

.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293289) 0:48:52.89
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293267) 0:48:52.67
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293267) 0:48:52.67
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"

```

```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"

```

Teardown (suite) Close Nms, Close All Connections

5.10 PASS Capture Collector State From The NMS (0.3s, 12 commands)

What the collectors actually hold: the tail of each router's syslog file and of snmptrapd's decrypted log. This is the management-plane evidence that used to come off h1.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS systemctl is-active rsyslog snmptrapd lab-trapcap

```

active
active
active

```

NMS systemctl is-active rsyslog snmptrapd lab-trapcap

```

active
active
active

```

NMS sudo ls -l /var/log/lab

```

total 5180
-rw-r--r-- 1 syslog syslog 283962 Sep  8 02:23 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 337492 Sep  8 02:23 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 233399 Sep  8 02:23 192.168.99.3.log
-rw-r--r-- 1 root  root  985767 Sep  8 02:23 snmptrapd.log
-rw-r--r-- 1 root  root  3436635 Sep  8 02:23 traps-raw.log

```

NMS sudo ls -l /var/log/lab

```

total 5180
-rw-r--r-- 1 syslog syslog 283962 Sep  8 02:23 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 337492 Sep  8 02:23 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 233399 Sep  8 02:23 192.168.99.3.log
-rw-r--r-- 1 root  root  985767 Sep  8 02:23 snmptrapd.log
-rw-r--r-- 1 root  root  3436635 Sep  8 02:23 traps-raw.log

```

NMS sudo tail -40 /var/log/lab/192.168.10.1.log

tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.10.1.log

tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.20.1.log

tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.20.1.log

tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.30.1.log

tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.30.1.log

tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/snmptrapd.log

```

2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:22.54 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.69 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.69 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.69.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.69.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.69.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.69.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.69.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.69.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.69.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.69.1 0
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:22.90 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.71 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.71 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.71.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.71.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.71.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.71.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.71.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.71.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.71.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.71.1 0

```

```
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:23.67 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708063 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708063 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:23.31 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708061 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708061 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:24.70 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.71 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.71 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.71 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.71 4
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.71 214 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.70 214
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.70 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.70 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.70 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.70 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.70 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.70 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.70 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.70 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.70 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.70 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.70 0
... 26 more lines
```

NMS sudo tail -40 /var/log/lab/snmptrapd.log

```
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:22.54 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.69 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.69 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.69.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.69.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.69.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.69.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.69.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.69.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.69.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.69.1 0
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:22.90 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.71 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.71 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.71.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.71.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.71.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.71.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.71.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.71.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.71.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.71.1 0
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:23.67 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708063 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708063 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:23.31 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708061 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708061 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:24.70 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.71 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.71 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.71 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.71 4
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.71 214 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.70 214
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.70 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.70 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.70 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.70 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.70 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.70 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.70 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.70 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.70 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.70 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.70 0
... 26 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.11 PASS Capture Config And Routing Table From Every Host (0.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **eth1** (checked 3×)
- output contains **default via** (checked 3×)

How it was checked

H1 ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 83564sec preferred_lft 72764sec
   inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
       valid_lft 86040sec preferred_lft 14040sec
   inet6 fe80::5054:ff:feaa:101/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines
```

H1 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 83564sec preferred_lft 72764sec
   inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
       valid_lft 86241sec preferred_lft 14241sec
   inet6 fe80::5054:ff:feaa:201/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines
```

H2 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
```

H3 ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 83565sec preferred_lft 72765sec
   inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
       valid_lft 86083sec preferred_lft 14083sec
   inet6 fe80::5054:ff:feaa:301/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines
```

H3 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

Teardown (suite) Close Nms, Close All Connections

5.12 PASS Every Router Routing Table Holds All Three LANs (0.2s, 3 commands)

A content check on what was just captured: in a working hub-and-spoke every router reaches every LAN, the spokes reaching each other's through the hub.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **#{R1_LAN_NET}** (checked 3×)
- output contains **#{R2_LAN_NET}** (checked 3×)
- output contains **#{R3_LAN_NET}** (checked 3×)

How it was checked

R1 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected
```

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

R2 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected
```

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

R3 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected
```

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

Teardown (suite) Close Nms, Close All Connections

06 Wire Encryption

6.01 PASS Protected Traffic Appears On The Link As ESP (15.7s, 9 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ESP**
- output matches `${R1_WAN_IP}\s+>\s+${R2_WAN_IP}`

How it was checked

R1 no monitor capture ROBOTCAP

Capture does not exist

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
\${output} =

R1 monitor capture ROBOTCAP match ipv4 any any
\${output} =

R1 monitor capture ROBOTCAP buffer size 5
\${output} =

R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP

R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
.....
Success rate is 0 percent (0/5)

R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP

R1 show monitor capture ROBOTCAP buffer brief

```
-----
#   size   timestamp      source          destination    dscp  protocol
-----
0  138     0.000000    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
1  138     0.016006    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
2  138     0.016006    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
3  138     0.091991    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
4  138     0.091991    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
5  138     0.179984    100.64.0.2      -> 100.64.0.1    48 CS6  ESP
6  138     0.179984    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
7  138     0.189993    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
8  138     0.189993    100.64.0.2      -> 100.64.0.1    48 CS6  ESP
9  138     0.393977    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
10 138     0.393977    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
... 272 more lines
```

R1 no monitor capture ROBOTCAP
\${output} =

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

6.02 PASS Protected Endpoint Addresses Never Appear In Cleartext On The Link (15.9s, 9 commands)

The strongest statement this lab can make about encryption: traffic between the loopbacks crosses the wire without either loopback address being visible, and with no cleartext ICMP.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain `${R1_LOOPBACK}`
- output does not contain `${R2_LOOPBACK}`
- output does not contain **ICMP**

How it was checked

R1 no monitor capture ROBOTCAP
Capture does not exist

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both

```

${output} =
R1 monitor capture ROBOTCAP match ipv4 any any
${output} =
R1 monitor capture ROBOTCAP buffer size 5
${output} =
R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP
R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
.....
Success rate is 0 percent (0/5)
R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP
R1 show monitor capture ROBOTCAP buffer brief
-----
#    size    timestamp    source            destination      dscp    protocol
-----
0    138      0.000000    100.64.0.1        -> 100.64.0.3      48 CS6   ESP
1    138      0.001999    100.64.0.3        -> 100.64.0.1      48 CS6   ESP
2    138      0.074993    100.64.0.1        -> 100.64.0.3      48 CS6   ESP
3    138      0.168998    100.64.0.2        -> 100.64.0.1      48 CS6   ESP
4    138      0.168998    100.64.0.1        -> 100.64.0.2      48 CS6   ESP
5    138      0.256991    100.64.0.2        -> 100.64.0.1      48 CS6   ESP
6    138      0.269991    100.64.0.1        -> 100.64.0.2      48 CS6   ESP
7    138      0.284989    100.64.0.1        -> 100.64.0.2      48 CS6   ESP
8    138      0.285996    100.64.0.2        -> 100.64.0.1      48 CS6   ESP
9    138      0.294983    100.64.0.3        -> 100.64.0.1      48 CS6   ESP
10   138      0.294983    100.64.0.1        -> 100.64.0.3      48 CS6   ESP
... 277 more lines
R1 no monitor capture ROBOTCAP
${output} =

```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

6.03 PASS Unprotected Link Traffic Is Visible As Cleartext ICMP (2.4s, 9 commands)

Positive control for the capture itself. Without this, the absence asserted above could just mean the capture was broken.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ICMP**
- output matches `${R1_WAN_IP}\s+->\s+${R2_WAN_IP}\s+\s+\s+\s+\s+ICMP`

How it was checked

```

R1 no monitor capture ROBOTCAP
Capture does not exist
R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
${output} =
R1 monitor capture ROBOTCAP match ipv4 any any
${output} =
R1 monitor capture ROBOTCAP buffer size 5
${output} =
R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP
R1 ping 100.64.0.2 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP
R1 show monitor capture ROBOTCAP buffer brief
-----
#    size    timestamp    source            destination      dscp    protocol
-----
0    114      0.000000    100.64.0.1        -> 100.64.0.2      0 BE     ICMP
1    114      0.000992    100.64.0.2        -> 100.64.0.1      0 BE     ICMP
2    114      0.000992    100.64.0.1        -> 100.64.0.2      0 BE     ICMP

```

```
3 114 0.001999 100.64.0.2 -> 100.64.0.1 0 BE ICMP
4 114 0.001999 100.64.0.1 -> 100.64.0.2 0 BE ICMP
5 114 0.002990 100.64.0.2 -> 100.64.0.1 0 BE ICMP
6 114 0.002990 100.64.0.1 -> 100.64.0.2 0 BE ICMP
7 114 0.003997 100.64.0.2 -> 100.64.0.1 0 BE ICMP
8 114 0.004989 100.64.0.1 -> 100.64.0.2 0 BE ICMP
9 114 0.004989 100.64.0.2 -> 100.64.0.1 0 BE ICMP
10 138 0.010985 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
... 48 more lines
```

R1 no monitor capture ROBOTCAP

`${output} =`

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

07 Ipsec Negative

7.01 **PASS** IKEv2 Refuses To Establish With A Mismatched Pre-Shared Key (13.3s, 13 commands)

Proves authentication is actually enforced. Without this, a tunnel that accepted any peer would still pass suite 04.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **Success rate is 100 percent**
- output does not contain **READY**

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 crypto ikev2 keyring LAB-KR

#{output} =

R2 peer ANY

#{output} =

R2 pre-shared-key WrongKeyOnPurpose999

#{output} =

R2 end

#{output} =

R1 clear crypto sa

#{output} =

R1 clear crypto ikev2 sa

#{output} =

R2 clear crypto sa

#{output} =

R2 clear crypto ikev2 sa

#{output} =

R3 clear crypto sa

#{output} =

R3 clear crypto ikev2 sa

#{output} =

R1 ping 172.16.2.1 source Loopback0 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:

Packet sent with a source address of 172.16.1.1

.....

Success rate is 0 percent (0/5)

R2 show crypto ikev2 sa

#{output} =

Teardown (suite) Restore Baseline Crypto, Close All Routers

7.02 **PASS** Tunnel Recovers Once The Correct Key Is Restored (18.3s, 13 commands)

Also guarantees the previous test cannot leave the lab broken.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2x)

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 crypto ikev2 keyring LAB-KR

#{output} =

R2 peer ANY

#{output} =

R2 pre-shared-key LabPreSharedKey123

#{output} =

R2 end

#{output} =

```

R1 clear crypto sa
${output} =
R1 clear crypto ikev2 sa
${output} =
R2 clear crypto sa
${output} =
R2 clear crypto ikev2 sa
${output} =
R3 clear crypto sa
${output} =
R3 clear crypto ikev2 sa
${output} =
R1 ping 10.20.2.1 source Loopback1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
.....
Success rate is 0 percent (0/5)
R1 ping 10.20.2.1 source Loopback1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

```

Teardown (suite) Restore Baseline Crypto, Close All Routers

7.03 PASS IPsec Rekeys And Traffic Still Flows Afterwards (95.2s, 37 commands)

Drops the SA lifetime to the 120s platform minimum, waits for the outbound SPI to rotate, then re-checks connectivity. Rekey is where real tunnels break.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)
 - **\${m}** is not empty (checked 11×)
 - **\${new}** differs from **\${old}** (checked 10×)
- SPI **\${new}** has not rotated yet

How it was checked

```

R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 crypto ipsec security-association lifetime seconds 120
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
R1 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 crypto ipsec security-association lifetime seconds 120
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
R2 end
${output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 crypto ipsec security-association lifetime seconds 120
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
R3 end
${output} =
R1 clear crypto sa
${output} =
R1 clear crypto ikev2 sa
${output} =
R2 clear crypto sa
${output} =
R2 clear crypto ikev2 sa
${output} =

```

R3 clear crypto sa

```
#{output} =
```

R3 clear crypto ikev2 sa

```
#{output} =
```

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
```

```
Packet sent with a source address of 10.20.1.1
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R1 show crypto ipsec sa

```
interface: Tunnel0
```

```
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1
```

```
protected vrf: (none)
```

```
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
```

```
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
```

```
current_peer 100.64.0.3 port 500
```

```
  PERMIT, flags={origin_is_acl,}
```

```
  #pkts encaps: 13, #pkts encrypt: 13, #pkts digest: 13
```

```
  #pkts decaps: 14, #pkts decrypt: 14, #pkts verify: 14
```

```
  #pkts compressed: 0, #pkts decompressed: 0
```

```
  #pkts not compressed: 0, #pkts compr. failed: 0
```

```
  #pkts not decompressed: 0, #pkts decompress failed: 0
```

```
  #send errors 0, #recv errors 0
```

```
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
```

```
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1
```

```
protected vrf: (none)
```

```
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
```

```
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
```

```
current_peer 100.64.0.3 port 500
```

```
  PERMIT, flags={origin_is_acl,}
```

```
  #pkts encaps: 13, #pkts encrypt: 13, #pkts digest: 13
```

```
  #pkts decaps: 14, #pkts decrypt: 14, #pkts verify: 14
```

```
  #pkts compressed: 0, #pkts decompressed: 0
```

```
  #pkts not compressed: 0, #pkts compr. failed: 0
```

```
  #pkts not decompressed: 0, #pkts decompress failed: 0
```

```
  #send errors 0, #recv errors 0
```

```
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
```

```
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1
```

```
protected vrf: (none)
```

```
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
```

```
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
```

```
current_peer 100.64.0.3 port 500
```

```
  PERMIT, flags={origin_is_acl,}
```

```
  #pkts encaps: 73, #pkts encrypt: 73, #pkts digest: 73
```

```
  #pkts decaps: 79, #pkts decrypt: 79, #pkts verify: 79
```

```
  #pkts compressed: 0, #pkts decompressed: 0
```

```
  #pkts not compressed: 0, #pkts compr. failed: 0
```

```
  #pkts not decompressed: 0, #pkts decompress failed: 0
```

```
  #send errors 0, #recv errors 0
```

```
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
```

```
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1
```

```
protected vrf: (none)
```

```
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
```

```
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
```

```
current_peer 100.64.0.3 port 500
```

```
  PERMIT, flags={origin_is_acl,}
```

```
  #pkts encaps: 132, #pkts encrypt: 132, #pkts digest: 132
```

```
  #pkts decaps: 138, #pkts decrypt: 138, #pkts verify: 138
```

```
  #pkts compressed: 0, #pkts decompressed: 0
```

```
  #pkts not compressed: 0, #pkts compr. failed: 0
```

```
  #pkts not decompressed: 0, #pkts decompress failed: 0
```

```
  #send errors 0, #recv errors 0
```

```
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
```

```
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1
```

```

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 198, #pkts encrypt: 198, #pkts digest: 198
#pkts decaps: 198, #pkts decrypt: 198, #pkts verify: 198
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 257, #pkts encrypt: 257, #pkts digest: 257
#pkts decaps: 257, #pkts decrypt: 257, #pkts verify: 257
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 317, #pkts encrypt: 317, #pkts digest: 317
#pkts decaps: 317, #pkts decrypt: 317, #pkts verify: 317
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 375, #pkts encrypt: 375, #pkts digest: 375
#pkts decaps: 375, #pkts decrypt: 375, #pkts verify: 375
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 435, #pkts encrypt: 435, #pkts digest: 435
#pkts decaps: 434, #pkts decrypt: 434, #pkts verify: 434
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

```

```

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 493, #pkts encrypt: 493, #pkts digest: 493
#pkts decaps: 498, #pkts decrypt: 498, #pkts verify: 498
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 549, #pkts encrypt: 549, #pkts digest: 549
#pkts decaps: 553, #pkts decrypt: 553, #pkts verify: 553
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 ping 10.20.2.1 source Loopback1 repeat 5

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

```

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 no crypto ipsec security-association lifetime seconds 120

```

${output} =

```

R1 end

```

${output} =

```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no crypto ipsec security-association lifetime seconds 120

```

${output} =

```

R2 end

```

${output} =

```

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 no crypto ipsec security-association lifetime seconds 120

```

${output} =

```

R3 end

```

${output} =

```

Teardown (this test) Restore Default SA Lifetime

08 Management Api

8.01 PASS RESTCONF Reports The Same Hostname As The CLI (0.8s, 3 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- output contains `${name}` (checked 3×)

How it was checked

```
R1 show running-config | include ^hostname
hostname c8000v-r1
```

```
R2 show running-config | include ^hostname
hostname c8000v-r2
```

```
R3 show running-config | include ^hostname
hostname c8000v-r3
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.02 PASS RESTCONF Interface State Agrees With The CLI (0.1s, 1 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${status}` equals up
- output matches `GigabitEthernet2` is `${R1_WAN_IP}` is YES is `IS` is `up` is `up`

How it was checked

```
R1 show ip interface brief GigabitEthernet2
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet2	100.64.0.1	YES	NVRAM	up	up

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.03 PASS Config Pushed Over RESTCONF Appears In The CLI (0.4s, 1 commands)

Write path plus cross-check: RESTCONF and the CLI must be two views of one datastore, not two datastores.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${resp.status_code}` in (201, 204) holds
- output contains `${RC_LOOPBACK_IP}`

How it was checked

```
R1 show running-config interface Loopback200
Building configuration...

Current configuration : 68 bytes
!
interface Loopback200
 ip address 198.51.100.1 255.255.255.0
end
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.04 PASS Config Deleted Over RESTCONF Disappears From The CLI (0.8s, 1 commands)

Runs after the PUT test, which creates the interface; a 404 here would mean the create silently failed.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${resp.status_code}` equals 204
- RESTCONF DELETE returned `${resp.status_code}`; expected 204 for the loopback created by the previous test
- output does not contain `${RC_LOOPBACK_IP}`

How it was checked

```
R1 show ip interface brief Loopback200
```

```
^  
% Invalid input detected at '^' marker.
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.05 **PASS** NETCONF Reports The Same Hostname As The CLI (0.9s, 1 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${name}` equals `${R1_NAME}`
- output contains `${name}`

How it was checked

```
R1 show running-config | include ^hostname  
hostname c8000v-r1
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.06 **PASS** NETCONF Advertises The Expected Base Capabilities (0.5s, 0 commands)

IOS-XE does not list per-module URIs such as Cisco-IOS-XE-native in the hello; modules are exposed via yang-library. That the native model is usable is proven by the hostname test above, which queries the native namespace directly.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- `${hit}` is not empty (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

09 Bgp

9.01 PASS BGP Sessions Are Established With The Expected Peer AS (0.1s, 4 commands)

The hub holds one session per spoke; each spoke holds exactly one.

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+ld+){5}\s+!S+\s+ld+` (checked 4x)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 190, main routing table version 190
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 87/78 prefixes, 99/90 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:51:02.025 ago)
```

```
Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 190, main routing table version 190
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 87/78 prefixes, 99/90 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:51:02.059 ago)
```

```
Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 202, main routing table version 202
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6168 total bytes of memory
BGP activity 96/87 prefixes, 105/96 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (00:51:01.094 ago)
```

```
Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

R3 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 202, main routing table version 202
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5952 total bytes of memory
BGP activity 102/93 prefixes, 105/96 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (00:51:01.132 ago)
```

```
Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.02 **PASS** Each Spoke Learns The Other Spoke Prefix Through The Hub (0.1s, 2 commands)

The defining property of this topology: spokes never peer with each other, so R3's prefix can only reach R2 via the hub, and its AS path must show both hops.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **#{via}** (checked 2×)
- output matches **#{R1_ASN}ls+#{far_asn}** (checked 2×)

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1

```
BGP routing table entry for 10.20.3.1/32, version 199
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:24:27 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1

```
BGP routing table entry for 10.20.2.1/32, version 200
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:24:31 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.03 **PASS** Every Peer Is Configured With MD5 Authentication (0.4s, 3 commands)

The negative test below proves a wrong password is rejected; this proves no peer was left without one in the first place.

Setup (suite) Establish BGP

Acceptance criteria

- **#{remote}** is not empty (checked 3×)
 - **#{remote}** matches **#{secured}** exactly (checked 3×)
- msg=#{r}: neighbours **#{remote}** but only **#{secured}** carry a password

How it was checked

R1 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.2 password LabBgpSecret
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.3 password LabBgpSecret
```

R2 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
```

R3 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.04 **PASS** A BFD Session Is Up For Every BGP Peer (0.1s, 3 commands)

One BFD session per peer, on the tunnel the peer address lives on -- the hub therefore holds two.

Setup (suite) Establish BGP

Acceptance criteria

- output contains `${R2_TUNNEL_LOCAL}`
- output contains `${R3_TUNNEL_LOCAL}`
- output does not contain **Down** (checked 3×)
- output contains `${r}_TUNNEL_HUB` (checked 2×)

How it was checked

R1 show bfd neighbors

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.2	4097/4097	Up	Up	Tu0
172.20.0.3	4098/4097	Up	Up	Tu0

R2 show bfd neighbors

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.1	4097/4097	Up	Up	Tu0

R3 show bfd neighbors

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.1	4097/4098	Up	Up	Tu0

Teardown (suite) Restore BGP Baseline, Close All Routers

9.05 PASS BGP Uses BFD For Fall-over On Every Peer (0.4s, 6 commands)

A BFD session that no protocol consumes would detect failures and change nothing, so this checks BGP is the registered client rather than just that BFD is running.

Setup (suite) Establish BGP

Acceptance criteria

- `${remote}` matches `${bfd}` exactly (checked 3×)
- msg=\${r}: neighbours \${remote} but only \${bfd} use BFD fall-over
- output does not contain **Down** (checked 3×)

How it was checked

R1 show running-config | include ^ neighbor .* remote-as|fall-over bfd

```
neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.2 fall-over bfd
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.3 fall-over bfd
```

R1 show bfd neighbors client bgp

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.2	4097/4097	Up	Up	Tu0
172.20.0.3	4098/4097	Up	Up	Tu0

R2 show running-config | include ^ neighbor .* remote-as|fall-over bfd

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 fall-over bfd
```

R2 show bfd neighbors client bgp

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.1	4097/4097	Up	Up	Tu0

R3 show running-config | include ^ neighbor .* remote-as|fall-over bfd

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 fall-over bfd
```

R3 show bfd neighbors client bgp

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.1	4097/4098	Up	Up	Tu0

Teardown (suite) Restore BGP Baseline, Close All Routers

9.06 PASS BFD Tears The Session Down Long Before The Hold Time (42.9s, 32 commands)

The test that makes BFD more than configuration. An ACL on the hub's tunnel drops only BFD control packets (UDP 3784 and 3785) and leaves BGP's TCP session untouched, so ordinary BGP liveness is unaffected. Without fall-over the session would survive until the 30s hold time expires; with it, BFD declares the peer down in about a second and BGP follows.

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+1d+){5}\s+1S+1s+1d+` (checked 4×)
- output does not match `^1d+$` (checked 3×)
- `${elapsed} < ${HOLD_TIME}` holds
- output contains **Known via "bgp** (checked 8×)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 190, main routing table version 190
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 87/78 prefixes, 99/90 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:51:03.123 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd
... 2 more lines
```

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 ip access-list extended BLOCK-BFD

`${output}` =

R1 deny udp any any eq 3784

`${output}` =

R1 deny udp any any eq 3785

`${output}` =

R1 permit ip any any

`${output}` =

R1 end

`${output}` =

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Tunnel0

`${output}` =

R1 ip access-group BLOCK-BFD in

`${output}` =

R1 end

`${output}` =

R1 show bgp ipv4 unicast summary | include 172.20.0.2

```
172.20.0.2      4      65002      18      21      190      0      0 00:01:48      3
```

R1 show bgp ipv4 unicast summary | include 172.20.0.2

```
172.20.0.2      4      65002      18      21      190      0      0 00:01:49      3
```

R1 show bgp ipv4 unicast summary | include 172.20.0.2

```
172.20.0.2      4      65002      0        0        1      0      0 00:00:00 Idle
```

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Tunnel0

`${output}` =

R1 no ip access-group BLOCK-BFD in

`${output}` =

R1 end

`${output}` =

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 no ip access-list extended BLOCK-BFD

```

${output} =
R1 end
${output} =
R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 196, main routing table version 196
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
1/1 BGP path/bestpath attribute entries using 296 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1448 total bytes of memory
BGP activity 87/84 prefixes, 99/96 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:51:05.638 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2    4      65002      0       0        1    0    0 00:00:00 Idle
172.20.0.3    4      65003      0       0        1    0    0 00:00:01 Idle

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 196, main routing table version 196
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
1/1 BGP path/bestpath attribute entries using 296 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1448 total bytes of memory
BGP activity 87/84 prefixes, 99/96 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:51:10.672 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2    4      65002      0       0        1    0    0 00:00:06 Idle
172.20.0.3    4      65003      0       0        1    0    0 00:00:06 Idle

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 202, main routing table version 202
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 93/84 prefixes, 105/96 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:51:15.760 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

R1 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:00 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:00 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32

```

```

Known via "bgp 65003", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.2 00:00:04 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.1, 00:00:04 ago
  opaque_ptr 0x7AED8E74CB50
  Route metric is 0, traffic share count is 1
  AS Hops 2
  Route tag 65001
  MPLS label: none

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.07 PASS Every Peer Has An Outbound Route-map Over Per-class Prefix-lists (0.7s, 13 commands)

One prefix-list per prefix class, and a route-map clause per class so each can carry its own community. The clauses are the whitelist: the route-map's implicit deny is what stops anything unnamed being advertised.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **RM-OUT- $\{peer\}$** (checked 4×)
- output contains **PL-OWN- $\{cls\}$** (checked 9×)

How it was checked

```

R2 show running-config | include route-map RM-OUT-R2 out
neighbor 172.20.0.2 route-map RM-OUT-R2 out

```

```

R3 show running-config | include route-map RM-OUT-R3 out
neighbor 172.20.0.3 route-map RM-OUT-R3 out

```

```

R1 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.1.1/32

```

```

R1 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.1.0/24

```

```

R1 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.10.0/24

```

```

R1 show running-config | include route-map RM-OUT-R1 out
neighbor 172.20.0.1 route-map RM-OUT-R1 out

```

```

R2 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.2.1/32

```

```

R2 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.2.0/24

```

```

R2 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.20.0/24

```

```

R1 show running-config | include route-map RM-OUT-R1 out
neighbor 172.20.0.1 route-map RM-OUT-R1 out

```

```

R3 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.3.1/32

```

```

R3 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.3.0/24

```

```

R3 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.30.0/24

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.08 PASS Every Peer Has An Inbound Policy Matching Community And AS Path (0.6s, 11

commands)

Both matches sit in one permit clause, so they are ANDed and the clause's implicit deny is the drop: a prefix with an unknown community *or* an unexpected AS path falls through.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **RM-IN- $\{peer\}$** (checked 4×)
- output contains **community (community-list filter): CL-VALID** (checked 4×)
- output matches **as-path $\{as-path\}$ filter): $\{as-path\}$** (checked 4×)
- **$\{entries\}$** has exactly 9 entries (checked 3×)

How it was checked

R2 show running-config | include route-map RM-IN-R2 in
neighbor 172.20.0.2 route-map RM-IN-R2 in

R2 show route-map RM-IN-R2
route-map RM-IN-R2, permit, sequence 10
Match clauses:
as-path (as-path filter): 10
community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes

R3 show running-config | include route-map RM-IN-R3 in
neighbor 172.20.0.3 route-map RM-IN-R3 in

R3 show route-map RM-IN-R3
route-map RM-IN-R3, permit, sequence 10
Match clauses:
as-path (as-path filter): 20
community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes

R1 show ip community-list CL-VALID
Named Community standard list CL-VALID
permit 65001:1
permit 65001:2
permit 65001:3
permit 65002:1
permit 65002:2
permit 65002:3
permit 65003:1
permit 65003:2
permit 65003:3

R1 show running-config | include route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-IN-R1 in

R1 show route-map RM-IN-R1
route-map RM-IN-R1, permit, sequence 10
Match clauses:
as-path (as-path filter): 10
community (community-list filter): CL-VALID
Set clauses:
Policy routing matches: 0 packets, 0 bytes

R2 show ip community-list CL-VALID
Named Community standard list CL-VALID
permit 65001:1
permit 65001:2
permit 65001:3
permit 65002:1
permit 65002:2
permit 65002:3
permit 65003:1
permit 65003:2
permit 65003:3

R1 show running-config | include route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-IN-R1 in

R1 show route-map RM-IN-R1
route-map RM-IN-R1, permit, sequence 10
Match clauses:
as-path (as-path filter): 10
community (community-list filter): CL-VALID
Set clauses:

Policy routing matches: 0 packets, 0 bytes

R3 show ip community-list CL-VALID

```
Named Community standard list CL-VALID
permit 65001:1
permit 65001:2
permit 65001:3
permit 65002:1
permit 65002:2
permit 65002:3
permit 65003:1
permit 65003:2
permit 65003:3
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.09 PASS Every Received Prefix Carries The Community Of Its Origin (0.6s, 18 commands)

A prefix is tagged by the router that owns it, so the community identifies both the origin AS and the kind of prefix. Checked on the receiver, which is the only place it proves the tag survived the wire.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${expected}** (checked 18×)

How it was checked

R1 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 202
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    31
  Refresh Epoch 1
  65002
    172.20.0.2 from 172.20.0.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC
```

R1 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 199
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    32
  Refresh Epoch 1
  65003
    172.20.0.3 from 172.20.0.3 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:24 UTC
```

R1 show bgp ipv4 unicast 10.30.2.0/24

```
BGP routing table entry for 10.30.2.0/24, version 201
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    31
  Refresh Epoch 1
  65002
    172.20.0.2 from 172.20.0.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC
```

R1 show bgp ipv4 unicast 10.30.3.0/24

```
BGP routing table entry for 10.30.3.0/24, version 198
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    32
  Refresh Epoch 1
  65003
    172.20.0.3 from 172.20.0.3 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:24 UTC
```

R1 show bgp ipv4 unicast 192.168.20.0/24

```

BGP routing table entry for 192.168.20.0/24, version 200
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    31
  Refresh Epoch 1
  65002
    172.20.0.2 from 172.20.0.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC

```

R1 show bgp ipv4 unicast 192.168.30.0/24

```

BGP routing table entry for 192.168.30.0/24, version 197
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    32
  Refresh Epoch 1
  65003
    172.20.0.3 from 172.20.0.3 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:24 UTC

```

R2 show bgp ipv4 unicast 10.20.1.1/32

```

BGP routing table entry for 10.20.1.1/32, version 211
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC

```

R2 show bgp ipv4 unicast 10.20.3.1/32

```

BGP routing table entry for 10.20.3.1/32, version 212
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC

```

R2 show bgp ipv4 unicast 10.30.1.0/24

```

BGP routing table entry for 10.30.1.0/24, version 210
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC

```

R2 show bgp ipv4 unicast 10.30.3.0/24

```

BGP routing table entry for 10.30.3.0/24, version 213
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC

```

R2 show bgp ipv4 unicast 192.168.10.0/24

```

BGP routing table entry for 192.168.10.0/24, version 209
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC

```

R2 show bgp ipv4 unicast 192.168.30.0/24

BGP routing table entry for 192.168.30.0/24, version 214
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:27 UTC

R3 show bgp ipv4 unicast 10.20.1.1/32

BGP routing table entry for 10.20.1.1/32, version 211
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:24 UTC

R3 show bgp ipv4 unicast 10.20.2.1/32

BGP routing table entry for 10.20.2.1/32, version 212
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:53 UTC

R3 show bgp ipv4 unicast 10.30.1.0/24

BGP routing table entry for 10.30.1.0/24, version 210
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:24 UTC

R3 show bgp ipv4 unicast 10.30.2.0/24

BGP routing table entry for 10.30.2.0/24, version 213
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:53 UTC

R3 show bgp ipv4 unicast 192.168.10.0/24

BGP routing table entry for 192.168.10.0/24, version 209
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:24 UTC

R3 show bgp ipv4 unicast 192.168.20.0/24

BGP routing table entry for 192.168.20.0/24, version 214
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 8 2026 02:26:53 UTC

Teardown (suite) Restore BGP Baseline, Close All Routers

9.10 **PASS** Re-advertised Prefixes Keep Their Origin Community Through The Hub (0.1s, 4 commands)

A spoke-to-spoke prefix crosses the hub, which re-advertises it. If the hub re-tagged it the community would say 65001 and the origin would be lost, so this asserts the far spoke's own community survives the hop.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **#{expected}** (checked 4x)

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 212
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC
```

R2 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 214
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:27 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 212
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:53 UTC
```

R3 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 214
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 8 2026 02:26:53 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.11 **PASS** A Prefix With An Unknown Community Is Rejected Inbound (31.1s, 41 commands)

*The prefix is explicitly permitted *outbound* by a temporary clause, so the outbound whitelist cannot be what stops it -- only the inbound community match can. The test asserts R2 really did advertise it before asserting R1 refused it, otherwise it would pass whenever the advertisement failed.*

Setup (suite) Establish BGP

Acceptance criteria

- output contains **#{BAD_COMM_ADDR}** (checked 7×)
R2 is not yet advertising the test prefix
- output contains **#{BAD_COMM_ADDR}**
R2 never advertised the prefix, so this proves nothing about the inbound policy
- output contains **Network not in table**
the hub accepted a prefix carrying community #{UNKNOWN_COMM}

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback98

#{output} =

R2 ip address 10.88.88.88 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 route-map RM-OUT-R1 permit 5

#{output} =

R2 match ip address prefix-list PL-TEST

#{output} =

R2 set community 65500:9

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 network 10.88.88.88 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 clear ip bgp * soft out

#{output} =

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
---------	----------	--------	--------	--------	------

```
*> 10.20.2.1/32      0.0.0.0          0          32768 i
*> 10.30.2.0/24      0.0.0.0          0          32768 i
*> 192.168.20.0      0.0.0.0          0          32768 i
```

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 215, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	10.88.88.88/32	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

```

... 1 more lines
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 215, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32    0.0.0.0             0         32768 i
*>  10.30.2.0/24    0.0.0.0             0         32768 i
*>  10.88.88.88/32  0.0.0.0             0         32768 i
*>  192.168.20.0   0.0.0.0             0         32768 i

```

```

... 1 more lines
R1 show bgp ipv4 unicast 10.88.88.88
% Network not in table

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 no network 10.88.88.88 mask 255.255.255.255
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no route-map RM-OUT-R1 permit 5
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip prefix-list PL-TEST
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no interface Loopback98
${output} =

R2 end
${output} =

```

Teardown (this test) Remove Test Prefix From R2

9.12 PASS Every Peer Has A Maximum-prefix Limit With Headroom (0.5s, 6 commands)

Filtering controls what a peer may send; this controls how much. The limit must sit above normal load or it would trip in steady state -- a hub holds 3 prefixes per spoke, a spoke 6 -- and far below anything that could exhaust memory.

Setup (suite) Establish BGP

Acceptance criteria

- `${remote}` matches `${limited}` exactly (checked 3×)
msg=\${r}: neighbours \${remote} but only \${limited} have a maximum-prefix
- `${n} < ${MAX_PREFIX}` holds (checked 3×)

How it was checked

```

R1 show running-config | include ^ neighbor .* remote-as|maximum-prefix
neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.2 maximum-prefix 15 80
neighbor 172.20.0.3 maximum-prefix 15 80

```


R1 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.2	4	65002	18	16	202	0	0	00:01:04	3
172.20.0.3	4	65003	13	16	202	0	0	00:01:07	3

R2 show running-config | include ^ neighbor .* remote-as|maximum-prefix

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 maximum-prefix 15 80
```

R2 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	16	18	215	0	0	00:01:04	6

R3 show running-config | include ^ neighbor .* remote-as|maximum-prefix

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 maximum-prefix 15 80
```

R3 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	16	14	214	0	0	00:01:07	6

Teardown (suite) Restore BGP Baseline, Close All Routers**9.13 PASS A Peer That Floods Prefixes Is Torn Down Rather Than Absorbed** (32.3s, 343

commands)

R2 originates 20 extra prefixes, tagged with a valid community and carrying a valid AS path, and explicitly permitted outbound -- so neither the whitelist, the community match nor the AS-path filter can be what stops them. Only the prefix count is left. The hub must drop the session instead of accepting an unbounded table.

Setup (suite) Establish BGP**Acceptance criteria**

- output does not match `^ld+$` (checked 6x)
- output contains **PfxCt**
the session went down, but not because of the prefix count: `$(state)`
- output contains `$(R2_TUNNEL_LOCAL)`
the hub logged no maximum-prefix violation for `$(R2_TUNNEL_LOCAL)`

How it was checked**R2 configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip prefix-list PL-FLOOD seq 5 permit 10.99.0.0/16 le 32`$(output) =`**R2 end**`$(output) =`**R2 configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

R2 route-map RM-OUT-R1 permit 7`$(output) =`**R2 match ip address prefix-list PL-FLOOD**`$(output) =`**R2 set community 65002:1**`$(output) =`**R2 end**`$(output) =`**R2 configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.0.0 255.255.255.0 Null0`$(output) =`**R2 end**`$(output) =`**R2 configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.1.0 255.255.255.0 Null0`$(output) =`**R2 end**`$(output) =`**R2 configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

```

R2 ip route 10.99.2.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.3.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.4.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.5.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.6.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.7.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.8.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.9.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.10.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.11.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.12.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal

```

```

Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.13.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.14.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.15.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.16.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.17.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.18.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.19.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.99.0.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.99.1.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =

```

```

R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.2.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.3.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.4.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.5.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.6.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.7.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end

```

```

    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.9.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.10.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.11.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.12.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.13.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.14.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =

```

```

R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.15.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.17.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.18.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.19.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      18      17      202      0      0 00:01:07      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      19      17      202      0      0 00:01:12      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      19      18      202      0      0 00:01:17      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      20      18      202      0      0 00:01:23      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      20      19      202      0      0 00:01:28      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      0      0      1      0      0 00:00:00 Idle (PfxCt)
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      0      0      1      0      0 00:00:00 Idle (PfxCt)
R1 show logging | include MAXPFXEXCEED
001028: Sep  8 02:27:59.362 UTC: %BGP-3-MAXPFXEXCEED: Number of prefixes received from 172.20.0.2 1/1
(afi/safi): 16 exceeds limit 15
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.

```

```

R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.0.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.0.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.1.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.1.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.2.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.2.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.3.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.3.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002

```

```

    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.4.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.4.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.5.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.5.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.6.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.6.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.7.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.7.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =

```



```

R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.8.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.9.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.9.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.10.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.10.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.11.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.11.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast

```

```

    ${output} =
R2 no network 10.99.12.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.12.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.13.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.13.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.14.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.14.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.15.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.15.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =

```

```

R2 no network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.16.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.17.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.17.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.18.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.18.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.19.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.19.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no route-map RM-OUT-R1 permit 7
  ${output} =
R2 end
  ${output} =
R2 configure terminal

```

Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip prefix-list PL-FL00D

#{output} =

R2 end

#{output} =

R1 clear ip bgp 172.20.0.2

#{output} =

Teardown (this test) Stop Flooding From R2

9.14 PASS The Session Recovers Once The Flood Stops (30.2s, 9 commands)

A maximum-prefix teardown is deliberately sticky -- no "restart" is configured -- so recovery takes an operator clearing it. This asserts the lab returns to a converged state rather than leaving the next suite a broken topology.

Setup (suite) Establish BGP

Acceptance criteria

- output matches `#{peer}ls+4ls+#{peer_asn}(ls+ld+){5}ls+ls+ls+ld+` (checked 2×)
- output contains **Known via "bgp"** (checked 7×)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 229, main routing table version 229
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
8/4 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
7 BGP route-map cache entries using 504 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5392 total bytes of memory
BGP activity 106/100 prefixes, 118/112 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:00:04.872 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 232, main routing table version 232
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
8/7 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
7 BGP route-map cache entries using 504 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6544 total bytes of memory
BGP activity 109/100 prefixes, 121/112 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:00:09.908 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:00 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:00 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

```
% Subnet not in table
R3 show ip route 10.20.2.1
% Subnet not in table
R3 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:04 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:04 ago
      opaque_ptr 0x7AED8E74CB50
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.15 PASS A Prefix With An Invalid AS Path Is Rejected Inbound (5.8s, 37 commands)

Same prefix, this time with a valid community but an AS path this topology cannot produce. Only the AS-path match differs, so a rejection here isolates that filter.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **#{BAD_COMM_ADDR}** (checked 2x)
R2 is not yet advertising the test prefix
- output contains **#{BAD_COMM_ADDR}**
R2 never advertised the prefix, so this proves nothing about the inbound policy
- output contains **Network not in table**
the hub accepted a prefix whose AS path contained **#{BOGUS_AS}**

How it was checked

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 interface Loopback98
  #{output} =
R2 ip address 10.88.88.88 255.255.255.255
  #{output} =
R2 end
  #{output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32
  #{output} =
R2 end
  #{output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 route-map RM-OUT-R1 permit 5
  #{output} =
R2 match ip address prefix-list PL-TEST
  #{output} =
R2 set community 65002:1
  #{output} =
R2 set as-path prepend 65099
  #{output} =
R2 end
  #{output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  #{output} =
R2 address-family ipv4 unicast
  #{output} =
R2 network 10.88.88.88 mask 255.255.255.255
```

```

${output} =
R2 end
${output} =
R2 clear ip bgp * soft out
${output} =
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 268, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32     0.0.0.0              0           32768 i
*>  10.30.2.0/24     0.0.0.0              0           32768 i
*>  192.168.20.0     0.0.0.0              0           32768 i

Total number of prefixes 3
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 269, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32     0.0.0.0              0           32768 i
*>  10.30.2.0/24     0.0.0.0              0           32768 i
*>  10.88.88.88/32   0.0.0.0              0           32768 i
*>  192.168.20.0     0.0.0.0              0           32768 i

... 1 more lines
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 269, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32     0.0.0.0              0           32768 i
*>  10.30.2.0/24     0.0.0.0              0           32768 i
*>  10.88.88.88/32   0.0.0.0              0           32768 i
*>  192.168.20.0     0.0.0.0              0           32768 i

... 1 more lines
R1 show bgp ipv4 unicast 10.88.88.88
% Network not in table
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.88.88.88 mask 255.255.255.255
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no route-map RM-OUT-R1 permit 5
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip prefix-list PL-TEST

```

```

${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no interface Loopback98
${output} =
R2 end
${output} =

```

Teardown (this test) Remove Test Prefix From R2

9.16 PASS Each Peer Is Advertised Exactly Its Whitelisted Prefixes (0.3s, 4 commands)

Not "at least" -- exactly. A whitelist that permits more than intended still passes a containment check, so the advertised set is compared for equality against what the peer should get: a spoke sends only its own prefixes, the hub sends its own plus the far spoke's.

Setup (suite) Establish BGP

Acceptance criteria

- **\${actual}** matches **\${expected}** exactly (checked 4×)
- msg=\${from} advertises \${actual} to \${to}, expected \${expected}

How it was checked

R2 show bgp ipv4 unicast neighbors 172.20.0.2 advertised-routes

BGP table version is 232, local router ID is 172.16.1.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3	0		0	65003 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.3.0/24	172.20.0.3	0		0	65003 i
*> 192.168.10.0	0.0.0.0	0		32768	i

... 3 more lines

R3 show bgp ipv4 unicast neighbors 172.20.0.3 advertised-routes

BGP table version is 232, local router ID is 172.16.1.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	172.20.0.2	0		0	65002 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	172.20.0.2	0		0	65002 i
*> 192.168.10.0	0.0.0.0	0		32768	i

... 3 more lines

R1 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 270, local router ID is 172.16.2.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R1 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```

BGP table version is 220, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.3.1/32	0.0.0.0	0		32768	i
*>	10.30.3.0/24	0.0.0.0	0		32768	i
*>	192.168.30.0	0.0.0.0	0		32768	i

Total number of prefixes 3

Teardown (suite) Restore BGP Baseline, Close All Routers

9.17 PASS A Prefix Outside The Whitelist Is Not Advertised (10.5s, 20 commands)

The point of a whitelist. A spoke originates a prefix nobody authorised; it appears in that spoke's own BGP table, so the origination worked, and must still never reach the hub.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${OFF_LIST_ADDR}**
R2 never originated the prefix, so this test proves nothing
- output does not contain **\${OFF_LIST_ADDR}/32**
R2 advertised an unlisted prefix to the hub
- output contains **Network not in table**
an unlisted prefix reached the hub

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback99

\${output} =

R2 ip address 10.77.77.77 255.255.255.255

\${output} =

R2 end

\${output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

\${output} =

R2 address-family ipv4 unicast

\${output} =

R2 network 10.77.77.77 mask 255.255.255.255

\${output} =

R2 end

\${output} =

R2 show bgp ipv4 unicast 10.77.77.77

BGP routing table entry for 10.77.77.77/32, version 271

Paths: (1 available, best #1, table default)

Flag: 0x8100

Not advertised to any peer

Refresh Epoch 1

Local

0.0.0.0 from 0.0.0.0 (172.16.2.1)

Origin IGP, metric 0, localpref 100, weight 32768, valid, sourced, local, best

rx pathid: 0, tx pathid: 0x0

Updated on Sep 8 2026 02:28:40 UTC

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 271, local router ID is 172.16.2.1

```

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,

```

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
--	---------	----------	--------	--------	--------	------


```
*> 10.20.2.1/32      0.0.0.0          0          32768 i
*> 10.30.2.0/24      0.0.0.0          0          32768 i
*> 192.168.20.0      0.0.0.0          0          32768 i
```

Total number of prefixes 3

R1 show bgp ipv4 unicast 10.77.77.77

% Network not in table

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 no network 10.77.77.77 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no interface Loopback99

#{output} =

R2 end

#{output} =

Teardown (suite) Restore BGP Baseline, Close All Routers

9.18 PASS Each Router Receives The Peer Prefix (0.1s, 2 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output matches `!*>ls+#{R2_BGP_PREFIX}/32ls+#{R2_TUNNEL_LOCAL}`
- output contains `#{R2_ASN}`
- output matches `!*>ls+#{R1_BGP_PREFIX}/32ls+#{R2_TUNNEL_HUB}`
- output contains `#{R1_ASN}`

How it was checked

R1 show bgp ipv4 unicast

BGP table version is 232, local router ID is 172.16.1.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	172.20.0.2	0		0	65002 i
*> 10.20.3.1/32	172.20.0.3	0		0	65003 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	172.20.0.2	0		0	65002 i

... 4 more lines

R2 show bgp ipv4 unicast

BGP table version is 272, local router ID is 172.16.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	172.20.0.1	0		0	65001 i
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3	0		0	65001 65003 i
*> 10.30.1.0/24	172.20.0.1	0		0	65001 i
*> 10.30.2.0/24	0.0.0.0	0		32768	i

... 4 more lines

Teardown (suite) Restore BGP Baseline, Close All Routers

9.19 PASS Peer Prefix Is Learned Via BGP And Not A Static Route (0.0s, 1 commands)

Distinguishes a genuinely BGP-installed route from one that merely happens to exist. eBGP is distance 20; a static is 1.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Known via "bgp \${R1_ASN}"**
- output contains **distance 20**
- output contains **type external**

How it was checked

R1 `show ip route 10.20.2.1`

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:42 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:42 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.20 PASS BGP Route Resolves Out The Encrypted Tunnel (0.0s, 1 commands)

The next hop is a tunnel address, so the forwarding path must recurse to Tunnel0 rather than the cleartext link.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **\${R2_TUNNEL_LOCAL}**
- output contains **Tunnel0**

How it was checked

R1 `show ip cef 10.20.2.1`

```
10.20.2.1/32
  nexthop 172.20.0.2 Tunnel0
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.21 PASS Traffic Over The BGP-Learned Route Reaches The Peer (0.0s, 1 commands)

End to end: this only succeeds because BGP installed the route.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

R1 `ping 10.20.2.1 source Loopback1 repeat 5`

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.22 PASS BGP-Routed Traffic Is Encrypted (5.7s, 7 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Success rate is 100 percent**
- **`\${r1_enc} >= \${expected}** holds (checked 2×)
- **`\${r1_dec} >= \${expected}** holds (checked 2×)
- **`\${r2_enc} >= \${expected}** holds (checked 2×)
- **`\${r2_dec} >= \${expected}** holds (checked 2×)

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 979, #pkts encrypt: 979, #pkts digest: 979
  #pkts decaps: 987, #pkts decrypt: 987, #pkts verify: 987
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2289, #pkts encrypt: 2289, #pkts digest: 2289
  #pkts decaps: 2270, #pkts decrypt: 2270, #pkts verify: 2270
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines
```

R1 ping 10.20.2.1 source Loopback1 repeat 20

```
Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (20/20), round-trip min/avg/max = 1/1/1 ms
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 979, #pkts encrypt: 979, #pkts digest: 979
  #pkts decaps: 987, #pkts decrypt: 987, #pkts verify: 987
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2289, #pkts encrypt: 2289, #pkts digest: 2289
  #pkts decaps: 2270, #pkts decrypt: 2270, #pkts verify: 2270
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
```

```

    #send errors 0, #recv errors 0
... 33 more lines
R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 1035, #pkts encrypt: 1035, #pkts digest: 1035
    #pkts decaps: 1048, #pkts decrypt: 1048, #pkts verify: 1048
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines

```

```

R2 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  current_peer 100.64.0.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 2344, #pkts encrypt: 2344, #pkts digest: 2344
    #pkts decaps: 2326, #pkts decrypt: 2326, #pkts verify: 2326
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.23 PASS BGP Control Plane Traffic Is Encrypted On The Wire (19.2s, 8 commands)

The reason for peering over the tunnel. Captures a keepalive window and asserts the session is invisible: no TCP, no peering addresses, no advertised prefixes -- only ESP.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **ESP**
- output does not contain **TCP**
- output does not contain **#{R2_TUNNEL_HUB}**
- output does not contain **#{R2_TUNNEL_LOCAL}**
- output does not contain **#{R1_BGP_PREFIX}**
- output does not contain **#{R2_BGP_PREFIX}**

How it was checked

R1 no monitor capture BGPCAP

Capture does not exist

R1 monitor capture BGPCAP interface GigabitEthernet2 both

#{output} =

R1 monitor capture BGPCAP match ipv4 any any

#{output} =

R1 monitor capture BGPCAP buffer size 5

#{output} =

R1 monitor capture BGPCAP start

Started capture point : BGPCAP

R1 monitor capture BGPCAP stop

Stopped capture point : BGPCAP

R1 show monitor capture BGPCAP buffer brief

#	size	timestamp	source	destination	dscp	protocol
0	138	0.000000	100.64.0.1	-> 100.64.0.3	48 CS6	ESP
1	138	0.000000	100.64.0.3	-> 100.64.0.1	48 CS6	ESP

```

2 138 0.018004 100.64.0.3 -> 100.64.0.1 48 CS6 ESP
3 138 0.018004 100.64.0.1 -> 100.64.0.3 48 CS6 ESP
4 154 0.025999 100.64.0.3 -> 100.64.0.1 48 CS6 ESP
5 138 0.167990 100.64.0.2 -> 100.64.0.1 48 CS6 ESP
6 138 0.167990 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
7 138 0.183981 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
8 138 0.183981 100.64.0.2 -> 100.64.0.1 48 CS6 ESP
9 138 0.221988 100.64.0.1 -> 100.64.0.3 48 CS6 ESP
10 138 0.339038 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
... 326 more lines

```

```

R1 no monitor capture BGPCAP
${output} =

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.24 PASS Withdrawing A Prefix Removes It From The Peer (60.5s, 23 commands)

Shuts the advertised interface on R2 and requires R1 to lose both the BGP path and the route, then restores it.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **% Subnet not in table** (checked 6×)
- output contains **Known via "bgp"** (checked 9×)

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback1

```

${output} =

```

R2 shutdown

```

${output} =

```

R2 end

```

${output} =

```

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:07 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:01:07 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

```

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:12 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:01:12 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

```

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:17 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:01:17 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

```

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external

```

```
Last update from 172.20.0.2 00:01:22 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.2, 00:01:22 ago
  opaque_ptr 0x71FE1BFD8DB8
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:27 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:27 ago
    opaque_ptr 0x71FE1BFD8DB8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface Loopback1

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:05 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:05 ago
    opaque_ptr 0x71FE1BFD8DB8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none
```

R2 show ip route 10.20.1.1

```
Routing entry for 10.20.1.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:02:08 ago
  Routing Descriptor Blocks:
  * 172.20.0.1, from 172.20.0.1, 00:02:08 ago
    opaque_ptr 0x7984168426C8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65001
    MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.25 PASS Session Recovers After A Hard Clear (15.2s, 6 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output matches `${peer}\s+4\s+${peer_asn}(\s+ld+){5}\s+lS+\s+ld+` (checked 4x)
- output contains **Known via "bgp"**

How it was checked

R1 `clear ip bgp *`

`${output} =`

R1 `show bgp ipv4 unicast summary`

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 113/110 prefixes, 125/122 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:02:17.374 ago)

Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 `show bgp ipv4 unicast summary`

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 113/110 prefixes, 125/122 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:02:22.414 ago)

Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 `show bgp ipv4 unicast summary`

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 113/110 prefixes, 125/122 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:02:27.448 ago)

Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 `show bgp ipv4 unicast summary`

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 119/110 prefixes, 131/122 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:02:32.488 ago)

Neighbor          V              AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 `show ip route 10.20.2.1`

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:03 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:03 ago
      opaque_ptr 0x71FE1BFD8DB8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.26 PASS Mismatched MD5 Password Prevents The Session Coming Up (50.4s, 19 commands)

Negative control on BGP authentication, mirroring the IPsec wrong-PSK test: without it, a session that accepted any peer would still pass everything above.

Setup (suite) Establish BGP

Acceptance criteria

- output does not match `${peer}{ls+ld+}{6}ls+S+ls+ld+ls*$`
- output matches `${peer}ls+4ls+${peer_asn}{ls+ld+}{5}ls+S+ls+ld+` (checked 2x)

How it was checked

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

`${output}` =

R2 neighbor 172.20.0.1 password WrongBgpSecret999

`${output}` =

R2 end

`${output}` =

R1 clear ip bgp *

`${output}` =

R2 clear ip bgp *

`${output}` =

R1 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5000 total bytes of memory
BGP activity 125/119 prefixes, 137/131 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:03:17.654 ago)

```

```

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

`${output}` =

R2 neighbor 172.20.0.1 password LabBgpSecret

`${output}` =

R2 end

`${output}` =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback1

`${output}` =

R2 no shutdown

`${output}` =

R2 end

`${output}` =


```

R1 clear ip bgp *
${output} =
R2 clear ip bgp *
${output} =
R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3848 total bytes of memory
BGP activity 128/125 prefixes, 140/137 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:03:17.848 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3848 total bytes of memory
BGP activity 128/125 prefixes, 140/137 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:03:22.887 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
Teardown (this test) Restore BGP Baseline

```

10 Host To Host

10.01 **PASS** Every Host Holds Its LAN Address (0.1s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains `${HOST_IP}[${h}]` (checked 3×)

How it was checked

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.02 **PASS** Every Host Routes The Other Two LANs Via Its Own Router (0.0s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output matches `${net}/24 via ${HOST_GW}[${h}]` (checked 9×)

How it was checked

H1 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
```

```
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
```

H3 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.03 PASS Every Host Can Reach Its Default Gateway (12.3s, 3 commands)

Isolates the host-to-router leg so a failure further along is not misread as a broken LAN.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss** (checked 3×)
ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H1 ping -c 5 192.168.10.1

```
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.429 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=0.512 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.447 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=0.457 ms
64 bytes from 192.168.10.1: icmp_seq=5 ttl=255 time=0.465 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4077ms
rtt min/avg/max/mdev = 0.429/0.462/0.512/0.027 ms
```

H2 ping -c 5 192.168.20.1

```
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=0.509 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=0.449 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=0.411 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=0.428 ms
64 bytes from 192.168.20.1: icmp_seq=5 ttl=255 time=0.428 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4079ms
rtt min/avg/max/mdev = 0.411/0.445/0.509/0.034 ms
```

H3 ping -c 5 192.168.30.1

```
PING 192.168.30.1 (192.168.30.1) 56(84) bytes of data.
64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=0.540 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=0.422 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=0.763 ms
64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=0.431 ms
64 bytes from 192.168.30.1: icmp_seq=5 ttl=255 time=1.04 ms

--- 192.168.30.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4081ms
rtt min/avg/max/mdev = 0.422/0.638/1.035/0.233 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.04 PASS Every Host Can Reach Every Other Host (24.1s, 6 commands)

The full mesh, including h2 <-> h3 which never touches a direct link and can only work if the hub re-advertises between spokes.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss** (checked 6×)
ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.18 ms  
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.27 ms  
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.62 ms  
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.25 ms  
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.15 ms  
  
--- 192.168.20.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.153/1.293/1.623/0.169 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.  
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=2.20 ms  
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.27 ms  
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.41 ms  
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.21 ms  
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.25 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.213/1.468/2.199/0.371 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.  
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.20 ms  
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.42 ms  
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=2.69 ms  
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.56 ms  
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.17 ms  
  
--- 192.168.10.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4007ms  
rtt min/avg/max/mdev = 1.171/1.607/2.694/0.561 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.  
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.34 ms  
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.16 ms  
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.44 ms  
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.25 ms  
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.25 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.159/1.285/1.435/0.093 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.  
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.30 ms  
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.26 ms  
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.16 ms  
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.30 ms  
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.16 ms  
  
--- 192.168.10.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.156/1.235/1.304/0.063 ms
```

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=2.14 ms  
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.33 ms  
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.57 ms  
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.38 ms  
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.35 ms  
  
--- 192.168.20.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.332/1.553/2.135/0.303 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.05 PASS Spoke To Spoke Traffic Is Routed Through The Hub (0.1s, 3 commands)

R2 must reach R3's LAN via the hub's tunnel address, not by any direct path -- there is no spoke-to-spoke link.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **Known via "bgp \${R2_ASN}"**
- output contains **\${R2_TUNNEL_HUB}**
- output contains **Tunnel0** (checked 2×)

How it was checked

R2 show ip route 192.168.30.0

```
Routing entry for 192.168.30.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:01:32 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:01:32 ago
    opaque_ptr 0x7984168426C8
    Route metric is 0, traffic share count is 1
    AS Hops 2
    Route tag 65001
    MPLS label: none
```

R2 show ip cef 192.168.30.10

```
192.168.30.0/24
  nexthop 172.20.0.3 Tunnel0
```

R3 show ip cef 192.168.20.10

```
192.168.20.0/24
  nexthop 172.20.0.2 Tunnel0
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.06 PASS Hub Reaches Both Spoke LANs On Separate Tunnels (0.0s, 2 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **\${DMVPN_TUNNEL}** (checked 2×)

How it was checked

R1 show ip cef 192.168.20.10

```
192.168.20.0/24
  nexthop 172.20.0.2 Tunnel0
```

R1 show ip cef 192.168.30.10

```
192.168.30.0/24
  nexthop 172.20.0.3 Tunnel0
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.07 PASS Host Traffic Is Encrypted In Transit (19.6s, 5 commands)

No host runs crypto: the routers encrypt for them, so the ESP counters must move by the traffic the hosts generated.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- **\${r1_enc} >= \${expected}** holds
- **\${r1_dec} >= \${expected}** holds
- **\${r2_enc} >= \${expected}** holds
- **\${r2_dec} >= \${expected}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1
```

```

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 2594, #pkts encrypt: 2594, #pkts digest: 2594
#pkts decaps: 2603, #pkts decrypt: 2603, #pkts verify: 2603
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 257, #pkts encrypt: 257, #pkts digest: 257
#pkts decaps: 257, #pkts decrypt: 257, #pkts verify: 257
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

H1 ping -c 20 192.168.20.10

```

PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.25 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.29 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.25 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.05 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.09 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.26 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.18 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.15 ms
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.27 ms
64 bytes from 192.168.20.10: icmp_seq=11 ttl=62 time=1.23 ms
64 bytes from 192.168.20.10: icmp_seq=12 ttl=62 time=2.23 ms
64 bytes from 192.168.20.10: icmp_seq=13 ttl=62 time=1.41 ms
... 11 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 2729, #pkts encrypt: 2729, #pkts digest: 2729
#pkts decaps: 2733, #pkts decrypt: 2733, #pkts verify: 2733
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 364, #pkts encrypt: 364, #pkts digest: 364
#pkts decaps: 371, #pkts decrypt: 371, #pkts verify: 371
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.08 PASS Host Addresses Never Appear In Cleartext On The Spoke Link (11.6s, 9 commands)

Capture the R1<->R2 link while the hosts talk: only ESP between the link addresses should be visible.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **ESP**
- output does not contain **\${H1_IP}**
- output does not contain **\${H2_IP}**
- output does not contain **ICMP**

How it was checked

R1 no monitor capture HOSTCAP

Capture does not exist

R1 monitor capture HOSTCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture HOSTCAP match ipv4 any any

\${output} =

R1 monitor capture HOSTCAP buffer size 5

\${output} =

R1 monitor capture HOSTCAP start

Started capture point : HOSTCAP

H1 ping -c 10 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
 64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=2.15 ms
 64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.23 ms
 64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.28 ms
 64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.06 ms
 64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.38 ms
 64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.33 ms
 64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.32 ms
 64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.26 ms
 64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.24 ms
 64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.19 ms

--- 192.168.20.10 ping statistics ---
 10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines
```

R1 monitor capture HOSTCAP stop

Stopped capture point : HOSTCAP

R1 show monitor capture HOSTCAP buffer brief

#	size	timestamp	source	destination	dscp	protocol
0	170	0.000000	100.64.0.1	-> 100.64.0.2	0 BE	ESP
1	170	0.001007	100.64.0.2	-> 100.64.0.1	0 BE	ESP
2	138	0.052000	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
3	138	0.053007	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
4	138	0.070004	100.64.0.3	-> 100.64.0.1	48 CS6	ESP
5	138	0.070004	100.64.0.1	-> 100.64.0.3	48 CS6	ESP
6	138	0.335041	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
7	138	0.335041	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
8	138	0.340031	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
9	138	0.358035	100.64.0.1	-> 100.64.0.3	48 CS6	ESP
10	138	0.358035	100.64.0.3	-> 100.64.0.1	48 CS6	ESP

... 220 more lines

R1 no monitor capture HOSTCAP

\${output} =

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.09 PASS Spoke To Spoke Traffic Is Encrypted On The Direct Path (11.7s, 11 commands)

The VTI lab this mirrors captured the hub's second link, because h2 to h3 crossed two tunnels and was decrypted and re-encrypted in between. Under DMVPN there is no second leg to capture: once the shortcut forms the hub is not on the path at all, so the capture moves to the spoke's own WAN interface -- the only place that traffic now appears. It should be ESP there, with neither host address visible.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **\${R3_WAN_IP}**
R2 has not resolved R3 directly, so no shortcut is in place
- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **ESP**
- output does not contain **\${H2_IP}**
- output does not contain **\${H3_IP}**
- output does not contain **ICMP**

How it was checked

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/9 ms
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:01:14, expire 00:03:45
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3
```

R2 no monitor capture HOSTCAP

Capture does not exist

R2 monitor capture HOSTCAP interface GigabitEthernet2 both

\${output} =

R2 monitor capture HOSTCAP match ipv4 any any

\${output} =

R2 monitor capture HOSTCAP buffer size 5

\${output} =

R2 monitor capture HOSTCAP start

Started capture point : HOSTCAP

H2 ping -c 10 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
 64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.32 ms
 64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.16 ms
 64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.23 ms
 64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.21 ms
 64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.24 ms
 64 bytes from 192.168.30.10: icmp_seq=6 ttl=62 time=1.18 ms
 64 bytes from 192.168.30.10: icmp_seq=7 ttl=62 time=1.72 ms
 64 bytes from 192.168.30.10: icmp_seq=8 ttl=62 time=2.26 ms
 64 bytes from 192.168.30.10: icmp_seq=9 ttl=62 time=1.30 ms
 64 bytes from 192.168.30.10: icmp_seq=10 ttl=62 time=1.09 ms
```

```
--- 192.168.30.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines
```

R2 monitor capture HOSTCAP stop

Stopped capture point : HOSTCAP

R2 show monitor capture HOSTCAP buffer brief

#	size	timestamp	source	destination	dscp	protocol
0	170	0.000000	100.64.0.2	-> 100.64.0.3	0 BE	ESP
1	170	0.001007	100.64.0.3	-> 100.64.0.2	0 BE	ESP
2	138	0.079006	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
3	138	0.079006	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
4	138	0.090999	100.64.0.2	-> 100.64.0.3	48 CS6	ESP
5	138	0.275987	100.64.0.3	-> 100.64.0.2	48 CS6	ESP
6	138	0.275987	100.64.0.2	-> 100.64.0.3	48 CS6	ESP
7	138	0.279984	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
8	138	0.346982	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
9	138	0.346982	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
10	138	0.385042	100.64.0.2	-> 100.64.0.3	48 CS6	ESP

... 218 more lines

R2 no monitor capture HOSTCAP

\${output} =

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.10 **PASS** The Hub Is Not On The Spoke To Spoke Path (0.0s, 2 commands)

The routing consequence of the shortcut, from the hosts' point of view rather than the routers'.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **#{R3_WAN_IP}**
R2 has not resolved R3 directly, so no shortcut is in place
- output does not contain **#{R1_DMVPN_IP}**
host traffic between the spokes still transits the hub:\n#{trace}

How it was checked

R2 `show ip nhrp 172.20.0.3`

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:01:26, expire 00:03:33
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3
```

H2 `traceroute -n -q 1 -w 1 192.168.30.10`

```
traceroute to 192.168.30.10 (192.168.30.10), 30 hops max, 60 byte packets
 1  192.168.20.1  0.934 ms
 2  172.20.0.3  2.530 ms
 3  192.168.30.10  2.554 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

11 Throughput

11.01 **PASS** Hub To Spoke Hosts Sustain The Target Throughput (32.2s, 2 commands)

h1 -> h2 crosses one tunnel.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- `${m}` is not empty
- `${mbps} >= ${TP_TARGET}` holds

How it was checked

```
H1 nc -l -p 5001 > /dev/null 2>&1 &
${output} =
```

```
H1 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=$(( $(date +%s) - S ))
SECS=31
```

Teardown (suite) Close All Connections

11.02 **PASS** Spoke To Spoke Hosts Sustain The Target Throughput (32.1s, 2 commands)

h2 -> h3 crosses two tunnels, being decrypted and re-encrypted at the hub, so it is the more demanding path.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- `${m}` is not empty
- `${mbps} >= ${TP_TARGET}` holds

How it was checked

```
H2 nc -l -p 5001 > /dev/null 2>&1 &
${output} =
```

```
H2 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.30.10 5001; echo SECS=$(( $(date +%s) - S ))
SECS=31
```

Teardown (suite) Close All Connections

11.03 **PASS** Bulk Throughput Traffic Is Encrypted (17.5s, 6 commands)

Rate alone says nothing about protection. A transfer of this size must move thousands of packets through ESP, so assert the counters rose in proportion to the bytes actually sent.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- `${m}` is not empty
- `${data_out} >= ${expected}` holds
- `${data_in} >= ${expected}` holds
- `${ack_in} > 0` holds
- `${ack_out} > 0` holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 53070, #pkts encrypt: 53070, #pkts digest: 53070
#pkts decaps: 28379, #pkts decrypt: 28379, #pkts verify: 28379
#pkts compressed: 0, #pkts decompressed: 0
```

```

#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 50716, #pkts encrypt: 50716, #pkts digest: 50716
#pkts decaps: 26229, #pkts decrypt: 26229, #pkts verify: 26229
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

H1 nc -l -p 5001 > /dev/null 2>&1 &

```
[1]+  Stopped (tty input) nc -l -p 5001 1>/dev/null 2>&1
```

H1 S=\$(date +%s); dd if=/dev/zero bs=64k count=512 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=\$((\$(date +%s) - S))

```
SECS=16
```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 78079, #pkts encrypt: 78079, #pkts digest: 78079
#pkts decaps: 40693, #pkts decrypt: 40693, #pkts verify: 40693
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 50814, #pkts encrypt: 50814, #pkts digest: 50814
#pkts decaps: 26326, #pkts decrypt: 26326, #pkts verify: 26326
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

Teardown (suite) Close All Connections

12 Nat

12.01 PASS Every Router Has A Static Source NAT Mapping For Its Host (0.3s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains `${HOST_IP}${h}}` (checked 3×)
- output contains `${HOST_NAT}${h}}` (checked 3×)
- output contains `route-map NAT-TO-DOMAIN` (checked 3×)

How it was checked

H1 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H2 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H3 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

Teardown (suite) Close All Connections

12.02 PASS NAT Inside And Outside Interfaces Are Marked Correctly (0.3s, 7 commands)

The LAN faces inside, the tunnels face outside. Getting this backwards silently disables translation rather than erroring.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains `ip nat inside` (checked 3×)
- output contains `ip nat outside` (checked 4×)

How it was checked

R1 show running-config interface GigabitEthernet3

Building configuration...

```
Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 negotiation auto
```

```

end

R1 show running-config interface Tunnel0
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines

R2 show running-config interface GigabitEthernet3
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 negotiation auto
end

R2 show running-config interface Tunnel0
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines

R3 show running-config interface GigabitEthernet3
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.30.1 255.255.255.0
 ip nat inside
 negotiation auto
end

R3 show running-config interface Tunnel0
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines

R1 show running-config interface Tunnel0
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub

```

```

ip address 172.20.0.1 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp redirect
ip tcp adjust-mss 1360
... 6 more lines

```

Teardown (suite) Close All Connections

12.03 **PASS** Mapped Prefixes Are Advertised Across The Fabric (0.1s, 3 commands)

Return traffic can only find its way home if each site's NAT range is reachable, so the mapped prefixes ride BGP like the real ones.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **Known via "bgp \${R2_ASN}"** (checked 2×)
- output contains **Known via "bgp \${R3_ASN}"**

How it was checked

R2 show ip route 10.30.1.0

```

Routing entry for 10.30.1.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:03:51 ago
  Routing Descriptor Blocks:
    * 172.20.0.1, from 172.20.0.1, 00:03:51 ago
      opaque_ptr 0x798416842BC8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none

```

R2 show ip route 10.30.3.0

```

Routing entry for 10.30.3.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:03:51 ago
  Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:03:51 ago
      opaque_ptr 0x798416842588
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

```

R3 show ip route 10.30.1.0

```

Routing entry for 10.30.1.0/24
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:03:51 ago
  Routing Descriptor Blocks:
    * 172.20.0.1, from 172.20.0.1, 00:03:51 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none

```

Teardown (suite) Close All Connections

12.04 **PASS** Every Host Reaches Every Other Host At Its Mapped Address (24.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss** (checked 6×)
- ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H2 ping -c 5 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.

```

```
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=2.51 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=2.06 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.61 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.35 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.269/1.758/2.509/0.465 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.19 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.16 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.29 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.14 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.13 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.130/1.182/1.289/0.056 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.21 ms
64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.28 ms
64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.24 ms
64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.13 ms
64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.38 ms

--- 10.30.1.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.132/1.248/1.381/0.082 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.37 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.88 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.28 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.24 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.29 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.244/1.412/1.882/0.238 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.13 ms
64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.19 ms
64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.23 ms
64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 10.30.1.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.130/1.184/1.227/0.031 ms
```

H2 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.18 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.37 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.12 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.115/1.205/1.373/0.089 ms
```

Teardown (suite) Close All Connections

12.05 PASS Translations Show The Host Rewritten To Its Mapped Address (4.1s, 2 commands)

The inside-local / inside-global pair is the direct evidence that the source address was rewritten.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}

- output matches `icmpseq=${H1_NAT_IP}::id+ls=${H1_IP}::id+ls=${H2_NAT_IP}`

How it was checked

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.33 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.11 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.110/1.225/1.326/0.070 ms
```

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.11	192.168.10.11	---	---
---	10.30.1.10	192.168.10.10	---	---
---	10.30.1.13	192.168.10.13	---	---
---	10.30.1.12	192.168.10.12	---	---
---	10.30.1.15	192.168.10.15	---	---
---	10.30.1.14	192.168.10.14	---	---
---	10.30.1.17	192.168.10.17	---	---
---	10.30.1.16	192.168.10.16	---	---
---	10.30.1.19	192.168.10.19	---	---
---	10.30.1.18	192.168.10.18	---	---
icmp	10.30.1.10:12369	192.168.10.10:12369	10.30.3.10:12369	10.30.3.10:12369
icmp	10.30.1.10:45085	192.168.10.10:45085	10.30.2.10:45085	10.30.2.10:45085
icmp	10.30.1.10:27844	192.168.10.10:27844	10.30.3.10:27844	10.30.3.10:27844

... 5 more lines

Teardown (suite) Close All Connections

12.06 PASS The Far Host Sees The Mapped Source Not The Real One (4.1s, 2 commands)

On the receiving router the peer appears as the sender's mapped address, which is the whole point of source NAT.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains `${H1_NAT_IP}`
- output does not contain `${H1_IP}`

How it was checked

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.34 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.09 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.39 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.41 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.093/1.273/1.412/0.135 ms
```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.11	192.168.20.11	---	---
---	10.30.2.10	192.168.20.10	---	---
---	10.30.2.13	192.168.20.13	---	---
---	10.30.2.12	192.168.20.12	---	---
---	10.30.2.15	192.168.20.15	---	---
---	10.30.2.14	192.168.20.14	---	---
---	10.30.2.17	192.168.20.17	---	---
---	10.30.2.16	192.168.20.16	---	---
---	10.30.2.19	192.168.20.19	---	---
---	10.30.2.18	192.168.20.18	---	---
icmp	10.30.2.10:47582	192.168.20.10:47582	10.30.1.10:47582	10.30.1.10:47582
icmp	10.30.2.10:18905	192.168.20.10:18905	10.30.3.10:18905	10.30.3.10:18905
icmp	10.30.2.10:4179	192.168.20.10:4179	10.30.3.10:4179	10.30.3.10:4179

... 7 more lines

Teardown (suite) Close All Connections

12.07 PASS Spoke To Spoke NAT Works Through The Hub (4.2s, 3 commands)

h2 -> h3 by mapped address crosses two tunnels and two translations, with no direct spoke-to-spoke path.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **\${H3_NAT_IP}**
- output contains **\${H2_NAT_IP}**

How it was checked

H2 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.  
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.26 ms  
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.24 ms  
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.17 ms  
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.09 ms  
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.24 ms  
  
--- 10.30.3.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4005ms  
rtt min/avg/max/mdev = 1.092/1.202/1.263/0.063 ms
```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.11	192.168.20.11	---	---
---	10.30.2.10	192.168.20.10	---	---
---	10.30.2.13	192.168.20.13	---	---
---	10.30.2.12	192.168.20.12	---	---
---	10.30.2.15	192.168.20.15	---	---
---	10.30.2.14	192.168.20.14	---	---
---	10.30.2.17	192.168.20.17	---	---
---	10.30.2.16	192.168.20.16	---	---
---	10.30.2.19	192.168.20.19	---	---
---	10.30.2.18	192.168.20.18	---	---
icmp	10.30.2.10:47582	192.168.20.10:47582	10.30.1.10:47582	10.30.1.10:47582
icmp	10.30.2.10:18905	192.168.20.10:18905	10.30.3.10:18905	10.30.3.10:18905
icmp	10.30.2.10:4179	192.168.20.10:4179	10.30.3.10:4179	10.30.3.10:4179
... 8 more lines				

R3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.11	192.168.30.11	---	---
---	10.30.3.10	192.168.30.10	---	---
---	10.30.3.13	192.168.30.13	---	---
---	10.30.3.12	192.168.30.12	---	---
---	10.30.3.15	192.168.30.15	---	---
---	10.30.3.14	192.168.30.14	---	---
---	10.30.3.17	192.168.30.17	---	---
---	10.30.3.16	192.168.30.16	---	---
---	10.30.3.19	192.168.30.19	---	---
---	10.30.3.18	192.168.30.18	---	---
icmp	10.30.3.10:43539	192.168.30.10:43539	10.30.2.10:43539	10.30.2.10:43539
icmp	10.30.3.10:55916	192.168.30.10:55916	10.30.2.10:55916	10.30.2.10:55916
icmp	10.30.3.10:12369	192.168.30.10:12369	10.30.1.10:12369	10.30.1.10:12369
... 7 more lines				

Teardown (suite) Close All Connections

12.08 PASS NATted Traffic Is Still Encrypted (19.6s, 5 commands)

Translation happens before encryption, so NATted traffic must still raise the ESP counters.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- **\${r1_enc} >= \${expected}** holds
- **\${r1_dec} >= \${expected}** holds

- `#{r2_enc} >= #{expected}` holds
- `#{r2_dec} >= #{expected}` holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 78406, #pkts encrypt: 78406, #pkts digest: 78406
  #pkts decaps: 41015, #pkts decrypt: 41015, #pkts verify: 41015
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 51117, #pkts encrypt: 51117, #pkts digest: 51117
  #pkts decaps: 26631, #pkts decrypt: 26631, #pkts verify: 26631
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

H1 ping -c 20 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.19 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 78537, #pkts encrypt: 78537, #pkts digest: 78537
  #pkts decaps: 41145, #pkts decrypt: 41145, #pkts verify: 41145
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 51226, #pkts encrypt: 51226, #pkts digest: 51226
```

```

#pkts decaps: 26745, #pkts decrypt: 26745, #pkts verify: 26745
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

Teardown (suite) Close All Connections

12.09 PASS Mapped Addresses Never Appear In Cleartext On The WAN (14.0s, 18 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss**
ping from \${alias} to \${target} lost packets: \${out}
- output contains **ESP**
- output does not contain **\${H1_NAT_IP}**
- output does not contain **\${H2_NAT_IP}**
- output does not contain **\${H1_IP}**

How it was checked

R1 no monitor capture NATCAP

Capture does not exist

R1 monitor capture NATCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture NATCAP match ipv4 any any

\${output} =

R1 monitor capture NATCAP buffer size 5

\${output} =

R1 monitor capture NATCAP start

Started capture point : NATCAP

R1 ping 100.64.0.2 repeat 2

Type escape sequence to abort.

Sending 2, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:

!!

Success rate is 100 percent (2/2), round-trip min/avg/max = 1/1/1 ms

R1 monitor capture NATCAP stop

Stopped capture point : NATCAP

R1 show monitor capture NATCAP buffer brief

```

-----
#  size  timestamp      source          destination    dscp  protocol
-----
0  138    0.000000    100.64.0.2      -> 100.64.0.1    48 CS6  ESP
1  138    0.000000    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
2  170    0.007004    100.64.0.2      -> 100.64.0.1    48 CS6  ESP
3  114    0.022002    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
4  114    0.022002    100.64.0.2      -> 100.64.0.1    0 BE    ICMP
5  114    0.023010    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
6  114    0.023010    100.64.0.2      -> 100.64.0.1    0 BE    ICMP
7  138    0.042006    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
8  138    0.042998    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
9  154    0.128000    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
10 138    0.265993    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
... 43 more lines

```

R1 no monitor capture NATCAP

\${output} =

R1 no monitor capture NATCAP

Capture does not exist

R1 monitor capture NATCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture NATCAP match ipv4 any any

\${output} =

R1 monitor capture NATCAP buffer size 5

\${output} =

R1 monitor capture NATCAP start

Started capture point : NATCAP

H1 ping -c 10 10.30.2.10

PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.

```

64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.77 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=2.13 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.54 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.30 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.31 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.35 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.31 ms

--- 10.30.2.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
... 1 more lines

```

R1 monitor capture NATCAP stop

Stopped capture point : NATCAP

R1 show monitor capture NATCAP buffer brief

```

-----
#   size  timestamp      source          destination    dscp    protocol
-----
0   170    0.000000    100.64.0.1     -> 100.64.0.2    0 BE    ESP
1   170    0.001007    100.64.0.2     -> 100.64.0.1    0 BE    ESP
2   138    0.092997    100.64.0.1     -> 100.64.0.2    48 CS6   ESP
3   138    0.092997    100.64.0.2     -> 100.64.0.1    48 CS6   ESP
4   138    0.262047    100.64.0.1     -> 100.64.0.3    48 CS6   ESP
5   138    0.262047    100.64.0.3     -> 100.64.0.1    48 CS6   ESP
6   138    0.275047    100.64.0.2     -> 100.64.0.1    48 CS6   ESP
7   138    0.275047    100.64.0.1     -> 100.64.0.2    48 CS6   ESP
8   138    0.394028    100.64.0.3     -> 100.64.0.1    48 CS6   ESP
9   138    0.394028    100.64.0.1     -> 100.64.0.3    48 CS6   ESP
10  138    0.487026    100.64.0.1     -> 100.64.0.2    48 CS6   ESP
... 220 more lines

```

R1 no monitor capture NATCAP

\${output} =

Teardown (suite) Close All Connections

12.10 PASS The Untranslated LAN Path Still Works (8.0s, 2 commands)

Guards the route-map scoping. If translation ever widened to all traffic, replies on this path would arrive from a mapped address the sender never contacted, and this would fail.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **0% packet loss** (checked 2×)
- ping from \${alias} to \${target} lost packets: \${out}

How it was checked

H1 ping -c 5 192.168.20.10

```

PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.14 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.12 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.37 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.13 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.15 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.124/1.181/1.367/0.093 ms

```

H2 ping -c 5 192.168.30.10

```

PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.25 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.15 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.43 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.26 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.24 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.147/1.264/1.425/0.089 ms

```

Teardown (suite) Close All Connections

13 Nat Pool

13.01 **PASS** Every Host Carries Its Full Block Of Addresses (0.1s, 6 commands)

10 statically mapped plus 30 pool-eligible, on one interface.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **\${lan}.\${o}/24** (checked 12×)
- **\${n}** equals **40** (checked 3×)

How it was checked

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H1 ip addr show eth1 | grep -c 'inet 192.168.10.'
40
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H2 ip addr show eth1 | grep -c 'inet 192.168.20.'
40
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H3 ip addr show eth1 | grep -c 'inet 192.168.30.'
40
```

Teardown (suite) Close All Connections

13.02 **PASS** Every Router Configures Ten Static Mappings And A Thirty Address Pool (0.4s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains = **10** (checked 3×)
- output contains **total addresses 30** (checked 3×)

How it was checked

H1 `show running-config | count ^ip nat inside source static`

Number of lines which match regexp = 10

H1 `show ip nat statistics`

Total active translations: 10 (10 static, 0 dynamic; 0 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 0 Misses: 0

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0

pool SITE-POOL: id 2, netmask 255.255.255.0

start 10.30.1.100 end 10.30.1.129

... 9 more lines

H2 `show running-config | count ^ip nat inside source static`

Number of lines which match regexp = 10

H2 `show ip nat statistics`

Total active translations: 10 (10 static, 0 dynamic; 0 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 0 Misses: 0

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0

pool SITE-POOL: id 2, netmask 255.255.255.0

start 10.30.2.100 end 10.30.2.129

... 9 more lines

H3 `show running-config | count ^ip nat inside source static`

Number of lines which match regexp = 10

H3 `show ip nat statistics`

Total active translations: 10 (10 static, 0 dynamic; 0 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 0 Misses: 0

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0

pool SITE-POOL: id 2, netmask 255.255.255.0

start 10.30.3.100 end 10.30.3.129

... 9 more lines

Teardown (suite) Close All Connections

13.03 **PASS** The Pool Access List Is One Range And Excludes The Static Block (0.1s, 3 commands)

A single /27 range rather than thirty host entries, and it must not overlap the statically mapped addresses -- otherwise which rule wins depends on evaluation order.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **#{permits}** has exactly **1** entries (checked 3×)
NAT-POOL-SRC should hold one range entry, found #{permits}
- output contains **#{lan}.#{DYN_OCTET_FIRST} 0.0.0.31** (checked 3×)
- output does not contain **host #{lan}.#{o}** (checked 6×)

How it was checked

H1 show ip access-lists NAT-POOL-SRC

```
Extended IP access list NAT-POOL-SRC
 10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
```

H2 show ip access-lists NAT-POOL-SRC

```
Extended IP access list NAT-POOL-SRC
 10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
```

H3 show ip access-lists NAT-POOL-SRC

```
Extended IP access list NAT-POOL-SRC
 10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
```

Teardown (suite) Close All Connections

13.04 PASS All Ten Static Addresses Translate To Their Own Fixed Mapping (11.5s, 2 commands)

Static NAT is one-to-one and deterministic: inside .10-.19 must appear outside as .10-.19, each to its own.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **#{static}** has exactly **10** entries
only #{static} of the 10 static addresses were translated
- **#{d}** equals **10**
the 10 static addresses shared outside addresses; they must be one-to-one
- **#{aligned}** holds

How it was checked

H1 for i in \$(seq 10 19) \$(seq 64 93); do ping -c 2 -W 2 -I 192.168.10.\$i 10.30.2.10 >/dev/null 2>&1 & done; wait; echo SENT

```
SENT
[40]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[39]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[38]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[37]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[36]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[35]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[34]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[33]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[32]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[31]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[30]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[29]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
[28]+ Done ping -c 2 -W 2 -I 192.168.10.#{i} 10.30.2.10 1>/dev/null 2>&1
... 27 more lines
```

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.76	---	---
---	10.30.1.129	192.168.10.87	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.79	---	---
---	10.30.1.116	192.168.10.81	---	---
---	10.30.1.119	192.168.10.84	---	---
---	10.30.1.123	192.168.10.89	---	---
---	10.30.1.120	192.168.10.85	---	---
---	10.30.1.126	192.168.10.92	---	---
---	10.30.1.125	192.168.10.91	---	---
---	10.30.1.101	192.168.10.65	---	---
---	10.30.1.102	192.168.10.66	---	---
---	10.30.1.111	192.168.10.75	---	---
---	...	68 more lines	---	---

Teardown (suite) Close All Connections

13.05 PASS All Thirty Pool Addresses Translate Into The Pool And Are Distinct (0.5s, 1

commands)

Without overload each client takes its own pool address, so 30 clients must yield 30 different outside addresses.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **\${dyn}** has exactly 30 entries
only \${dyn} of the 30 pool clients were translated
- **\${d}** equals 30
pool clients shared outside addresses; that is overload, not a 1:1 pool
- **\${within}** holds

How it was checked

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.76	---	---
---	10.30.1.129	192.168.10.87	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.79	---	---
---	10.30.1.116	192.168.10.81	---	---
---	10.30.1.119	192.168.10.84	---	---
---	10.30.1.123	192.168.10.89	---	---
---	10.30.1.120	192.168.10.85	---	---
---	10.30.1.126	192.168.10.92	---	---
---	10.30.1.125	192.168.10.91	---	---
---	10.30.1.101	192.168.10.65	---	---
---	10.30.1.102	192.168.10.66	---	---
---	10.30.1.111	192.168.10.75	---	---
... 68 more lines				

Teardown (suite) Close All Connections

13.06 PASS The Pool Is Exactly Consumed With No Misses (0.0s, 1 commands)

30 clients against a 30 address pool: full allocation and no misses is the evidence that none of them shared or failed.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- output matches **misses 0**

How it was checked

R1 show ip nat statistics

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 120 Misses: 40
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
  pool SITE-POOL: id 2, netmask 255.255.255.0
    start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

Teardown (suite) Close All Connections

13.07 PASS A Thirty-first Client Gets No Translation Once The Pool Is Exhausted (4.3s, 7

commands)

The boundary the pool sizing never otherwise reaches. The pool holds exactly as many addresses as there are clients, which means the exhaustion path is never exercised by a normal run -- and an off-by-one in pool bounds would look identical to a correct configuration. So bring up one more client inside the pool ACL and require the router to refuse it: no translation, no borrowing of another client's address, and the miss counted. Without "overload" there is nothing to share, so silently reusing an address would be a correctness bug, not graceful degradation. Run against R1/H1, because it is R1's pool that the tests above fill: the suite drives traffic from H1 first and only reaches the other routers later. Asserting exhaustion on a router whose pool is still empty would test nothing.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- **`\${n}`** is not empty (checked 2×)
- output contains **100% packet loss**
the thirty-first client was translated even though the pool was exhausted
- **`\${xlate}`** is empty
- **`\${after}`** > **`\${before}`** holds

How it was checked

R1 show ip nat statistics | include allocated

type generic, total addresses 30, allocated 30 (100%), misses 0

R1 show ip nat statistics

Total active translations: 80 (10 static, 70 dynamic; 40 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 120 Misses: 40

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60

pool SITE-POOL: id 2, netmask 255.255.255.0

start 10.30.1.100 end 10.30.1.129

... 9 more lines

H1 sudo ip addr add 192.168.10.94/24 dev eth1

\${output} =

H1 ping -c 3 -W 2 -I 192.168.10.94 10.30.2.10

PING 10.30.2.10 (10.30.2.10) from 192.168.10.94 : 56(84) bytes of data.

From 192.168.10.1 icmp_seq=2 Destination Host Unreachable

From 192.168.10.1 icmp_seq=3 Destination Host Unreachable

--- 10.30.2.10 ping statistics ---

3 packets transmitted, 0 received, +2 errors, 100% packet loss, time 2042ms

R1 show ip nat translations | include 192.168.10.94

\${output} =

R1 show ip nat statistics

Total active translations: 80 (10 static, 70 dynamic; 40 extended)

Outside interfaces:

Tunnel0

Inside interfaces:

GigabitEthernet3

Hits: 120 Misses: 40

Reserved port setting disabled provisioned no

Dynamic overload mapping configured: 0

Expired translations: 0

Dynamic mappings:

-- Inside Source

[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60

pool SITE-POOL: id 2, netmask 255.255.255.0

start 10.30.1.100 end 10.30.1.129

... 9 more lines

H1 sudo ip addr del 192.168.10.94/24 dev eth1

\${output} =

Teardown (this test) Remove The Spare Client From H1

13.08 PASS The Pool Is Still Intact After The Exhaustion Attempt (0.5s, 2 commands)

The refusal must cost nothing: the thirty legitimate clients keep the addresses they held, and none was evicted to make room for the one that was turned away.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- **`\${dyn}`** has exactly **30** entries
only `\${dyn}` of the original thirty pool clients still hold a translation
- **`\${d}`** equals **30**
the refused client caused two survivors to share an address

How it was checked

R1 show ip nat statistics | include allocated

type generic, total addresses 30, allocated 30 (100%), misses 1

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.76	---	---
---	10.30.1.129	192.168.10.87	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.79	---	---
---	10.30.1.116	192.168.10.81	---	---
---	10.30.1.119	192.168.10.84	---	---
---	10.30.1.123	192.168.10.89	---	---
---	10.30.1.120	192.168.10.85	---	---
---	10.30.1.126	192.168.10.92	---	---
---	10.30.1.125	192.168.10.91	---	---
---	10.30.1.101	192.168.10.65	---	---
---	10.30.1.102	192.168.10.66	---	---
---	10.30.1.111	192.168.10.75	---	---
...	68 more lines			

Teardown (suite) Close All Connections

13.09 PASS Static And Pool Translations Coexist Across The Whole Block (0.5s, 1 commands)

All 40 of a host's addresses in flight, translated by two different mechanisms, with no outside address reused.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **`\${all}`** has exactly **40** entries
expected 40 translated addresses, found `\${all}`
- **`\${d}`** equals **40**
the 40 inside addresses did not receive 40 distinct outside addresses

How it was checked

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.76	---	---
---	10.30.1.129	192.168.10.87	---	---
---	10.30.1.113	192.168.10.78	---	---
---	10.30.1.114	192.168.10.79	---	---
---	10.30.1.116	192.168.10.81	---	---
---	10.30.1.119	192.168.10.84	---	---
---	10.30.1.123	192.168.10.89	---	---
---	10.30.1.120	192.168.10.85	---	---
---	10.30.1.126	192.168.10.92	---	---
---	10.30.1.125	192.168.10.91	---	---
---	10.30.1.101	192.168.10.65	---	---
---	10.30.1.102	192.168.10.66	---	---
---	10.30.1.111	192.168.10.75	---	---
...	68 more lines			

Teardown (suite) Close All Connections

13.10 PASS Both Mechanisms Work On Every Router (23.6s, 4 commands)

The spokes carry the same scheme, not just the hub.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **static** has exactly **10** entries (checked 2×)
- **dyn** has exactly **30** entries (checked 2×)
- **aligned** holds (checked 2×)

How it was checked

```
H2 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.20.$i 10.30.1.10 >/dev/null 2>&1 & done; wait;
echo SENT
```

```
SENT
[40]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+ Done ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines
```

H2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.75	---	---
---	10.30.2.129	192.168.20.88	---	---
---	10.30.2.106	192.168.20.70	---	---
---	10.30.2.105	192.168.20.69	---	---
---	10.30.2.111	192.168.20.76	---	---
---	10.30.2.108	192.168.20.72	---	---
---	10.30.2.101	192.168.20.65	---	---
---	10.30.2.102	192.168.20.66	---	---
---	10.30.2.126	192.168.20.92	---	---
---	10.30.2.125	192.168.20.91	---	---
---	10.30.2.123	192.168.20.89	---	---
---	10.30.2.120	192.168.20.85	---	---
---	10.30.2.116	192.168.20.81	---	---
...	108 more lines			

```
H3 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.30.$i 10.30.1.10 >/dev/null 2>&1 & done; wait;
echo SENT
```

```
SENT
[40]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+ Done ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines
```

H3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.104	192.168.30.68	---	---
---	10.30.3.107	192.168.30.71	---	---
---	10.30.3.109	192.168.30.73	---	---
---	10.30.3.110	192.168.30.74	---	---
---	10.30.3.103	192.168.30.67	---	---
---	10.30.3.100	192.168.30.64	---	---
---	10.30.3.124	192.168.30.89	---	---
---	10.30.3.127	192.168.30.93	---	---
---	10.30.3.121	192.168.30.86	---	---
---	10.30.3.122	192.168.30.87	---	---
---	10.30.3.118	192.168.30.83	---	---
---	10.30.3.117	192.168.30.82	---	---
---	10.30.3.115	192.168.30.80	---	---

... 68 more lines

Teardown (suite) Close All Connections

13.11 PASS Only The Static Mappings Live In The Configuration (0.1s, 1 commands)

Static mappings are configuration and survive a reload; pool allocations are runtime state and do not. Confusing the two is how NAT surprises people after a reboot.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **static** `${H1_IP} ${H1_NAT_IP}`
- output does not contain `${NAT24}[R1].${POOL_OCTET_FIRST}`
- output contains **ip nat inside source list** `NAT-POOL-SRC pool SITE-POOL`

How it was checked

R1 **show running-config | include ip nat inside source**

```
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

Teardown (suite) Close All Connections

13.12 PASS Traffic From Both Mechanisms Is Encrypted (38.7s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **0% packet loss** (checked 2×)
ping from `${source}` on `${alias}` to `${target}` lost packets
- `${r1_enc} >= ${expected}` holds
- `${r1_dec} >= ${expected}` holds
- `${r2_enc} >= ${expected}` holds
- `${r2_dec} >= ${expected}` holds

How it was checked

R1 **show crypto ipsec sa**

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 79100, #pkts encrypt: 79100, #pkts digest: 79100
    #pkts decaps: 41709, #pkts decrypt: 41709, #pkts verify: 41709
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 **show crypto ipsec sa**

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
  current_peer 100.64.0.3 port 500
    PERMIT, flags={origin_is_acl,}
```

```

#pkts encaps: 51605, #pkts encrypt: 51605, #pkts digest: 51605
#pkts decaps: 27124, #pkts decrypt: 27124, #pkts verify: 27124
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

H1 ping -c 20 -I 192.168.10.10 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) from 192.168.10.10 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.11 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.53 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.73 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.35 ms
... 11 more lines

```

H1 ping -c 20 -I 192.168.10.64 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.50 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.33 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.29 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.31 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.57 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.20 ms
... 11 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 79366, #pkts encrypt: 79366, #pkts digest: 79366
#pkts decaps: 41973, #pkts decrypt: 41973, #pkts verify: 41973
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 51820, #pkts encrypt: 51820, #pkts digest: 51820
#pkts decaps: 27339, #pkts decrypt: 27339, #pkts verify: 27339
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

Teardown (suite) Close All Connections

13.13 PASS Neither Real Nor Translated Addresses Appear In Cleartext (11.6s, 9 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **0% packet loss**
ping from \${source} on \${alias} to \${target} lost packets
- output contains **ESP**
- output does not contain \${LAN24}[H1].\${DYN_OCTET_FIRST}
- output does not contain \${NAT24}[R1].\${POOL_OCTET_FIRST}

How it was checked

R1 no monitor capture POOLCAP

Capture does not exist

R1 monitor capture POOLCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture POOLCAP match ipv4 any any

\${output} =

R1 monitor capture POOLCAP buffer size 5

\${output} =

R1 monitor capture POOLCAP start

Started capture point : POOLCAP

H1 ping -c 10 -I 192.168.10.64 10.30.2.10

PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.

```
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=2.64 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.77 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.31 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.32 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.30 ms
```

--- 10.30.2.10 ping statistics ---

10 packets transmitted, 10 received, 0% packet loss, time 9013ms

... 1 more lines

R1 monitor capture POOLCAP stop

Stopped capture point : POOLCAP

R1 show monitor capture POOLCAP buffer brief

```
-----
#   size  timestamp      source          destination    dscp  protocol
-----
0   138    0.000000    100.64.0.1     -> 100.64.0.3    48 CS6  ESP
1   138    0.000000    100.64.0.3     -> 100.64.0.1    48 CS6  ESP
2   170    0.003998    100.64.0.1     -> 100.64.0.2    0 BE    ESP
3   170    0.004989    100.64.0.2     -> 100.64.0.1    0 BE    ESP
4   138    0.125985    100.64.0.1     -> 100.64.0.3    48 CS6  ESP
5   138    0.171988    100.64.0.3     -> 100.64.0.1    48 CS6  ESP
6   138    0.171988    100.64.0.1     -> 100.64.0.3    48 CS6  ESP
7   138    0.246981    100.64.0.2     -> 100.64.0.1    48 CS6  ESP
8   138    0.246981    100.64.0.1     -> 100.64.0.2    48 CS6  ESP
9   138    0.368984    100.64.0.1     -> 100.64.0.2    48 CS6  ESP
10  138    0.369976    100.64.0.2     -> 100.64.0.1    48 CS6  ESP
... 221 more lines
```

R1 no monitor capture POOLCAP

\${output} =

Teardown (suite) Close All Connections

14 Ntp

14.01 PASS Every Router Is An NTP Client Of The NMS Over The Management VRF (0.3s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ntp source \${OOB_INTF}[\${r}]** (checked 3×)
- output contains **ntp server vrf \${MGMT_VRF} \${NMS_OOB_IP}** (checked 3×)
- output does not contain **pool.ntp.org** (checked 3×)

How it was checked

R1 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

R2 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

R3 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

Teardown (suite) Close Nms Session, Close All Connections

14.02 PASS The NMS Itself Synchronises To The Public Pool (2.4s, 2 commands)

The precondition the rest of this suite rests on. chrony is configured "local stratum 10", so it serves time whether or not it has an upstream -- meaning the routers can agree perfectly on time that came from nowhere. This requires a real upstream: a reference that is not the local clock, and a stratum that proves it came from somewhere else.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **7F7F0101**
the NMS is serving its own local clock, so lab time is fabricated
- **\${m}** is not empty
- **0 < \${m}[0] < \${MAX_STRATUM}** holds
- output matches **\^*s+ld+\\.ld+\\.ld+\\.ld+**
the NMS has selected no upstream source: \${sources}

How it was checked

NMS chronyc tracking

```
Reference ID      : C0CF37FE (192.207.55.254)
Stratum           : 2
Ref time (UTC)    : Tue Sep 08 02:30:33 2026
System time       : 0.000395503 seconds slow of NTP time
Last offset       : -0.000212829 seconds
RMS offset        : 0.000697437 seconds
Frequency         : 8.097 ppm fast
Residual freq     : +0.001 ppm
Skew              : 0.362 ppm
Root delay        : 0.050906945 seconds
Root dispersion   : 0.001151797 seconds
Update interval   : 519.4 seconds
Leap status       : Normal
```

NMS chronyc -n sources

MS Name/IP address	Stratum	Poll	Reach	LastRx	Last sample
^+ 96.19.94.82	1	9	377	481	-2512us[-2724us] +/- 36ms
^- 172.233.157.223	2	9	377	477	+107us[-105us] +/- 119ms
^- 66.118.229.14	2	9	377	24	+1394us[+1394us] +/- 157ms
^+ 38.45.64.130	2	9	377	413	+3335us[+3335us] +/- 28ms
^* 192.207.55.254	1	9	377	474	-1412us[-1624us] +/- 29ms
^- 69.10.208.170	2	9	377	475	+3743us[+3530us] +/- 81ms

Teardown (suite) Close Nms Session, Close All Connections

14.03 PASS Every Router Appears As A Client Of The NMS (0.1s, 1 commands)

Seen from the server, which is where a missing client shows up as an absence rather than as a router quietly using something else.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${{r}}_OOB_IP` (checked 3×)

How it was checked

NMS `sudo chronyc clients`

Hostname	NTP	Drop	Int	IntL	Last	Cmd	Drop	Int	Last
192.168.99.2	56	0	7	-	70	0	0	-	-
192.168.99.3	56	0	7	-	59	0	0	-	-
192.168.99.1	56	0	7	-	0	0	0	-	-
localhost	0	0	-	-	-	8	0	-5	0

Teardown (suite) Close Nms Session, Close All Connections

14.04 PASS Every Router Forms Associations With Pool Servers (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- `${rows}` is not empty (checked 3×)

How it was checked

R1 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    0    128  377 0.916  2.641 6.959
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    70   128  377 0.896  3.892 3.919
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    59   128  377 0.104  0.972 4.637
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.05 PASS Associations Become Reachable (0.1s, 3 commands)

A non-zero reach register means packets are actually coming back, which distinguishes a working client from one that has merely been configured.

Setup (suite) Open All Routers

Acceptance criteria

- `${up}` is not empty (checked 3×)

How it was checked

R1 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    0    128  377 0.916  2.641 6.959
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    70   128  377 0.896  3.892 3.919
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 `show ntp associations`

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    59   128  377 0.104  0.972 4.637
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.06 PASS Every Router Selects A System Peer (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- `${peer}` differs from `$(None)` (checked 3×)

How it was checked

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    0    128  377  0.916  2.641  6.959
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    70   128  377  0.896  3.892  3.919
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    60   128  377  0.104  0.972  4.637
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.07 PASS Every Router Clock Is Synchronised (0.1s, 3 commands)

The end state: IOS declares the clock synchronised. This lags real accuracy by a long way. IOS holds the flag back until its loopfilter leaves the 'FREQ' drift-measurement phase, measured on this platform at about 17 minutes from a cold NTP start -- while the offset is already within milliseconds well before that. The wait below is bounded to cover a freshly provisioned lab; on a lab that has been up it returns on the first poll and costs nothing.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Clock is synchronized** (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49F188.E41895E8 (02:29:28.891 UTC Tue Sep 8 2026)
clock offset is 2.6417 msec, root delay is 51.84 msec
root dispersion is 19.62 msec, peer dispersion is 6.95 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007759 s/s
system poll interval is 128, last update was 539 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49F14C.03D70A48 (02:28:28.015 UTC Tue Sep 8 2026)
clock offset is 3.8927 msec, root delay is 51.84 msec
root dispersion is 20.59 msec, peer dispersion is 3.91 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007576 s/s
system poll interval is 128, last update was 599 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49EDE7.4A3D7170 (02:13:59.290 UTC Tue Sep 8 2026)
clock offset is 0.9724 msec, root delay is 50.87 msec
root dispersion is 45.70 msec, peer dispersion is 4.63 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007218 s/s
system poll interval is 128, last update was 1468 sec ago.
```

Teardown (suite) Close Nms Session, Close All Connections

14.08 PASS Stratum Is Sane On Every Router (0.1s, 3 commands)

Stratum 16 means unsynchronised. A client of a stratum 1-2 pool server should land well below the limit.

Setup (suite) Open All Routers

Acceptance criteria

- `${m}` is not empty (checked 3×)
- `${m}[0] < ${MAX_STRATUM}` holds (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49F188.E41895E8 (02:29:28.891 UTC Tue Sep 8 2026)
clock offset is 2.6417 msec, root delay is 51.84 msec
root dispersion is 19.62 msec, peer dispersion is 6.95 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007759 s/s
system poll interval is 128, last update was 539 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49F14C.03D70A48 (02:28:28.015 UTC Tue Sep 8 2026)
clock offset is 3.8927 msec, root delay is 51.84 msec
root dispersion is 20.59 msec, peer dispersion is 3.91 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007576 s/s
system poll interval is 128, last update was 599 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49EDE7.4A3D7170 (02:13:59.290 UTC Tue Sep 8 2026)
clock offset is 0.9724 msec, root delay is 50.87 msec
root dispersion is 45.70 msec, peer dispersion is 4.63 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007218 s/s
system poll interval is 128, last update was 1468 sec ago.
```

Teardown (suite) Close Nms Session, Close All Connections

14.09 PASS The Reference Is One Of The Peers The Router Actually Polls (0.1s, 6 commands)

Guards against a router calling itself synchronised to something that is not in its association list.

Setup (suite) Open All Routers

Acceptance criteria

- `${ref}` is not empty (checked 3×)
- output contains `${ref}[0]` (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49F188.E41895E8 (02:29:28.891 UTC Tue Sep 8 2026)
clock offset is 2.6417 msec, root delay is 51.84 msec
root dispersion is 19.62 msec, peer dispersion is 6.95 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007759 s/s
system poll interval is 128, last update was 539 sec ago.
```

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    0    128  377  0.916  2.641  6.959
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0019 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49F14C.03D70A48 (02:28:28.015 UTC Tue Sep 8 2026)
clock offset is 3.8927 msec, root delay is 51.84 msec
root dispersion is 20.59 msec, peer dispersion is 3.91 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007576 s/s
system poll interval is 128, last update was 599 sec ago.
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 192.207.55.254 2    70   128  377  0.896  3.892  3.919
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```

Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0018 Hz, precision is 2**10
ntp uptime is 382700 (1/100 of seconds), resolution is 4000
reference time is EE49EDE7.4A3D7170 (02:13:59.290 UTC Tue Sep 8 2026)
clock offset is 0.9724 msec, root delay is 50.87 msec
root dispersion is 45.70 msec, peer dispersion is 4.63 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007218 s/s
system poll interval is 128, last update was 1468 sec ago.

```

R3 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  192.207.55.254  2    60   128  377  0.104  0.972  4.637
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

Teardown (suite) Close Nms Session, Close All Connections

14.10 PASS Clock Offset Is Small On Every Router (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- $\{\text{peer}\}$ differs from $\{\text{None}\}$ (checked 3×)
- $\{\text{offset}\} < \{\text{MAX_OFFSET_MS}\}$ holds (checked 3×)

How it was checked

R1 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  192.207.55.254  2    0   128  377  0.916  2.641  6.959
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

R2 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  192.207.55.254  2    70   128  377  0.896  3.892  3.919
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

R3 show ntp associations

```

address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  192.207.55.254  2    60   128  377  0.104  0.972  4.637
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```

Teardown (suite) Close Nms Session, Close All Connections

14.11 PASS All Routers Agree On The Time (0.0s, 3 commands)

The practical payoff: three independently synchronised clocks should read the same, which is what makes logs across the fabric correlatable.

Setup (suite) Open All Routers

Acceptance criteria

- $\{\text{s12}\} < \{\text{MAX_SKEW_SECONDS}\}$ holds
- $\{\text{s13}\} < \{\text{MAX_SKEW_SECONDS}\}$ holds

How it was checked

R1 show clock

```
02:38:27.653 UTC Tue Sep 8 2026
```

R2 show clock

```
02:38:27.663 UTC Tue Sep 8 2026
```

R3 show clock

```
02:38:27.681 UTC Tue Sep 8 2026
```

Teardown (suite) Close Nms Session, Close All Connections

14.12 PASS The Clock Is No Longer At The Boot Epoch (0.1s, 6 commands)

An unsynchronised IOS device reports a 1993 or 1900 clock; a synchronised one reports now. This is the crude check that catches a clock that never moved at all.

Setup (suite) Open All Routers

Acceptance criteria

- `${year} >= 2026` holds (checked 3×)
- output does not contain `*` (checked 3×)

How it was checked

R1 show clock

02:38:27.690 UTC Tue Sep 8 2026

R1 show clock

02:38:27.701 UTC Tue Sep 8 2026

R2 show clock

02:38:27.714 UTC Tue Sep 8 2026

R2 show clock

02:38:27.727 UTC Tue Sep 8 2026

R3 show clock

02:38:27.745 UTC Tue Sep 8 2026

R3 show clock

02:38:27.759 UTC Tue Sep 8 2026

Teardown (suite) Close Nms Session, Close All Connections

15 Syslog

15.01 **PASS** The Collector Is A Service On The NMS Serving Every Router On One Port (0.1s, 2 commands)

What the move bought: one rsyslog listener for all three routers instead of one busybox listener per router, and a system service rather than something a suite starts -- so no run can leave a stale listener swallowing another's messages.

Setup (suite) Start Collector And Routers

Acceptance criteria

- `${state}` equals **active**
- output contains `:${SYSLOG_PORT}`
- output contains **rsyslogd**

How it was checked

NMS `systemctl is-active rsyslog`
active

NMS `sudo ss -ltnp | grep ':514 ' || echo NONE`

UNCONN	0	0	0.0.0.0:514	0.0.0.0:*	users:("rsyslogd",pid=1126,fd=6))
UNCONN	0	0	:::514	:::*	users:("rsyslogd",pid=1126,fd=7))

Teardown (suite) Close Collector, Close All Connections

15.02 **PASS** Every Router Is Configured To Log To The Collector (0.4s, 3 commands)

Setup (suite) Start Collector And Routers

Acceptance criteria

- output contains **logging host** `${SYSLOG_HOST}` (checked 3×)
- output contains **logging source-interface** `${OOB_INTF}${r}` vrf `${MGMT_VRF}` (checked 3×)
- output contains **service sequence-numbers** (checked 3×)

How it was checked

R1 `show running-config | include ^logging|^service (timestamps|sequence)`

```
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

R2 `show running-config | include ^logging|^service (timestamps|sequence)`

```
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

R3 `show running-config | include ^logging|^service (timestamps|sequence)`

```
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

Teardown (suite) Close Collector, Close All Connections

15.03 **PASS** Every Router Reports The Collector As An Active Logging Host (0.2s, 3 commands)

The router's own view: it should name the collector and show a non-zero count of messages sent to it.

Setup (suite) Start Collector And Routers

Acceptance criteria

- output contains `${SYSLOG_HOST}` (checked 3×)
- output contains **level** `${TRAP_LEVEL}` (checked 3×)

How it was checked

```
R1 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 1120 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
R2 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 1248 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
R3 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 951 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
```

Teardown (suite) Close Collector, Close All Connections

15.04 **PASS** The Hub Message Reaches The Collector (4.0s, 1 commands)

End to end over the LAN: emit a uniquely marked message and find it at the collector.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\${n} > 0$ holds

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-HUB-223828' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

Teardown (suite) Close Collector, Close All Connections

15.05 **PASS** Each Spoke Message Reaches The Collector Across The Tunnel (8.1s, 2 commands)

The spokes have no direct path to h1's LAN; their messages are carried by the tunnels and the hub's routing.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\${n} > 0$ holds (checked 2×)

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-R2-223828' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
```

```
NMS sudo grep -c -- 'LABSYSLOG-R3-223828' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
```

Teardown (suite) Close Collector, Close All Connections

15.06 **PASS** Received Messages Carry Facility Severity And Mnemonic (4.1s, 2 commands)

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\${n} > 0$ holds
- $\${m}[\text{facility}]$ equals **FACCHECK**
- $\${m}[\text{mnemonic}]$ equals **MNEMCHECK**
- $\${m}[\text{sev}]$ equals **5**

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-FIELDS-223828' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
<189>207: 000203: *Sep  7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep  7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep  7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep  7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep  7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep  7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
```

```

0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 2025 more lines

```

Teardown (suite) Close Collector, Close All Connections

15.07 PASS Received Messages Carry Sequence Numbers (0.1s, 1 commands)

'service sequence-numbers' is configured, so every message should be numbered -- which is how a collector detects loss.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\{msgs\}$ is not empty
- $\{uniq\} > 1$ holds
- $\min(\$seqs) \geq 0$ holds

How it was checked

NMS `sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true`

```

<189>207: 000203: *Sep 7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep 7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep 7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep 7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep 7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep 7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 2025 more lines

```

Teardown (suite) Close Collector, Close All Connections

15.08 PASS Message Timestamps Agree With The Router Clock (4.1s, 3 commands)

Ties syslog to NTP: a message's timestamp should match the sending router's clock. Skewed timestamps make a collector's correlation across devices worthless.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\${n} > 0$ holds
- $\${skew} < \${MAX_TS_SKEW}$ holds

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-TIME-223828' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

R1 show clock

```
02:38:51.220 UTC Tue Sep 8 2026
```

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
```

```
<189>207: 000203: *Sep 7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep 7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep 7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep 7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep 7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep 7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0 (10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel: 172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel: 172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel: 172.20.0.3 NBMA: 100.64.0.1)
... 2026 more lines
```

Teardown (suite) Close Collector, Close All Connections

15.09 PASS Messages Below The Trap Level Are Not Forwarded (14.0s, 1 commands)

Negative control. Trap level is informational (6), so a debug-severity (7) message must not reach the collector. If this ever passes traffic, the level filter is not working and the positive tests above prove much less than they appear to.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\${n}$ equals 0
- a debug-severity message reached the collector despite trap level $\${TRAP_LEVEL}$

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-DEBUG-223828' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
0
0
```

Teardown (suite) Close Collector, Close All Connections

15.10 PASS All Three Routers Are Represented At The Collector (12.2s, 3 commands)

Messages from every router, distinguishable at the collector, which is the point of a central log host.

Setup (suite) Start Collector And Routers

Acceptance criteria

- $\${n} > 0$ holds (checked 3×)

How it was checked


```
NMS sudo grep -c -- 'LABSYSLOG-ALL-R1-223828' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R2-223828' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R3-223828' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
```

Teardown (suite) Close Collector, Close All Connections

16 Snmpv3

16.01 PASS Both Trap Receivers Are Services On The NMS (0.1s, 2 commands)

snmptrapd on the standard port and the raw capture on a second one, both system services rather than listeners a suite starts -- so no run can leave one behind, which is how a negative test once passed while nothing was arriving.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `${state}` equals **active active**
- output contains `:${TRAP_PORT}`
- output contains `:${TRAP_RAW_PORT}`

How it was checked

```
NMS systemctl is-active snmptrapd lab-trapcap | tr '\n' ' '
active active

NMS sudo ss -ltnp | grep -E ':(162|1162) ' || echo NONE
UNCONN 0      0      0.0.0.0:1162    0.0.0.0:*      users:(("python3",pid=1288,fd=3))
UNCONN 0      0      0.0.0.0:162    0.0.0.0:*
users:(("snmptrapd",pid=1247,fd=3),("systemd",pid=1,fd=101))
UNCONN 0      0      [::]:162      [::]:*
users:(("snmptrapd",pid=1247,fd=4),("systemd",pid=1,fd=102))
```

Teardown (suite) Close Trap Collector, Close All Connections

16.02 PASS Every Router Has An SNMPv3 User With Authentication And Privacy (0.1s, 3 commands)

v3 without auth and priv is barely better than v2c, so the protocols in use are asserted, not just the user's existence.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **User name: \${SNMP_USER}** (checked 3×)
- output contains **Authentication Protocol: SHA** (checked 3×)
- output contains **Privacy Protocol: AES128** (checked 3×)
- output contains **Group-name: \${SNMP_GROUP}** (checked 3×)

How it was checked

```
R1 show snmp user
User name: labmon
Engine ID: 800000090300525400C80101
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

R2 show snmp user
User name: labmon
Engine ID: 800000090300525400C80201
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

R3 show snmp user
User name: labmon
Engine ID: 800000090300525400C80301
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

Teardown (suite) Close Trap Collector, Close All Connections

16.03 PASS Every Router Has A v3 Group Requiring Privacy (0.1s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **groupname: \${SNMP_GROUP}** (checked 3×)
- output contains **security model:v3 priv** (checked 3×)

How it was checked

R1 show snmp group

```
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

R2 show snmp group

```
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

R3 show snmp group

```
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.04 PASS No Router Accepts SNMPv1 Or v2c Communities (0.3s, 3 commands)

Security posture: configuring v3 is pointless if a community string still gets you in.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **\${out}** is empty (checked 3×)

How it was checked

R1 show running-config | include ^snmp-server community
\${output} =

R2 show running-config | include ^snmp-server community
\${output} =

R3 show running-config | include ^snmp-server community
\${output} =

Teardown (suite) Close Trap Collector, Close All Connections

16.05 PASS Every Router Targets The Collector With v3 Privacy (0.1s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **\${COLLECTOR}** (checked 3×)
- output contains **security model: v3 priv** (checked 3×)
- output contains **user: \${SNMP_USER}** (checked 3×)

How it was checked

R1 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

```
Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R2 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

```
Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

R3 show snmp host

```
Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

```
Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv
```

Teardown (suite) Close Trap Collector, Close All Connections

16.06 PASS Global Trap Sending Is Enabled (0.1s, 3 commands)

Enabling individual trap types leaves the global switch off: the router builds trap PDUs and never sends them, which looks exactly like a broken collector.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **SNMP global trap: enabled** (checked 3×)

How it was checked

R1 show snmp | include global trap

```
SNMP global trap: enabled
```

R2 show snmp | include global trap

```
SNMP global trap: enabled
```

R3 show snmp | include global trap

```
SNMP global trap: enabled
```

Teardown (suite) Close Trap Collector, Close All Connections

16.07 PASS Traps From Every Router Reach The Collector (13.3s, 51 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **\${traps}** is not empty (checked 3×)

How it was checked

R1 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 interface Loopback270

```
${output} =
```

R1 ip address 10.99.97.1 255.255.255.255

```
${output} =
```

R1 snmp trap link-status

```

    ${output} =
R1 end
    ${output} =
R1 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Loopback270
    ${output} =
R1 shutdown
    ${output} =
R1 end
    ${output} =
R1 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Loopback270
    ${output} =
R1 no shutdown
    ${output} =
R1 end
    ${output} =
R1 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R1 no interface Loopback270
    ${output} =
R1 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
    ${output} =
R2 ip address 10.99.97.1 255.255.255.255
    ${output} =
R2 snmp trap link-status
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
    ${output} =
R2 shutdown
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
    ${output} =
R2 no shutdown
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no interface Loopback270
    ${output} =
R2 end
    ${output} =
R3 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R3 interface Loopback270
    ${output} =
R3 ip address 10.99.97.1 255.255.255.255
    ${output} =
R3 snmp trap link-status
    ${output} =

```

```
R3 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Loopback270
${output} =

R3 shutdown
${output} =

R3 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Loopback270
${output} =

R3 no shutdown
${output} =

R3 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 no interface Loopback270
${output} =

R3 end
${output} =

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb62948440836307965026c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6dda4f4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdfc96
8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aedd6db
b3ee8cb9bfc314d080afa0b0fb48addca5f604a787508df6374629f81bf9c5e8ed322c4fbc9fd5daf2e665dcbc312cc01027c101b15d
95d74b6c51ef9faf06c0670481620af82d566719918aab054f908d5415c32688a26757db8041390a1fd440c938835bf8a171ee3321ef
59f0d1a550a2e55083266bba89add3ddb3176843bbfc92d3de40fb901ba83f7d8c7124f27c7fe6db1dce
1788819441.982 192.168.99.1 3081f1020103300d020138020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c54e0b089dfb6d3286639954d04089b184efb1b39c21a0481a38399f66019ba392443b719d26a
fd64cb17b4f7e47e6efc18540d7a161bb4df34a5d637d8baf88d7959b126d7f786e1521239899f10d3d27949f04880309d9a8e456a78
4b512b0558a7215ca4ba9844e8fac890bae9434f944d06cc273d3d6bb9fd842b28c4aa4b6f7723acfc5a8d378b7f6d6019f1c750dbd
2b44a96e3e05c10c5df1deb3c7c8f3b1022054494304e4456fad68c0f554b6016aa90535c790e85d921b
1788819442.019 192.168.99.2 3081f1020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cc5e940312135bbb74315f07f0408bdf4dc46c97074950481a3d122cd69208ea63a33c15acf8
7a45102a3d13ff4011a6fe0077d1190d97e78309f972e8bd391feaf8f439f142fa831bc51566d9eb8017d4cb97e7436f0ff3c8ef79ef
ea5d0c4c1aab31cc9b7f8cfff78355da4f4fcfa92bdd9281133bec7fae933da59e755762b226e899cb3c6998c3fb887f3397554d5152
dcbea7abdb77818b788f02536a47b9893019f437a968d65cf16414aaecf588c7acf31617d6f2d26c240f
1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c97f59d812586c8cbc82aeecc80408a7c7be9c6efe40590481a3944f05c35a695e8ab698a77fc0
679d4817a25d2867b284b3553e5b9560d1abeb0194ef14399db882e9fa2c2b7a2e5a3a8776478141012d370690d28f11c26bfc8aa1fd
53215e87e900d9dda1de7d41080b83bf95c9c0743949bd5c4af9ea11abc6a118939fad85eb0735991d6b9d8e891ccf0bd9c9b99abf172
805a8c9b5e5d778a279e1a0384713bb102e8ba0b0be1d56a9527722ef00de7e6baabfe3aae5f497a44f4
1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8010102
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... 6713 more lines
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
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```



```
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... 6714 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.08 PASS Received Traps Are SNMP Version 3 (0.4s, 3 commands)

Reads the version field out of the datagram itself rather than trusting the configuration.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `${traps}` is not empty (checked 3×)
- `${versions}` equals `${[[3]]}` (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788819441.732 192.168.99.1 308f1f020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
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... 6714 more lines

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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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... 6715 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.09 PASS Received Traps Carry The Configured User And Privacy Flags (0.4s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `${users}` equals `${['${SNMP_USER}']}` (checked 3×)
- `${all_priv}` holds (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
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```



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... 6716 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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```

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1788819442.306 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
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3389b312

```
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1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
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f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bf23d6a00a98a878bc9f8f206e82d8a3606be17431
... 6717 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.10 PASS Trap Payloads Are Encrypted (0.4s, 3 commands)

With privacy in force the scoped PDU arrives as an opaque OCTET STRING. If it were a readable SEQUENCE the varbinds would be on the wire in clear.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- `#{encrypted}` holds (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
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1788819442.267 192.168.99.2 3081f1020103300d020142020205dc04010302010304373035040c800000090300525400c8020102
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```

1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c73a76bc49e552aab0d08f64d04089b184efb1b39c21e0481a3b1a6978d7875aef47484d41426
fcac7bd01a45e729fb041e4e0053bfc6e1e7b4453849a0950cb51754d5aa8bb1e683b91e18dadd70c8eeb400fd5392669c3139475a2e
f9cea2e5f556ab286affc2bd299f7257f7f9aaa3362d185225a4037aefee8a1fb964fcd2e3cf759405bf4911bb9c8e340b9c3c32a39f9b
caf86df01c6728b43aa549379361193e7dcca3f3f0abd1c6f25b53985eb169cb05adfaa725d29e505e5
1788819442.518 192.168.99.2 3081f1020103300d020144020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040ce22c3e1cccc3a0816b2de3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5df6e6a36ba9
061eb6d5f8e4cea2b22f8283e04ff27fe21962da8a88d91b1c4fbd6c6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a923e4766fa1263512636869017b82cec5982625Feded62ba54a93598fd3604d189683a16346138220d45f3a5
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
```

```
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcd825ab5152edf654545b7ab79bde2ffad6f60703003bf05fd372b55c6598a7edececf536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babd3b4f4ce0ebb07f01f9f9fd45e6fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecdcd96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67978f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbbbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be04256f622
03359df9f4b249e253bfc47820d97178378d92b51f24de8d455b5342f2d76797f484d9471253faea500058a89ac76482836cd483314
b637ad7fc13ded93b704de011795cd6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd550b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbeb0d2965d0cedde29f5e6919fa5f14ad44ce1b2fffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 6717 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.11 PASS snmptrapd Decrypts Traps From Every Router (0.2s, 3 commands)

The strongest single statement this suite makes: a real SNMP manager, holding only the configured user and each router's engine ID, authenticated and decrypted the traps. That cannot happen unless the keys, the engine IDs and the privacy protocol all match on both sides.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **{entries}** is not empty (checked 3×)
- **{oids}** is not empty (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true`

```
NET-SNMP version 5.9.4.pre2
2026-09-08 01:45:03 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:32.39 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.29 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.30 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.66 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
... 6659 more lines
```

NMS `sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true`

```
NET-SNMP version 5.9.4.pre2
2026-09-08 01:45:03 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:32.39 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
```

```

iso.3.6.1.4.1.9.10.62.1.2.3.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.29 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.30 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.66 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
... 6659 more lines

```

NMS `sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true`

```

NET-SNMP version 5.9.4.pre2
2026-09-08 01:45:03 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:32.39 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.29 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.30 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.66 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
... 6659 more lines

```

Teardown (suite) Close Trap Collector, Close All Connections

16.12 PASS One Raw Listener Receives Every Router (0.1s, 1 commands)

The property the per-router-port workaround existed to dodge: a single socket now serves all three senders.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains \${R1_OOB_IP}
- output contains \${R2_OOB_IP}
- output contains \${R3_OOB_IP}

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788819441.732 192.168.99.1 308f1f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
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1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
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aef92a4fd087aa
1788819442.733 192.168.99.1 308f1f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
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d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfe88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
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```



```
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 6717 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.13 PASS Each Router Identifies Itself By Its Own Engine ID (0.4s, 6 commands)

The engine ID in the received trap must match the one the router reports, and must differ between routers -- that is what lets a collector attribute a trap to a device.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **`\${local}`** is not empty (checked 3×)
- **`\${ids}`** has exactly **1** entries (checked 3×)
- **`\${ids}[0]`** equals **`\${expected}`** (checked 3×)
- **`\${distinct}`** equals **3**

the three routers did not present three distinct engine IDs: **`\${seen}`**

How it was checked

R1 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80101
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
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158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
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8ee7b7c9e197db9da0102a62640bcd82abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aed6db
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1788819442.019 192.168.99.2 3081f1020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
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1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
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1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8010102
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1788819442.267 192.168.99.2 3081f1020103300d020142020205dc04010302010304373035040c800000090300525400c8020102
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1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
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1788819442.518 192.168.99.2 3081f1020103300d020144020205dc04010302010304373035040c800000090300525400c8020102
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```
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... 6717 more lines
```

R2 show snmp engineID | include Local

```
Local snmp engineID: 800000090300525400C80201
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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... 6717 more lines
```

R3 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80301
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c3bb1f3582527a19034ec5d610408a7c7be9c6efe405b0481a3fa3381e2249898d509d85e1d9b
3389b3128c7d34aaa0145cd77854357e6d69cd388db0d86c165a4fdd8a3d87f9bb961854ba187a1984c45dbbf58adca5541a5485c344
583a562913db9d44704b325f28ab90c3c74d18b4c3bd08fc9130c87f7165dc4c6d22a6e4226cd8546bac4ed78bd313ac2d76bfad8165bf
d95c8ff1fe7981815276c5c818ab3f027acbecb251449958991c430e2af57b98530410546f515e89124
1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c73a76bc49e552aab0d08f64d04089b184efb1b39c21e0481a3b1a6978d7875aef47484d41426
fcac7bd01a45e729fb041e4e0053bcf6e1e7b4453849a0950cb51754d5aa8bb1e683b91e18dedd70c8eeb400fd5392669c3139475a2e
f9cea2e5f556ab286affc24b299f7257af79aaa3362d185225a4037aef86b1fb964fcd2e3f759405bf4911bf9c8e3a40b9c3c32a39f9b
caf86df01c6728b43aa549379361193e7dccea3f3f0abd1c6f25b53985eb169cb05adfaa725d29e505e5
1788819442.518 192.168.99.2 3081f1020103300d020144020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040ce22c3e1cccc3a081b62d3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5dfe6a36ba9
061eb6d5f8e4cea2b22f8283e04ff27fe21962da8a88d91b1c4fbdcb6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a92447466fa1263512636869017b82cec5982625Feded62ba54a93598f2360d4189683a1634613822d04df53fa5
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
```

020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcd825ab5152edf654545b7ab79bde2ffad60703003bf05fd372b55c6598a7edececd536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babd3db4f4ce0ebb07f01f9fffd45e6fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecdcd96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f622
03359df9f4b249e253bfc47820d97178378d92b51f24de8d8455b5342f2d76797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbe0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 6717 more lines

Teardown (suite) Close Trap Collector, Close All Connections

17 Snmp Polling

17.01 **PASS** The NMS Runs Real Net-SNMP Tooling (0.1s, 2 commands)

The point of this VM: unlike the cirros hosts it has a genuine SNMP client, so the polling below is the real thing rather than something hand rolled for the test.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output contains **NET-SNMP version**
- output contains **snmpget**
- output contains **snmpwalk**

How it was checked

NMS `snmpget --version 2>&1 | head -2; snmpwalk --version 2>&1 | head -1`

```
NET-SNMP version: 5.9.4.pre2
NET-SNMP version: 5.9.4.pre2
```

NMS `command -v snmpget snmpwalk snmptrapd`

```
/usr/bin/snmpget
/usr/bin/snmpwalk
/usr/sbin/snmptrapd
```

Teardown (suite) Close Nms, Close All Connections

17.02 **PASS** Every Router Answers An SNMPv3 authPriv Poll (0.2s, 3 commands)

The agent must be reachable and must accept the credentials the provisioner configured.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

Teardown (suite) Close Nms, Close All Connections

17.03 **PASS** Polled sysName Matches The Router's Own Hostname (0.4s, 6 commands)

Cross-check: the value SNMP returns is compared against what the device says on its CLI, so a plausible-looking answer from the wrong box cannot pass.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **`\${polled}`** starts with **`\${expected}`[0]** (checked 3×)
- `msg=SNMP sysName `${polled}` does not match CLI hostname `${expected}`[0]`

How it was checked

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

R1 `show running-config | include ^hostname`

```
hostname c8000v-r1
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
R2 show running-config | include ^hostname
hostname c8000v-r2
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

```
R3 show running-config | include ^hostname
hostname c8000v-r3
```

Teardown (suite) Close Nms, Close All Connections

17.04 PASS Polled sysDescr Identifies The Platform (0.2s, 6 commands)

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output matches Cisco IOS[-]XE Software|Cisco IOS Software \[IOSXE] (checked 3×)

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R1 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R2 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R3 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

Teardown (suite) Close Nms, Close All Connections

17.05 PASS sysUpTime Advances Between Two Polls (5.1s, 2 commands)

Proves the answers come from a live agent rather than a cached or replayed response.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- \${second} > \${first} holds

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
```

```
.1.3.6.1.2.1.1.3.0 = Timeticks: (392575) 1:05:25.75
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
```

```
.1.3.6.1.2.1.1.3.0 = Timeticks: (393077) 1:05:30.77
```

Teardown (suite) Close Nms, Close All Connections

17.06 PASS Walking The Interface Table Returns Every Interface The CLI Shows (0.2s, 6 commands)

A walk, not a get: the agent must return the whole subtree. Every physical interface named by the CLI must appear in it.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- `${walked}` contains `${intf}` (checked 12×)
msg=\${r}: ifDescr walk is missing `${intf}`

How it was checked

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

R1 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	100.64.0.1	YES	NVRAM	up	up
GigabitEthernet3	192.168.10.1	YES	NVRAM	up	up
GigabitEthernet4	192.168.99.1	YES	NVRAM	up	up
Loopback0	172.16.1.1	YES	NVRAM	up	up
Loopback1	10.20.1.1	YES	NVRAM	up	up
Tunnel0	172.20.0.1	YES	NVRAM	up	up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

R2 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	100.64.0.2	YES	NVRAM	up	up
GigabitEthernet3	192.168.20.1	YES	NVRAM	up	up
GigabitEthernet4	192.168.99.2	YES	NVRAM	up	up
Loopback0	172.16.2.1	YES	NVRAM	up	up
Loopback1	10.20.2.1	YES	NVRAM	up	up
Tunnel0	172.20.0.2	YES	NVRAM	up	up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

R3 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	100.64.0.3	YES	NVRAM	up	up
GigabitEthernet3	192.168.30.1	YES	NVRAM	up	up
GigabitEthernet4	192.168.99.3	YES	NVRAM	up	up
Loopback0	172.16.3.1	YES	NVRAM	up	up
Loopback1	10.20.3.1	YES	NVRAM	up	up
Tunnel0	172.20.0.3	YES	NVRAM	up	up

Teardown (suite) Close Nms, Close All Connections

17.07 PASS Polled Interface Count Agrees With The Interface Table (0.3s, 6 commands)

ifNumber is a scalar the agent maintains separately from the table, so the two disagreeing would mean a broken agent.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **`\${number}` equals `\${rows}`** (checked 3×)

msg=\${r}: ifNumber says `\${number}` but the table has `\${rows}` rows

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.1.0
```

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.1.0
```

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.1.0
```

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

Teardown (suite) Close Nms, Close All Connections

17.08 PASS The LAN Interface Is Reported Up By Both SNMP And The CLI (0.4s, 9 commands)

ifOperStatus 1 is up(1). Checked against the CLI line protocol state for the same interface.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- `${index}` differs from **-1** (checked 3×)
msg=\${r}: no Gi3 in the walk
- output contains **1** (checked 3×)
msg=\${r}: SNMP says Gi3 is not up
- output contains **line protocol is up** (checked 3×)

How it was checked

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"  
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"  
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"  
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"  
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"  
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"  
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"  
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"  
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

```
R1 show interfaces GigabitEthernet3 | include line protocol
```

```
GigabitEthernet3 is up, line protocol is up
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"  
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"  
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"  
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"  
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"  
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"  
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"  
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"  
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

```
R2 show interfaces GigabitEthernet3 | include line protocol
```

```
GigabitEthernet3 is up, line protocol is up
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"  
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"  
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"  
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"  
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"  
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"  
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"  
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"  
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1  
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
```

```
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

R3 show interfaces GigabitEthernet3 | include line protocol

```
GigabitEthernet3 is up, line protocol is up
```

Teardown (suite) Close Nms, Close All Connections

17.09 PASS Spokes Are Polled Across The Encrypted Tunnel (1.2s, 16 commands)

The NMS shares a LAN with the hub only. Traffic to R2 and R3 crosses the IPsec tunnels, so a burst of polling must show up on the ESP counters at both ends.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- `${a_enc} >= ${expected}` holds
- `${a_dec} >= ${expected}` holds
- `${b_enc} >= ${expected}` holds
- `${b_dec} >= ${expected}` holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 80030, #pkts encrypt: 80030, #pkts digest: 80030
  #pkts decaps: 42644, #pkts decrypt: 42644, #pkts verify: 42644
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 52454, #pkts encrypt: 52454, #pkts digest: 52454
  #pkts decaps: 27967, #pkts decrypt: 27967, #pkts verify: 27967
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 80036, #pkts encrypt: 80036, #pkts digest: 80036
  #pkts decaps: 42649, #pkts decrypt: 42649, #pkts verify: 42649
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 52459, #pkts encrypt: 52459, #pkts digest: 52459
  #pkts decaps: 27972, #pkts decrypt: 27972, #pkts verify: 27972
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines

```

Teardown (suite) Close Nms, Close All Connections

17.10 PASS A Wrong Authentication Password Is Refused (0.1s, 2 commands)

Negative control. The agent answers the correct credentials first, so a refusal here cannot be a dead router.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A NotTheRightPassword -x AES -X LabPrivPass123 -t 2 -r 0
192.168.99.1 1.3.6.1.2.1.1.5.0
snmpget: Authentication failure (incorrect password, community or key)

```

Teardown (suite) Close Nms, Close All Connections

17.11 PASS A Wrong Privacy Password Is Refused (2.2s, 3 commands)

Authentication succeeds but the payload will not decrypt, so the router drops the request silently and the client times out. Bracketed by a good poll to prove it is still alive.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c800v-r1.lab.local"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X NotTheRightPassword -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
```

```
Timeout: No Response from 192.168.99.1.
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c800v-r1.lab.local"
```

Teardown (suite) Close Nms, Close All Connections

17.12 PASS Polling Without Privacy Is Refused By Every Router (0.3s, 6 commands)

The group is configured "v3 priv". Correct credentials at a lower security level must still be denied, otherwise the privacy requirement is decorative.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
```

```
Error in packet
```

```
Reason: authorizationError (access denied to that object)
```

```
NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
```

```
Error in packet
```

```
Reason: authorizationError (access denied to that object)
```

```
NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0
```

```
Error in packet
```

```
Reason: authorizationError (access denied to that object)
```

```
NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0
```

```
Error in packet
```

```
Reason: authorizationError (access denied to that object)
```

```
NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0
```

```
Error in packet
```

```
Reason: authorizationError (access denied to that object)
```

```
NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0
```

```
Error in packet
```

```
Reason: authorizationError (access denied to that object)
```

Teardown (suite) Close Nms, Close All Connections

17.13 PASS An Unknown User Is Refused (0.1s, 1 commands)

Only the provisioned user exists in the USM table.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u intruder -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
```

```
snmpget: Unknown user name
```

Teardown (suite) Close Nms, Close All Connections

17.14 PASS SNMPv2c Gets No Answer At The Wire (6.3s, 6 commands)

Suite 16 asserts no community is configured. This asserts the consequence on the network: a v2c poll gets nothing back, while a v3 poll to the same address succeeds.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output contains **Timeout** (checked 3×)

How it was checked

NMS `snmpget -On -v2c -c public -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0`

Timeout: No Response from 192.168.99.1.

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0`

.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS `snmpget -On -v2c -c public -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0`

Timeout: No Response from 192.168.99.2.

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS `snmpget -On -v2c -c public -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0`

Timeout: No Response from 192.168.99.3.

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0`

.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

Teardown (suite) Close Nms, Close All Connections

18 Mtu

18.01 PASS Every Tunnel Reports The Expected Transport MTU (0.3s, 9 commands)

The tunnel MTU must be below the physical one by the ESP overhead. Equal would mean encryption costs nothing, which would mean it is not happening.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- `${tunnels}` is not empty (checked 3×)
- output contains `${TUNNEL_MTU} bytes` (checked 3×)
- output contains `MTU is ${TUNNEL_MTU} bytes` (checked 3×)
- `${TUNNEL_MTU} < ${PHYSICAL_MTU}` holds

How it was checked

```
R1 show ip interface brief | include ^Tunnel
Tunnel0          172.20.0.1      YES NVRAM  up          up

R1 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1472 bytes

R1 show ip interface Tunnel0 | include MTU
MTU is 1472 bytes

R2 show ip interface brief | include ^Tunnel
Tunnel0          172.20.0.2      YES NVRAM  up          up

R2 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1472 bytes

R2 show ip interface Tunnel0 | include MTU
MTU is 1472 bytes

R3 show ip interface brief | include ^Tunnel
Tunnel0          172.20.0.3      YES NVRAM  up          up

R3 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1472 bytes

R3 show ip interface Tunnel0 | include MTU
MTU is 1472 bytes
```

Teardown (suite) Close All Connections

18.02 PASS The Overhead Is What GRE With A Tunnel Key Costs (0.0s, 0 commands)

Records the number rather than assuming it: 20 bytes of outer IP, 4 of GRE and 4 for the tunnel key. If the encapsulation changes -- dropping the key, or moving IPsec from transport to tunnel mode -- this is the test that should fail first.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- `${overhead}` equals 28

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

18.03 PASS A Packet Exactly At The Tunnel MTU Crosses With DF Set (0.1s, 2 commands)

The permissive half of the boundary. Sourced from Loopback1 so the traffic is inside the protected selector. Warmed up first: the first packet between two spokes is consumed establishing the shortcut and its security association, so a cold measurement here reports 80 percent and says nothing about the MTU. That first-packet loss is DMVPN working, not failing.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 100 percent**
a packet of exactly the tunnel MTU (`${TUNNEL_MTU}`) did not cross

How it was checked

```
R2 ping 10.20.3.1 source Loopback1 repeat 5
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

R2 ping 10.20.3.1 source Loopback1 size 1472 df-bit repeat 5

```
Type escape sequence to abort.
Sending 5, 1472-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

Teardown (suite) Close All Connections

18.04 **PASS** One Byte Over The Tunnel MTU Does Not Cross With DF Set (10.0s, 1 commands)

The restrictive half. Without this the test above would pass just as well against a tunnel with no MTU limit at all.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 0 percent**
a packet one byte over the tunnel MTU crossed with DF set

How it was checked

R2 ping 10.20.3.1 source Loopback1 size 1473 df-bit repeat 5

```
Type escape sequence to abort.
Sending 5, 1473-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
.....
Success rate is 0 percent (0/5)
```

Teardown (suite) Close All Connections

18.05 **PASS** Oversize Traffic Still Crosses When Fragmentation Is Allowed (0.1s, 2 commands)

Distinguishes an MTU limit from a blackhole: the same sizes that fail with DF set must succeed without it.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)
a \${size}-byte packet failed even with fragmentation allowed

How it was checked

R2 ping 10.20.3.1 source Loopback1 size 1500 repeat 5

```
Type escape sequence to abort.
Sending 5, 1500-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 size 2000 repeat 5

```
Type escape sequence to abort.
Sending 5, 2000-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (suite) Close All Connections

18.06 **PASS** Hosts Pass Sub-MTU Traffic With DF Set (4.0s, 1 commands)

End to end across two encrypt/decrypt hops, from machines that know nothing about the tunnel.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains , 0% packet loss
- a host could not pass \${HOST_PAYLOAD} bytes with DF set

How it was checked

```
H2 ping -c 5 -M do -s 1444 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 1444(1472) bytes of data.
1452 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=2.76 ms
1452 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.77 ms
1452 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.50 ms
1452 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.51 ms
1452 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.38 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.382/1.783/2.760/0.504 ms
```

Teardown (suite) Close All Connections

18.07 PASS An End Host Learns The Tunnel Path MTU (2.0s, 1 commands)

The test that matters most in practice. The host is two hops and an encryption boundary away from the constraint, yet it must end up knowing the real limit -- otherwise a client retransmits into a black hole instead of backing off. The host reporting the exact tunnel MTU is proof that path MTU discovery survived the tunnel.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **mtu=\${TUNNEL_MTU}**
- the host did not learn the \${TUNNEL_MTU}-byte path MTU: \${out}

How it was checked

```
H2 ping -c 2 -M do -s 1472 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
From 192.168.20.1 icmp_seq=1 Frag needed and DF set (mtu = 1472)
ping: local error: message too long, mtu=1472

--- 192.168.30.10 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1015ms
```

Teardown (suite) Close All Connections

18.08 PASS Hosts Pass Oversize Traffic Without DF Set (4.0s, 1 commands)

The counterpart: the same payload that was refused with DF set gets through when it may be fragmented.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains , 0% packet loss
- oversize host traffic failed even with fragmentation allowed

How it was checked

```
H2 ping -c 5 -s 1472 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
1480 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.56 ms
1480 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.64 ms
1480 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.40 ms
1480 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.55 ms
1480 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.51 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.403/1.533/1.644/0.078 ms
```

Teardown (suite) Close All Connections

19 Oob

19.01 PASS Every Router Has A Management VRF (0.0s, 3 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **MGMT_VRF** (checked 3×)
- output contains **ipv4** (checked 3×)

How it was checked

R1 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R2 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R3 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.02 PASS Only The OOB Interface Is In The Management VRF (0.1s, 3 commands)

A VRF containing a data interface would defeat the point, so this asserts membership is exactly one interface.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **MGMT** (checked 3×)
- output does not contain **data** (checked 9×)

How it was checked

R1 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R2 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R3 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.03 PASS Every Router Holds Its OOB Address On The Flat Network (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **192.168.99.1** (checked 3×)
- output contains **up** (checked 3×)

How it was checked

R1 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.1	YES	NVRAM	up	up
------------------	--------------	-----	-------	----	----

R2 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.2	YES	NVRAM	up	up
------------------	--------------	-----	-------	----	----

R3 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.3	YES	NVRAM	up	up
------------------	--------------	-----	-------	----	----

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.04 PASS The Management VRF Routing Table Holds Only The Flat Network (0.1s, 3

commands)

The isolation, stated as routing: the VRF must know how to reach the management network and nothing else. A default route or a data prefix here would be a leak.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **#{OOB_NET}** (checked 3×)
- output does not contain **#{data}** (checked 12×)

How it was checked

R1 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

R2 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

R3 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.05 PASS The Global Table Cannot Reach The Management Network (0.1s, 3 commands)

The other direction. Traffic in the global table must have no path to the OOB network, otherwise the VRF is decorative.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **not in table** (checked 3×)

How it was checked

```

R1 show ip route 192.168.99.0
% Network not in table

R2 show ip route 192.168.99.0
% Network not in table

R3 show ip route 192.168.99.0
% Network not in table

```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.06 PASS Every Router Answers SNMP On Its OOB Address (0.4s, 6 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- `${name}` starts with `${expected}[0]` (checked 3×)
- polling `${OOB_IP}${r}` answered as `'${name}'`, not `${r}`

How it was checked

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

R1 show running-config | include ^hostname
hostname c8000v-r1

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

R2 show running-config | include ^hostname
hostname c8000v-r2

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

R3 show running-config | include ^hostname
hostname c8000v-r3

```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.07 PASS Management Traffic Never Touches The Data Path (3.3s, 28 commands)

The negative property that makes this out-of-band. Polling a spoke used to cross an IPsec tunnel and was asserted to advance the ESP counters; now it must do the opposite. The comparison is against an idle window of the same length, because BGP and BFD tick those counters continuously regardless -- so an absolute delta proves nothing.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- `${attributable} < ${POLL_BURST}` holds

How it was checked

```

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 80244, #pkts encrypt: 80244, #pkts digest: 80244
    #pkts decaps: 42860, #pkts decrypt: 42860, #pkts verify: 42860
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 78 more lines

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"  
  
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3  
1.3.6.1.2.1.1.5.0  
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```



```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
```

```
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 80256, #pkts encrypt: 80256, #pkts digest: 80256
#pkts decaps: 42865, #pkts decrypt: 42865, #pkts verify: 42865
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 80256, #pkts encrypt: 80256, #pkts digest: 80256
#pkts decaps: 42865, #pkts decrypt: 42865, #pkts verify: 42865
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 80262, #pkts encrypt: 80262, #pkts digest: 80262
#pkts decaps: 42877, #pkts decrypt: 42877, #pkts verify: 42877
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.08 PASS Syslog Reaches The Collector From The OOB Address (12.2s, 4 commands)

rsyslog files are named for the sender. A file named after a LAN address would mean the router is logging from the data path despite the configuration. Each router is made to emit a marked message rather than waiting on ambient traffic: whether a router happens to have logged recently is not what this test is about, and depending on it makes the result a matter of timing.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains `${OOB_IP}${r}` (checked 3×)
- `${n} > 0` holds (checked 3×)
- output does not contain `${r}_LAN_IP.log` (checked 3×)

How it was checked

```
NMS sudo grep -c -- 'LAB00B-SRC-R1-224043' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

```
NMS sudo grep -c -- 'LAB00B-SRC-R2-224043' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
```

```
NMS sudo grep -c -- 'LAB00B-SRC-R3-224043' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
```

1

```
NMS sudo ls -l /var/log/lab
total 5920
-rw-r--r-- 1 syslog syslog 318916 Sep  8 02:40 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 381321 Sep  8 02:40 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 255866 Sep  8 02:40 192.168.99.3.log
-rw-r--r-- 1 root  root 1235305 Sep  8 02:40 snmptrapd.log
-rw-r--r-- 1 root  root 3843161 Sep  8 02:40 traps-raw.log
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.09 PASS A Spoke Stays Manageable With Its Tunnel Down (14.4s, 15 commands)

The reason out-of-band management exists. Shutting R2's tunnel removes every data path between the hub and the spoke -- BGP drops, the LAN becomes unreachable -- and the spoke must still answer SNMP and keep logging, because none of that ever used the tunnel.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **not in table** (checked 2×)
the hub can still reach R2's LAN, so the data path is not down yet
- **#{before}** equals **#{after}**
R2 stopped answering SNMP once its tunnel went down, so management is not out-of-band
- **#{n} > 0** holds
- output matches **#{peer}ls+4ls+#{peer_asn}(ls+ld+){5}ls+S+ls+ld+** (checked 2×)

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Tunnel0

#{output} =

R2 shutdown

#{output} =

R2 end

#{output} =

R1 show ip route 192.168.20.0

```
Routing entry for 192.168.20.0/24
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:08:34 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:08:34 ago
    opaque_ptr 0x71FE1BFD9178
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none
```

R1 show ip route 192.168.20.0

% Network not in table

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
NMS sudo grep -c -- 'LAB00B-224100' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Tunnel0

#{output} =

R2 no shutdown

#{output} =

R2 end

#{output} =

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001

```
BGP table version is 13, main routing table version 13
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 137/131 prefixes, 149/143 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:13:05.687 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 16, main routing table version 16
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 140/131 prefixes, 152/143 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (00:13:10.726 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

Teardown (this test) Restore Tunnels

20 Tacacs

20.01 **PASS** The TACACS Server Runs On The NMS (0.1s, 2 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- `${state}` equals **active**
tac_plus is not running on the NMS
- output contains `:${TAC_PORT}`
- output contains **tac_plus**

How it was checked

```
NMS systemctl is-active tacacs-lab
active
```

```
NMS sudo ss -ltnp | grep ':49 ' || echo NONE
LISTEN 0      128          0.0.0.0:49      0.0.0.0:*      users: (("tac_plus",pid=1460,fd=4))
LISTEN 0      128          [::]:49         [::]:*         users: (("tac_plus",pid=1460,fd=5))
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.02 **PASS** The Server Holds Both Remote Users And The Automation Account (0.1s, 1 commands)

The automation account has to exist on the server as well as locally on each router. Without it the first router to get AAA becomes unmanageable by this very test suite, while the server sits there perfectly healthy.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains `${ADMIN_USER}`
 - output contains `${OPS_USER}`
 - output contains `${USERNAME}`
- the automation account is missing from the server, which would lock the suite out

How it was checked

```
NMS sudo grep -oE '^user = [A-Za-z0-9_-]+' /etc/tacacs+/tac_plus.conf
user = netadmin
user = netops
user = lab
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.03 **PASS** Every Router Points At The Server Through The Management VRF (0.4s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **aaa new-model** (checked 3×)
- output contains **address ipv4 \${NMS_OOB_IP}** (checked 3×)
- output contains **ip vrf forwarding \${MGMT_VRF}** (checked 3×)
- output contains **ip tacacs source-interface \${OOB_INTF}[\${r}]** (checked 3×)

How it was checked

```
R1 show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
```

```

aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10

```

R2 `show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source`

```

aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10

```

R3 `show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source`

```

aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10

```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.04 **PASS** Every Method List Falls Back To The Local Database (0.4s, 3 commands)

A method list with no local fallback locks the lab out the moment the server is unreachable -- including from the console, which is the one way back in.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **login default group \${TAC_GROUP} local** (checked 3×)
- output contains **exec default group \${TAC_GROUP} local** (checked 3×)

How it was checked

R1 `show running-config | include ^aaa authentication login|^aaa authorization exec`

```

aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated

```

R2 `show running-config | include ^aaa authentication login|^aaa authorization exec`

```

aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated

```

R3 `show running-config | include ^aaa authentication login|^aaa authorization exec`

```

aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated

```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.05 **PASS** Every Router Reports The Server Alive (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **Server Status: Alive** (checked 3×)
- output contains **\${NMS_OOB_IP}** (checked 3×)

How it was checked

R1 `show tacacs`

```

Tacacs+ Server - public :
Server name: LAB-TAC
Server address: 192.168.99.10
Server port: 49

```

```

        Socket opens:      824
        Socket closes:     824
        Socket aborts:     0
        Socket errors:     0
        Socket Timeouts:   0
    Failed Connect Attempts: 0
    Total Packets Sent:     575
    Total Packets Recv:     575
        Server Status: Alive
    Continous Authc fail count: 3
    ... 1 more lines

```

R2 show tacacs

```

Tacacs+ Server - public :
    Server name: LAB-TAC
    Server address: 192.168.99.10
    Server port: 49
        Socket opens:      1186
        Socket closes:     1186
        Socket aborts:     0
        Socket errors:     0
        Socket Timeouts:   0
    Failed Connect Attempts: 0
    Total Packets Sent:     980
    Total Packets Recv:     980
        Server Status: Alive
    Continous Authc fail count: 3
    ... 1 more lines

```

R3 show tacacs

```

Tacacs+ Server - public :
    Server name: LAB-TAC
    Server address: 192.168.99.10
    Server port: 49
        Socket opens:      623
        Socket closes:     623
        Socket aborts:     0
        Socket errors:     0
        Socket Timeouts:   0
    Failed Connect Attempts: 0
    Total Packets Sent:     413
    Total Packets Recv:     413
        Server Status: Alive
    Continous Authc fail count: 3
    ... 1 more lines

```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.06 PASS A Remote Administrator Authenticates And Is Given Privilege 15 (0.2s, 0 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- `${priv}` equals `${ADMIN_PRIV}` (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.07 PASS A Remote Operator Authenticates And Is Held At Privilege 1 (0.2s, 0 commands)

The pair that proves the level comes from the server. Two users, same router, same method list, different levels -- which cannot happen if the router is applying a default.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- `${priv}` equals `${OPS_PRIV}` (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.08 PASS A Wrong Password Is Refused (2.1s, 0 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.09 PASS An Unknown User Is Refused (2.1s, 0 commands)

The server rejects the user and the router honours that rather than quietly trying the local database.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.10 PASS The Server Records Authentication In Its Accounting Log (0.1s, 1 commands)

Accounting is what makes remote authentication auditable; without it the router knows who logged in and nobody else ever finds out.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true
Sep  8 02:38:21 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=4661      timezone=UTC
service=shell  start_time=1788835101 priv-lvl=15 cmd=show monitor capture POOLCAP buffer brief <cr>
Sep  8 02:38:24 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=4662      timezone=UTC
service=shell  start_time=1788835104 priv-lvl=15 cmd=no monitor capture POOLCAP <cr>
Sep  8 02:38:24 192.168.99.1 lab      tty435 10.0.2.2      start   task_id=4664      timezone=UTC
service=shell  start_time=1788835104
Sep  8 02:38:24 192.168.99.2 lab      tty434 10.0.2.2      stop    task_id=5005      timezone=UTC
service=shell  start_time=1788835011 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=93 stop_time=1788835104
Sep  8 02:38:24 192.168.99.3 lab      tty434 10.0.2.2      stop    task_id=4473      timezone=UTC
service=shell  start_time=1788835011 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=93 stop_time=1788835104
Sep  8 02:38:24 192.168.99.1 lab      tty434 10.0.2.2      stop    task_id=4643      timezone=UTC
service=shell  start_time=1788835011 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=93 stop_time=1788835104
Sep  8 02:38:24 192.168.99.2 lab      tty435 10.0.2.2      start   task_id=5013      timezone=UTC
service=shell  start_time=1788835104
Sep  8 02:38:24 192.168.99.3 lab      tty434 10.0.2.2      start   task_id=4479      timezone=UTC
service=shell  start_time=1788835104
Sep  8 02:38:24 192.168.99.1 lab      tty435 10.0.2.2      stop    task_id=4664      timezone=UTC
service=shell  start_time=1788835104 priv-lvl=15 cmd=show running-config <cr>
Sep  8 02:38:24 192.168.99.2 lab      tty435 10.0.2.2      stop    task_id=5013      timezone=UTC
service=shell  start_time=1788835104 priv-lvl=15 cmd=show running-config <cr>
Sep  8 02:38:24 192.168.99.3 lab      tty434 10.0.2.2      stop    task_id=4479      timezone=UTC
service=shell  start_time=1788835104 priv-lvl=15 cmd=show running-config <cr>
Sep  8 02:38:27 192.168.99.1 lab      tty435 10.0.2.2      stop    task_id=4664      timezone=UTC
service=shell  start_time=1788835104 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788835107
Sep  8 02:38:27 192.168.99.2 lab      tty435 10.0.2.2      stop    task_id=5013      timezone=UTC
service=shell  start_time=1788835104 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788835107
Sep  8 02:38:27 192.168.99.3 lab      tty434 10.0.2.2      stop    task_id=4479      timezone=UTC
service=shell  start_time=1788835104 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788835107
... 186 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.11 PASS Accounting Records Carry The Privilege Level And Come From The OOB Address (0.1s, 1 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **service=shell**
- output contains **\${R3_OOB_IP}**

the record did not come from R3's out-of-band address

How it was checked

NMS `sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true`

```
Sep  8 02:38:21 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4661 timezone=UTC
service=shell start_time=1788835101 priv-lvl=15 cmd=show monitor capture POOLCAP buffer brief <cr>
Sep  8 02:38:24 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4662 timezone=UTC
service=shell start_time=1788835104 priv-lvl=15 cmd=no monitor capture POOLCAP <cr>
Sep  8 02:38:24 192.168.99.1 lab tty435 10.0.2.2 start task_id=4664 timezone=UTC
service=shell start_time=1788835104
Sep  8 02:38:24 192.168.99.2 lab tty434 10.0.2.2 stop task_id=5005 timezone=UTC
service=shell start_time=1788835011 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=93 stop_time=1788835104
Sep  8 02:38:24 192.168.99.3 lab tty434 10.0.2.2 stop task_id=4473 timezone=UTC
service=shell start_time=1788835011 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=93 stop_time=1788835104
Sep  8 02:38:24 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4643 timezone=UTC
service=shell start_time=1788835011 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=93 stop_time=1788835104
Sep  8 02:38:24 192.168.99.2 lab tty435 10.0.2.2 start task_id=5013 timezone=UTC
service=shell start_time=1788835104
Sep  8 02:38:24 192.168.99.3 lab tty434 10.0.2.2 start task_id=4479 timezone=UTC
service=shell start_time=1788835104
Sep  8 02:38:24 192.168.99.1 lab tty435 10.0.2.2 stop task_id=4664 timezone=UTC
service=shell start_time=1788835104 priv-lvl=15 cmd=show running-config <cr>
Sep  8 02:38:24 192.168.99.2 lab tty435 10.0.2.2 stop task_id=5013 timezone=UTC
service=shell start_time=1788835104 priv-lvl=15 cmd=show running-config <cr>
Sep  8 02:38:24 192.168.99.3 lab tty434 10.0.2.2 stop task_id=4479 timezone=UTC
service=shell start_time=1788835104 priv-lvl=15 cmd=show running-config <cr>
Sep  8 02:38:27 192.168.99.1 lab tty435 10.0.2.2 stop task_id=4664 timezone=UTC
service=shell start_time=1788835104 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788835107
Sep  8 02:38:27 192.168.99.2 lab tty435 10.0.2.2 stop task_id=5013 timezone=UTC
service=shell start_time=1788835104 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788835107
Sep  8 02:38:27 192.168.99.3 lab tty434 10.0.2.2 stop task_id=4479 timezone=UTC
service=shell start_time=1788835104 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788835107
... 186 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.12 PASS The Local Account Still Works When The Server Is Unreachable (19.6s, 21

commands)

The resilience property. R2's out-of-band interface is shut, which is the only path to the server, so TACACS becomes unreachable rather than merely unwilling. The automation account must still get in through the local database, and a remote-only user must not -- it exists nowhere else. The precondition is a ping inside the VRF, not "show tacacs". That field reports cached state -- it keeps saying Alive until a transaction actually fails -- so waiting on it would either hang or, worse, let this test run with the server still reachable and prove nothing.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output matches **Success rate is 0 percent**|does not have a usable source address
 - **\${priv}** equals **15**
- the local account could not get in with the server unreachable
- output contains **Success rate is 100 percent** (checked 4×)

How it was checked

R2 **configure** **terminal**

Enter configuration commands, one per line. End with CNTL/Z.

R2 **interface** **GigabitEthernet4**

`${output}` =

R2 **shutdown**

`${output}` =

R2 **end**


```

${output} =
R2 ping vrf MGMT 192.168.99.10 repeat 2
% VRF MGMT does not have a usable source address
R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 interface GigabitEthernet4
${output} =
R1 no shutdown
${output} =
R1 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 interface GigabitEthernet4
${output} =
R2 no shutdown
${output} =
R2 end
${output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 interface GigabitEthernet4
${output} =
R3 no shutdown
${output} =
R3 end
${output} =
R2 ping vrf MGMT 192.168.99.10 repeat 5
% VRF MGMT does not have a usable source address
R2 ping vrf MGMT 192.168.99.10 repeat 5
% VRF MGMT does not have a usable source address
R2 ping vrf MGMT 192.168.99.10 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
R2 ping vrf MGMT 192.168.99.10 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

```

Teardown (this test) Restore Oob Interfaces

21 Vty Acl

21.01 **PASS** Every Router Applies The ACL Inbound On All VTY Lines (0.3s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **access-class \${VTY_ACL} in vrf-also** (checked 3×)

How it was checked

R1 show running-config | section ^line vty

```
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
```

R2 show running-config | section ^line vty

```
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
```

R3 show running-config | section ^line vty

```
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.02 **PASS** Only SSH Is Accepted On The VTY Lines (0.3s, 3 commands)

An ACL restricting who may connect is worth less if telnet is still an option for those who may.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **transport input ssh** (checked 3×)
- output does not contain **transport input all** (checked 3×)
- output does not contain **transport input telnet** (checked 3×)

How it was checked

R1 show running-config | section ^line vty

```
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
```

R2 show running-config | section ^line vty

```
line vty 0 4
  access-class VTY-MGMT in vrf-also
```

```

exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh

```

R3 show running-config | section ^line vty

```

line vty 0 4
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh
line vty 5 15
access-class VTY-MGMT in vrf-also
exec-timeout 0 0
activation-character 13
transport input ssh

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.03 PASS The ACL Permits The Management Network And Denies The Rest (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **permit** **\$(OOB_NET)**, wildcard bits **0.0.0.255** (checked 3×)
 - output contains **deny** (checked 3×)
- any log

How it was checked

R1 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (68 matches)
 30 deny any log

```

R2 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (82 matches)
 30 deny any log

```

R3 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (68 matches)
 30 deny any log

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.04 PASS The Documented Exception Is Present And Is Only The Hypervisor Path (0.1s, 3 commands)

Pinned deliberately. This permit is the one concession in the policy, so it should be visible in a test rather than only in a comment -- and widening it should break something.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **permit** **\$(MGMT_NAT_NET)**, wildcard bits **0.0.0.255** (checked 3×)
- **\$(permits)** has exactly **2** entries (checked 3×)

How it was checked

R1 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (68 matches)
 30 deny any log

```

R2 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
10 permit 192.168.99.0, wildcard bits 0.0.0.255
20 permit 10.0.2.0, wildcard bits 0.0.0.255 (82 matches)
30 deny any log
```

R3 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
10 permit 192.168.99.0, wildcard bits 0.0.0.255
20 permit 10.0.2.0, wildcard bits 0.0.0.255 (68 matches)
30 deny any log
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.05 PASS SSH From The Management Network Is Allowed (0.0s, 1 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- `${banner}` starts with SSH-

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.99.10',0))
try:
    s.connect(('192.168.99.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

SSH-2.0-Cisco-1.25

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.06 PASS SSH From Outside The Management Network Is Refused (0.1s, 1 commands)

The same machine, a different source address, and a target it is directly connected to -- so nothing but the ACL can account for the difference. Before the access-class was applied this path returned a banner just like the one above.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

REFUSED ConnectionRefusedError: [Errno 111] Connection refused

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.07 PASS Every Router Refuses A Login From Outside The Management Network (0.2s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
```

```
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

REFUSED ConnectionRefusedError: [Errno 111] Connection refused

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.20.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

REFUSED ConnectionRefusedError: [Errno 111] Connection refused

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.30.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

REFUSED ConnectionRefusedError: [Errno 111] Connection refused

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.08 PASS The Refusal Is Logged And Reaches The Collector (0.1s, 2 commands)

"deny any log" makes a rejected connection auditable rather than merely blocked, and the message travels the OOB network to the collector like every other log.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output matches **SEC-6-IPACCESSLOGS***
no access-list denial logged by \${router} reached the collector

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
```

REFUSED ConnectionRefusedError: [Errno 111] Connection refused

NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true

```
<189>207: 000203: *Sep  7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep  7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep  7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep  7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep  7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep  7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep  7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep  7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep  7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep  7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep  7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
```

```

'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep  7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep  7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep  7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 2107 more lines

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.09 PASS The Harness Path Still Works (0.1s, 3 commands)

The reason the second permit exists. If this fails the lab is unreachable by every other suite, so it is asserted here rather than discovered in the next run. Asserted from the ACL's own hit counter rather than from "show users". That column prints a resolved name when reverse DNS answers -- inside the packaged appliance the same session shows as "_gateway" instead of 10.0.2.2 -- so matching on the rendered address made this depend on name resolution rather than on the thing being tested. A hit on that ACE means a session was admitted by exactly the permit under discussion.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- `${hits}` is not empty (checked 3×)
- `${hits}[0] > 0` holds (checked 3×)

How it was checked

R1 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (2 matches)
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (68 matches)
 30 deny any log (3 matches)

```

R2 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (82 matches)
 30 deny any log (1 match)

```

R3 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (68 matches)
 30 deny any log (1 match)

```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

22 Banner

22.01 **PASS** Every Router Has A Login Banner Configured (0.3s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **banner login** (checked 3×)

How it was checked

R1 `show running-config | include ^banner`
banner login ^C

R2 `show running-config | include ^banner`
banner login ^C

R3 `show running-config | include ^banner`
banner login ^C

Teardown (suite) Close All Connections

22.02 **PASS** The Banner Is Presented Before Authentication (6.2s, 0 commands)

Captured from a login that is refused, so the banner demonstrably reaches an unauthenticated client.

Setup (suite) Open All Routers

Acceptance criteria

- `${banner}` is not empty (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.03 **PASS** The Presented Banner Matches The Source File Exactly (6.1s, 0 commands)

Line for line, not a substring match: a device carrying a stale or truncated copy should fail rather than pass on one memorable phrase still being present.

Setup (suite) Open All Routers

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.04 **PASS** The Banner States The Warning It Exists To Give (2.1s, 0 commands)

A banner that does not actually warn is decoration. These are the elements that make it one: restricted access, a statement of monitoring, and consent following from use.

Setup (suite) Open All Routers

Acceptance criteria

- output contains `${phrase}` (checked 4×)
the banner does not state: `${phrase}`

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.05 **PASS** The Banner Discloses Nothing About The Platform (6.2s, 0 commands)

An unauthenticated visitor should learn that they are not welcome, and nothing else. A banner naming the vendor, model, software or hostname helps whoever should not be there more than it helps anyone who should.

Setup (suite) Open All Routers

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.06 **PASS** The Banner Survives A Configuration Save (0.1s, 3 commands)

Guards the case where a banner is applied to the running configuration and never written -- present until the next reload, and absent exactly when it would be needed.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **banner login** (checked 3×)

How it was checked

R1 `show startup-config | include ^banner`
banner login ^C

R2 `show startup-config | include ^banner`
banner login ^C

R3 `show startup-config | include ^banner`
banner login ^C

Teardown (suite) Close All Connections

23 Destructive

23.01 PASS A Shortcut Outlives The Hub, But Its Routing Does Not (67.7s, 30 commands)

The claim worth testing rather than assuming. A spoke-to-spoke tunnel is genuinely direct: the NHRP mapping and the IPsec session are between the spokes and the hub is not on the path. So the hub going away should not break it. Except that it does, one layer up. Both spokes learn each other's prefixes from BGP, and both peer only with the hub. Lose the hub and BFD tears those sessions down in about two seconds, the prefixes withdraw, and there is no route to put on the tunnel -- which is still sitting there, perfectly healthy, carrying nothing. So this asserts both halves: the tunnel address still answers, and the loopback behind it does not. The data plane is hub-independent here; the control plane is not.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **peer WAN IP**
- output contains **not in table** (checked 2×)
- output contains **R3 WAN IP**
the direct NHRP mapping did not survive the hub, so the tunnel was never direct
- output contains **R3 WAN IP**
the spoke-to-spoke IPsec session did not survive the hub
- output contains **Success rate is 100 percent**
the direct tunnel stopped carrying traffic when the hub went away
- output contains **Success rate is 0 percent**
the spoke still routes to the far loopback with the hub down, so the
- output contains **WAN IP** (checked 8×)
- output contains **Known via** (checked 9×)
- output contains **Success rate is 100 percent**
the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

R2 ping 10.20.3.1 source Loopback1 repeat 10

```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!!!!!!
Success rate is 90 percent (9/10), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:58
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3
```

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Tunnel0

```
{output} =
```

R1 shutdown

```
{output} =
```

R1 end

```
{output} =
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:01:01 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:01:01 ago
    opaque_ptr 0x798416842808
    Route metric is 0, traffic share count is 1
    AS Hops 2
    Route tag 65001
    MPLS label: none
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:07, expire 00:04:52
```

```
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3
```

R2 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
100.64.0.3 Tu0              100.64.0.3          00:00:07 UA
100.64.0.1 Tu0              00:00:00 DN
```

R2 ping 172.20.0.3 source Tunnel0 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.20.0.3, timeout is 2 seconds:
Packet sent with a source address of 172.20.0.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.....
Success rate is 0 percent (0/5)
```

R1 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R1 interface Tunnel0

```
{output} =
```

R1 no shutdown

```
{output} =
```

R1 end

```
{output} =
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====
```

```
Interface: Tunnel0, IPv4 NHRP Details
Type:Unknown, NHRP Peers:1,
```

```
# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 1 more lines
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====
```

```
Interface: Tunnel0, IPv4 NHRP Details
Type:Unknown, NHRP Peers:1,
```

```
# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 2 more lines
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
```

NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

#	Ent	Peer	NBMA	Addr	Peer	Tunnel	Add	State	UpDn	Tm	Attrb
... 2 more lines											

R2 show ip route 10.20.3.1

% Subnet not in table

R2 show ip route 10.20.3.1

% Subnet not in table

R2 show ip route 10.20.3.1

Routing entry for 10.20.3.1/32
Known via "bgp 65002", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.3 00:00:05 ago
Routing Descriptor Blocks:
* 172.20.0.3, from 172.20.0.1, 00:00:05 ago
opaque_ptr 0x798416842808
Route metric is 0, traffic share count is 1
AS Hops 2
Route tag 65001
MPLS label: none

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

% Subnet not in table

R3 show ip route 10.20.2.1

Routing entry for 10.20.2.1/32
Known via "bgp 65003", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.2 00:00:03 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.1, 00:00:03 ago
opaque_ptr 0x7AED8E74CB50
Route metric is 0, traffic share count is 1
AS Hops 2
Route tag 65001
MPLS label: none

R2 ping 10.20.3.1 source Loopback1 repeat 5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

Teardown (this test) Restore Hub Tunnel

23.02 PASS The Fabric Comes Back When The Hub Does (0.2s, 5 commands)

Recovery without operator action beyond restoring the hub: registration, BGP and the shortcut all return on their own.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **\${R}_WAN_IP** (checked 2×)
- output contains **Known via**
- output contains **\${peer}_WAN_IP**
- output contains **\${R3_DMVPN_IP}**
- output does not contain **\${R1_DMVPN_IP}**

the shortcut did not re-form after the hub returned

How it was checked

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel

=====
```

```
Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,
```

```
# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
Known via "bgp 65002", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.3 00:00:30 ago
Routing Descriptor Blocks:
* 172.20.0.3, from 172.20.0.1, 00:00:30 ago
  opaque_ptr 0x798416842808
  Route metric is 0, traffic share count is 1
  AS Hops 2
  Route tag 65001
  MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 10

```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!
Success rate is 100 percent (10/10), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
Tunnel0 created 00:01:07, expire 00:03:52
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3
```

R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric

```
Type escape sequence to abort.
Tracing the route to 10.20.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.20.0.3 2 msec
```

Teardown (suite) Restore Everything, Close All Connections

23.03 PASS Blocking The Spokes At The Underlay Falls Back Through The Hub (23.6s, 27 commands)

The realistic partial failure: two sites that can both reach the hub but not each other, which is what happens whenever a firewall between branches permits the head end and nothing else. Resolution cannot complete, so no shortcut forms -- and connectivity must survive anyway, through the hub. The suite's other fallback test only clears the cache, which is the easy version: nothing is stopping the shortcut from forming a moment later. Here it genuinely cannot.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- `#{rate}[0] > 0` holds
- output does not contain `#{R3_WAN_IP}`
a direct mapping formed even though the underlay between spokes is blocked
- output contains `#{R1_DMVPN_IP}`
traffic is not transiting the hub, so it is not using the fallback:\n#{trace}
- output contains `#{#{r}_WAN_IP}` (checked 2×)
- output contains **Known via** (checked 2×)
- output contains **Success rate is 100 percent** (checked 2×)
the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

```

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip access-list extended BLOCK-SPOKE
${output} =

R2 deny ip host 100.64.0.2 host 100.64.0.3
${output} =

R2 deny ip host 100.64.0.3 host 100.64.0.2
${output} =

R2 permit ip any any
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 interface GigabitEthernet2
${output} =

R2 ip access-group BLOCK-SPOKE in
${output} =

R2 end
${output} =

R2 clear ip nhrp
${output} =

R2 ping 10.20.3.1 source Loopback1 repeat 10
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.....
Success rate is 80 percent (8/10), round-trip min/avg/max = 1/1/2 ms

R2 show ip nhrp 172.20.0.3
172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:04, expire 00:03:00
    Type: dynamic, Flags: temporary
    NBMA address: 100.64.0.1

R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric
Type escape sequence to abort.
Tracing the route to 10.20.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.20.0.1 0 msec
 2 172.20.0.3 3 msec

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 interface GigabitEthernet2
${output} =

R2 no ip access-group BLOCK-SPOKE in
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip access-list extended BLOCK-SPOKE
${output} =

R2 end
${output} =

R2 clear ip nhrp
${output} =

R1 show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel

=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----

```

... 2 more lines

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:46 ago
  Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:00:46 ago
      opaque_ptr 0x798416842808
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:19 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:19 ago
      opaque_ptr 0x7AED8E74CB50
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Remove Spoke Block

23.04 PASS An NHRP Authentication Mismatch Is Refused (12.4s, 22 commands)

The authentication string must match across the cloud. A spoke that gets it wrong is refused registration and simply never appears at the hub. Both caches have to be cleared to see it. The hub holds its registration entry for the full holdtime, so clearing only the spoke leaves the hub answering from a stale cache -- and the test reads "still registered" and concludes, wrongly, that the mismatch had no effect.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output does not contain **`\${peer}\_WAN\_IP`**
the hub still holds a registration for **`\${peer}`**
- output contains **`\${r}\_WAN\_IP`** (checked 4×)
- output contains **Known via** (checked 2×)
- output contains **Success rate is 100 percent** (checked 2×)
the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Tunnel0

`\${output}` =

R3 ip nhrp authentication WrongKey

`\${output}` =

R3 end

`\${output}` =

R1 clear ip nhrp

`\${output}` =

R2 clear ip nhrp

`\${output}` =

```

R3 clear ip nhrp
${output} =

R1 show dmvpn | include 100.64.0
${output} =

R3 configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.

R3 interface Tunnel0
${output} =

R3 tunnel key 100
${output} =

R3 ip nhrp authentication LabNhrp
${output} =

R3 end
${output} =

R1 clear ip nhrp
${output} =

R2 clear ip nhrp
${output} =

R3 clear ip nhrp
${output} =

R1 show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====

R1 show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
... 2 more lines

R2 show ip route 10.20.3.1
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:59 ago
  Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:00:59 ago
      opaque_ptr 0x798416842808
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

R3 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:32 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:32 ago
      opaque_ptr 0x7AED8E74CB50
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

R2 ping 10.20.3.1 source Loopback1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!

```

Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms

R2 ping 10.20.3.1 source Loopback1 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:

Packet sent with a source address of 10.20.2.1

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

Teardown (this test) Restore Spoke Tunnel Settings

23.05 PASS A Tunnel Key Mismatch Fails Silently Instead (47.7s, 30 commands)

The same outcome by a different route, and the reason both are worth having. A wrong authentication string is a refusal. A wrong tunnel key is not refused by anything -- the GRE headers simply do not match, the packets are discarded, and the tunnel interface goes on reporting itself up with nothing passing through it.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output does not contain **\${peer}_WAN_IP**
the hub still holds a registration for \${peer}
- output matches **upls+up**
the interface reported itself down, which would make this failure the
- output contains **\${r}_WAN_IP** (checked 6×)
- output contains **Known via** (checked 8×)
- output contains **Success rate is 100 percent** (checked 2×)
the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Tunnel0

\${output} =

R3 tunnel key 999

\${output} =

R3 end

\${output} =

R1 clear ip nhrp

\${output} =

R2 clear ip nhrp

\${output} =

R3 clear ip nhrp

\${output} =

R1 show dmvpn | include 100.64.0

\${output} =

R3 show ip interface brief | include Tunnel0

Tunnel0	172.20.0.3	YES	NVRAM	up	up
---------	------------	-----	-------	----	----

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Tunnel0

\${output} =

R3 tunnel key 100

\${output} =

R3 ip nhrp authentication LabNhrp

\${output} =

R3 end

\${output} =

R1 clear ip nhrp

\${output} =

R2 clear ip nhrp

\${output} =

R3 clear ip nhrp

\${output} =

R1 show dmvpn

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete


```

N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

```

R1 show dmvpn

```

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

```

```

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:1,

```

```

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
... 1 more lines

```

R1 show dmvpn

```

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

```

```

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

```

```

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
... 2 more lines

```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```

Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:04 ago
  Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:00:04 ago
      opaque_ptr 0x798416842948
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:00 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:00 ago
      opaque_ptr 0x7AED8E74CB50
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Restore Spoke Tunnel Settings

23.06 PASS Blocking The Too-Big Message Blackholes Large Traffic (7.3s, 18 commands)

The classic silent MTU failure. The router tells a host "fragmentation needed, MTU 1472" and the host adapts. Take that message away and nothing announces the fault: small packets keep working, ping keeps working, and anything bulk stalls with no diagnostic anywhere. The message is suppressed with "no ip unreachable" on the interface it would leave by -- R2's LAN side, facing the host. An outbound ACL was the obvious first choice and it does not work: an outbound ACL does not filter traffic the router originates, and this message is originated by R2. Applying it to Tunnel0 does nothing either, for the same reason -- the constraint is on the tunnel, but the message is emitted towards the host. Packet loss on its own proves nothing here: the oversized ping fails either way, because the host is forbidden to fragment it. What changes is whether anything says why, so the assertions are about the diagnostic and not the loss. This is what makes the boundary the MTU suite measures matter, and it is why that suite asserts the host learns the path MTU rather than merely that big packets fail.

Setup (this test) Wait For Converged Fabric

Acceptance criteria

- output contains **\${r}_WAN_IP** (checked 2×)
- output contains **Known via** (checked 2×)
- output contains **Success rate is 100 percent**
the spokes cannot reach each other end to end, so the fabric is not converged
- output contains **mtu = \${TUNNEL_MTU}**
the host did not learn the path MTU even before anything was blocked
- output contains **100% packet loss**
large traffic got through even with the too-big message suppressed
- output does not contain **mtu = \${TUNNEL_MTU}**
the host still learned the MTU, so the message was not actually suppressed
- output does not contain **Frag needed**
the host was still told fragmentation was needed, so the failure is not
- output contains **, 0% packet loss**
small packets failed too, so this is an outage rather than a blackhole

How it was checked

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel

=====
```

```
Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,
```

```
# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
Known via "bgp 65002", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.3 00:00:37 ago
Routing Descriptor Blocks:
* 172.20.0.3, from 172.20.0.1, 00:00:37 ago
```

```
opaque_ptr 0x798416842948
Route metric is 0, traffic share count is 1
AS Hops 2
Route tag 65001
MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:08 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:08 ago
      opaque_ptr 0x7AED8E74CB50
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

H2 sudo ip route flush cache

```
${output} =
```

H2 ping -c 2 -W 2 -M do -s 1472 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
From 192.168.20.1 icmp_seq=1 Frag needed and DF set (mtu = 1472)
ping: local error: message too long, mtu=1472

--- 192.168.30.10 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1009ms
```

R2 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R2 interface GigabitEthernet3

```
${output} =
```

R2 no ip unreachable

```
${output} =
```

R2 end

```
${output} =
```

H2 sudo ip route flush cache

```
${output} =
```

H2 ping -c 3 -W 2 -M do -s 1472 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.

--- 192.168.30.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2039ms

H2 ping -c 2 -W 2 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.40 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.25 ms

--- 192.168.30.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.250/1.323/1.396/0.073 ms
```

R2 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R2 interface GigabitEthernet3

```
${output} =
```

R2 ip unreachable

```
${output} =
```

R2 end

```
${output} =
```

H2 sudo ip route flush cache

```
${output} =
```

Teardown (this test) Remove Pmtu Block

23.07 PASS NHRP Mappings Are Soft State And Age Out (132.1s, 48 commands)

A shortcut is a cache entry, not configuration: it exists because traffic justified it and it should disappear when that stops being true. Proven with a shortened holdtime -- the configured 300s would add five minutes to every run for a property that is no truer at 300 than at 60. Expiry is not punctual. NHRP ages entries lazily, and a 60s holdtime was measured clearing at about 170s, so the wait here is generous rather than exact.

Setup (this test) Wait For Converged Fabric

Acceptance criteria

- output contains `${r}_WAN_IP` (checked 6×)
 - output contains **Known via** (checked 4×)
 - output contains **Success rate is 100 percent** (checked 3×)
- the spokes cannot reach each other end to end, so the fabric is not converged
- output contains `${peer}_WAN_IP` (checked 3×)
 - output does not contain `${peer}_WAN_IP` (checked 11×)

How it was checked

R1 show dmvpn

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

#	Ent	Peer	NBMA	Addr	Peer	Tunnel	Add	State	UpDn	Tm	Attrb
...	2	more	lines								

R2 show ip route 10.20.3.1

Routing entry for 10.20.3.1/32
Known via "bgp 65002", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.3 00:00:44 ago
Routing Descriptor Blocks:
* 172.20.0.3, from 172.20.0.1, 00:00:44 ago
opaque_ptr 0x798416842948
Route metric is 0, traffic share count is 1
AS Hops 2
Route tag 65001
MPLS label: none

R3 show ip route 10.20.2.1

Routing entry for 10.20.2.1/32
Known via "bgp 65003", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.2 00:00:15 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.1, 00:00:15 ago
opaque_ptr 0x7AED8E74CB50
Route metric is 0, traffic share count is 1
AS Hops 2
Route tag 65001
MPLS label: none

R2 ping 10.20.3.1 source Loopback1 repeat 5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Tunnel0

`${output}` =

R2 ip nhrp holdtime 60

`${output}` =

R2 end

`${output}` =

```

R3 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.

R3 interface Tunnel0
  ${output} =

R3 ip nhrp holdtime 60
  ${output} =

R3 end
  ${output} =

R1 clear ip nhrp
  ${output} =

R2 clear ip nhrp
  ${output} =

R3 clear ip nhrp
  ${output} =

R2 ping 10.20.3.1 source Loopback1 repeat 10
  Type escape sequence to abort.
  Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  .!!!!!!!
  Success rate is 90 percent (9/10), round-trip min/avg/max = 1/1/1 ms

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:02, expire 00:00:58
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:02, expire 00:00:57
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:12, expire 00:00:47
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:22, expire 00:00:37
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:32, expire 00:00:27
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:42, expire 00:00:54
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:00:52, expire 00:00:44
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:01:02, expire 00:00:34
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:01:12, expire 00:00:24
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
    Tunnel0 created 00:01:22, expire 00:00:14
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3

```

```

Tunnel0 created 00:01:32, expire 00:00:04
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
${output} =

R2 ping 10.20.3.1 source Loopback1 repeat 10
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!!!!
Success rate is 90 percent (9/10), round-trip min/avg/max = 1/1/2 ms

R2 show ip nhrp 172.20.0.3
172.20.0.3/32 via 172.20.0.3
Tunnel0 created 00:00:02, expire 00:00:58
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3

R2 show ip nhrp 172.20.0.3
172.20.0.3/32 via 172.20.0.3
Tunnel0 created 00:00:02, expire 00:00:57
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Tunnel0
${output} =

R2 ip nhrp holdtime 300
${output} =

R2 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Tunnel0
${output} =

R3 ip nhrp holdtime 300
${output} =

R3 end
${output} =

R1 clear ip nhrp
${output} =

R2 clear ip nhrp
${output} =

R3 clear ip nhrp
${output} =

R1 show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

R1 show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 2 more lines

R2 show ip route 10.20.3.1
Routing entry for 10.20.3.1/32
Known via "bgp 65002", distance 20, metric 0
Tag 65001, type external

```

```
Last update from 172.20.0.3 00:02:49 ago
Routing Descriptor Blocks:
* 172.20.0.3, from 172.20.0.1, 00:02:49 ago
  opaque_ptr 0x798416842948
  Route metric is 0, traffic share count is 1
  AS Hops 2
  Route tag 65001
  MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:02:20 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:02:20 ago
    opaque_ptr 0x7AED8E74CB50
    Route metric is 0, traffic share count is 1
    AS Hops 2
    Route tag 65001
    MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Teardown (this test) Restore Holdtime

Appendix — captured device state

Written by suite *05 Capture State* during the run. Routing tables and protocol state first, then what the NMS sees for each router -- polled over SNMPv3, and received as syslog and traps -- then the full running configuration of each router.

R1 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [254/0] via 10.0.2.2
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C 10.0.2.0/24 is directly connected, GigabitEthernet1
L 10.0.2.15/32 is directly connected, GigabitEthernet1
C 10.20.1.1/32 is directly connected, Loopback1
B 10.20.2.1/32 [20/0] via 172.20.0.2, 00:06:57
B 10.20.3.1/32 [20/0] via 172.20.0.3, 00:06:56
S 10.30.1.0/24 is directly connected, Null0
B 10.30.2.0/24 [20/0] via 172.20.0.2, 00:06:57
B 10.30.3.0/24 [20/0] via 172.20.0.3, 00:06:56
100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 100.64.0.0/24 is directly connected, GigabitEthernet2
L 100.64.0.1/32 is directly connected, GigabitEthernet2
172.16.0.0/32 is subnetted, 1 subnets
C 172.16.1.1 is directly connected, Loopback0
172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.20.0.0/24 is directly connected, Tunnel0
L 172.20.0.1/32 is directly connected, Tunnel0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet3
L 192.168.10.1/32 is directly connected, GigabitEthernet3
B 192.168.20.0/24 [20/0] via 172.20.0.2, 00:06:57
B 192.168.30.0/24 [20/0] via 172.20.0.3, 00:06:56
```

R2 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [254/0] via 10.0.2.2
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C 10.0.2.0/24 is directly connected, GigabitEthernet1
L 10.0.2.15/32 is directly connected, GigabitEthernet1
B 10.20.1.1/32 [20/0] via 172.20.0.1, 00:06:57
C 10.20.2.1/32 is directly connected, Loopback1
B 10.20.3.1/32 [20/0] via 172.20.0.3, 00:06:27
B 10.30.1.0/24 [20/0] via 172.20.0.1, 00:06:57
S 10.30.2.0/24 is directly connected, Null0
B 10.30.3.0/24 [20/0] via 172.20.0.3, 00:06:27
100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 100.64.0.0/24 is directly connected, GigabitEthernet2
L 100.64.0.2/32 is directly connected, GigabitEthernet2
172.16.0.0/32 is subnetted, 1 subnets
```



```

C      172.16.2.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.20.0.0/24 is directly connected, Tunnel0
L      172.20.0.2/32 is directly connected, Tunnel0
B      192.168.10.0/24 [20/0] via 172.20.0.1, 00:06:57
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.20.0/24 is directly connected, GigabitEthernet3
L      192.168.20.1/32 is directly connected, GigabitEthernet3
B      192.168.30.0/24 [20/0] via 172.20.0.3, 00:06:27

```

R3 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```

S*    0.0.0.0/0 [254/0] via 10.0.2.2
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C      10.0.2.0/24 is directly connected, GigabitEthernet1
L      10.0.2.15/32 is directly connected, GigabitEthernet1
B      10.20.1.1/32 [20/0] via 172.20.0.1, 00:06:56
B      10.20.2.1/32 [20/0] via 172.20.0.2, 00:06:56
C      10.20.3.1/32 is directly connected, Loopback1
B      10.30.1.0/24 [20/0] via 172.20.0.1, 00:06:56
B      10.30.2.0/24 [20/0] via 172.20.0.2, 00:06:56
S      10.30.3.0/24 is directly connected, Null0
      100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      100.64.0.0/24 is directly connected, GigabitEthernet2
L      100.64.0.3/32 is directly connected, GigabitEthernet2
      172.16.0.0/32 is subnetted, 1 subnets
C      172.16.3.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.20.0.0/24 is directly connected, Tunnel0
L      172.20.0.3/32 is directly connected, Tunnel0
B      192.168.10.0/24 [20/0] via 172.20.0.1, 00:06:56
B      192.168.20.0/24 [20/0] via 172.20.0.2, 00:06:56
      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.30.0/24 is directly connected, GigabitEthernet3
L      192.168.30.1/32 is directly connected, GigabitEthernet3

```

R1 — bgp summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 184, main routing table version 184
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 84/75 prefixes, 96/87 paths, scan interval 60 secs
9 networks peaked at 01:35:12 Sep 8 2026 UTC (00:48:10.065 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.2	4	65002	51	53	184	0	0	00:06:57	3
172.20.0.3	4	65003	51	54	184	0	0	00:06:56	3

R2 — bgp summary

BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 190, main routing table version 190
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory

2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 90/81 prefixes, 99/90 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (00:48:09.146 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	53	51	190	0	0	00:06:57	6

R3 — bgp summary

BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 196, main routing table version 196
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 99/90 prefixes, 102/93 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (00:48:09.233 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	54	51	196	0	0	00:06:56	6

R1 — bgp table

BGP table version is 184, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	172.20.0.2	0		0	65002 i
*> 10.20.3.1/32	172.20.0.3	0		0	65003 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	172.20.0.2	0		0	65002 i
*> 10.30.3.0/24	172.20.0.3	0		0	65003 i
*> 192.168.10.0	0.0.0.0	0		32768	i
*> 192.168.20.0	172.20.0.2	0		0	65002 i
*> 192.168.30.0	172.20.0.3	0		0	65003 i

R2 — bgp table

BGP table version is 190, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	172.20.0.1	0		0	65001 i
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3	0		0	65001 65003 i
*> 10.30.1.0/24	172.20.0.1	0		0	65001 i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 10.30.3.0/24	172.20.0.3	0		0	65001 65003 i
*> 192.168.10.0	172.20.0.1	0		0	65001 i
*> 192.168.20.0	0.0.0.0	0		32768	i
*> 192.168.30.0	172.20.0.3	0		0	65001 65003 i

R3 — bgp table

BGP table version is 196, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

```

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-state,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	172.20.0.1	0		0	65001 i
*>	10.20.2.1/32	172.20.0.2			0	65001 65002 i
*>	10.20.3.1/32	0.0.0.0	0		32768	i
*>	10.30.1.0/24	172.20.0.1	0		0	65001 i
*>	10.30.2.0/24	172.20.0.2			0	65001 65002 i
*>	10.30.3.0/24	0.0.0.0	0		32768	i
*>	192.168.10.0	172.20.0.1	0		0	65001 i
*>	192.168.20.0	172.20.0.2			0	65001 65002 i
*>	192.168.30.0	0.0.0.0	0		32768	i

R1 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
1	100.64.0.1/500	100.64.0.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1073, Session-id: 73				
Local spi: 6F52660E0B4ECD39		Remote spi: CA8260705A3473F7		

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
2	100.64.0.1/500	100.64.0.3/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1074, Session-id: 74				
Local spi: AEF5C6DE6B24D6A3		Remote spi: 68C218A7F9527196		

IPv6 Crypto IKEv2 SA

R2 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
1	100.64.0.2/500	100.64.0.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1070, Session-id: 68				
Local spi: CA8260705A3473F7		Remote spi: 6F52660E0B4ECD39		

IPv6 Crypto IKEv2 SA

R3 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
1	100.64.0.3/500	100.64.0.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/128 sec				
CE id: 1068, Session-id: 66				
Local spi: 68C218A7F9527196		Remote spi: AEF5C6DE6B24D6A3		

IPv6 Crypto IKEv2 SA

H1 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 83564sec preferred_lft 72764sec

```

```

inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
    valid_lft 86040sec preferred_lft 14040sec
inet6 fe80::5054:ff:feaa:101/64 scope link
    valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.15/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.16/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.17/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.18/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.19/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.64/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.65/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.66/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.67/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.68/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.69/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.70/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.71/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.72/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.73/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.74/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.75/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.76/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.77/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.78/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.79/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.80/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.81/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.82/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.83/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.84/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.85/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.86/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.87/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.88/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.89/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.90/24 scope global secondary eth1

```

```

    valid_lft forever preferred_lft forever
inet 192.168.10.91/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.92/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.93/24 scope global secondary eth1
    valid_lft forever preferred_lft forever

```

H1 — ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1

```

H2 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83564sec preferred_lft 72764sec
    inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
        valid_lft 86241sec preferred_lft 14241sec
    inet6 fe80::5054:ff:feaa:201/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.15/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.16/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.17/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.18/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.19/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.64/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.65/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.66/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.67/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.68/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.69/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.70/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.71/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.72/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.73/24 scope global secondary eth1
        valid_lft forever preferred_lft forever

```

```

inet 192.168.20.74/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.75/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.76/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.77/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.78/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.79/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.80/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.81/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.82/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.83/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.84/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.85/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.86/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.87/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.88/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.89/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.90/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.91/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.92/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.93/24 scope global secondary eth1
    valid_lft forever preferred_lft forever

```

H2 — ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1

```

H3 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83565sec preferred_lft 72765sec
    inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
        valid_lft 86083sec preferred_lft 14083sec
    inet6 fe80::5054:ff:feaa:301/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever

```

[illegible]

H3 — ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
```

```
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

NMS — network

```
lo UNKNOWN 127.0.0.1/8 ::1/128
ens3 UP 10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4 UP 192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5 UP 192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS — snmp version

NET-SNMP version: 5.9.4.pre2

NMS — collectors

active
active
active

NMS — collector files

```
total 5180
-rw-r--r-- 1 syslog syslog 283962 Sep 8 02:23 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 337492 Sep 8 02:23 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 233399 Sep 8 02:23 192.168.99.3.log
-rw-r--r-- 1 root root 985767 Sep 8 02:23 snmptrapd.log
-rw-r--r-- 1 root root 3436635 Sep 8 02:23 traps-raw.log
```

NMS — R1 syslog received

tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory

NMS — R2 syslog received

tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory

NMS — R3 syslog received

tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory

NMS — snmptrapd

```
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:22.54 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.69 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.69 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.69.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.69.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.69.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.69.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.69.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.69.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.69.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.69.1 0
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:22.90 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.71 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.71 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.71.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.71.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.71.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.71.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.71.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.71.1 "64 40 00 03 "
```


iso.3.6.1.4.1.9.9.171.1.3.3.1.6.71.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.71.1 0
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:23.67 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708063 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708063 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:54 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:23.31 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708061 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708061 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:24.70 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.71 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.71 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.71 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.71 4
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.71 214 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.70 214
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.70 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.70 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.70 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.70 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.70 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.70 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.70 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.70 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.70 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.70 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.70 0
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:24.70 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.69 215 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.68 215
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.68 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.68 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.68 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.68 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.68 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.68 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.68 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.68 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.68 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.68 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.68 0
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.05 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.2
iso.3.6.1.4.1.9.10.137.1.2.1.8.987167 3 iso.3.6.1.4.1.9.10.137.1.2.1.8.987167 3 iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:24.70 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.06 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.73 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.73 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.73 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.73 6
2026-09-08 02:22:56 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:24.71 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.2
iso.3.6.1.4.1.9.10.137.1.2.1.8.987165 3 iso.3.6.1.4.1.9.10.137.1.2.1.8.987165 3 iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-08 02:22:57 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.99 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.53 "DMVPN" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.53 4
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.53 "DMVPN_NHRP_ERROR" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.53 " Tunnel0: NHRP
Encap Error for Purge Reply , Reason: protocol generic error (7) on (Tunnel: 172.20.0.2 NBMA: 100.64.0.2)"
iso.3.6.1.4.1.9.9.41.1.2.3.1.6.53 0:0:48:25.98
2026-09-08 02:22:57 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:25.63 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.54 "DMVPN" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.54 4
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.54 "DMVPN_NHRP_ERROR" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.54 " Tunnel0: NHRP
Encap Error for Purge Request , Reason: protocol generic error (7) on (Tunnel: 172.20.0.3 NBMA:
100.64.0.3)" iso.3.6.1.4.1.9.9.41.1.2.3.1.6.54 0:0:48:25.62
2026-09-08 02:23:06 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:35.16 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.435.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.41954 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.41954 3560
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.41954 3504
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.41954 52339 iso.3.6.1.4.1.9.2.9.2.1.18.435 "lab"
2026-09-08 02:23:06 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:34.82 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.46468 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.46468 3549
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.46468 2784
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.46468 29731 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-08 02:23:06 <UNKNOWN> [UDP: [192.168.99.1]:53525->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:34.92 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.435.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.51866 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.51866 3572
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.51866 2688
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.51866 28707 iso.3.6.1.4.1.9.2.9.2.1.18.435 "lab"
2026-09-08 02:23:09 <UNKNOWN> [UDP: [192.168.99.1]:53525->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:37.86 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1

```

iso.3.6.1.4.1.9.9.43.1.1.6.1.3.53 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.53 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.53 2      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.53 "lab"
2026-09-08 02:23:13 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:42.84          iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.110 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.110 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.110 2      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.110 "lab"
2026-09-08 02:23:18 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:48:46.61          iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.76 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.76 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.76 2      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.76 "lab"

```

NMS — R1 polled over SNMPv3: snmp system

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293250) 0:48:52.50
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

NMS — R1 polled over SNMPv3: snmp interfaces

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"

```

NMS — R2 polled over SNMPv3: snmp system

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293289) 0:48:52.89
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

NMS — R2 polled over SNMPv3: snmp interfaces

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"

```

NMS — R3 polled over SNMPv3: snmp system

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (293267) 0:48:52.67

```

NMS — R3 polled over SNMPv3: snmp interfaces

R1 — running config

3000V IPsec Testbed - test evidence


```

604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28
quit
!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 9VRWIY571LW
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootstrap level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$uia9E1anibovxE$Qyklcz.SLvH8epz2w2TjV6d9TggTO3p9KSxEGgTTD2
username lab privilege 15 secret 9 $9$ksvY7W/rHjFg0.$AdIeocLbIuwaSOCYSowG0l/6LNagQ58URuynu0Dbtq6
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14
!
crypto ikev2 policy LAB-POL
 proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
 peer ANY
 address 0.0.0.0 0.0.0.0
 pre-shared-key LabPreSharedKey123
!
!
!
crypto ikev2 profile LAB-PROF
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
 mode transport
!
crypto ipsec profile LAB-IPSEC
 set transform-set LAB-TS
 set ikev2-profile LAB-PROF
!
!
!
!
!
!
```

```

!
!
!
interface Loopback0
description Protected endpoint
ip address 172.16.1.1 255.255.255.255
!
interface Loopback1
description BGP-advertised prefix
ip address 10.20.1.1 255.255.255.255
!
interface Tunnel0
description DMVPN hub
ip address 172.20.0.1 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp redirect
ip tcp adjust-mss 1360
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode gre multipoint
tunnel key 100
tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
description MGMT - QEMU user-mode net (DHCP)
ip address dhcp
negotiation auto
!
interface GigabitEthernet2
description WAN - shared provider segment, DMVPN underlay
ip address 100.64.0.1 255.255.255.0
negotiation auto
!
interface GigabitEthernet3
description LAN toward the local host
ip address 192.168.10.1 255.255.255.0
ip nat inside
negotiation auto
!
interface GigabitEthernet4
description OOB management - NTP, syslog, SNMP
vrf forwarding MGMT
ip address 192.168.99.1 255.255.255.0
negotiation auto
!
router bgp 65001
bgp router-id 172.16.1.1
bgp log-neighbor-changes
neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.2 password LabBgpSecret
neighbor 172.20.0.2 timers 10 30
neighbor 172.20.0.2 fall-over bfd
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.3 password LabBgpSecret
neighbor 172.20.0.3 timers 10 30
neighbor 172.20.0.3 fall-over bfd
!
address-family ipv4
network 10.20.1.1 mask 255.255.255.255
network 10.30.1.0 mask 255.255.255.0
network 192.168.10.0
neighbor 172.20.0.2 activate
neighbor 172.20.0.2 send-community both
neighbor 172.20.0.2 route-map RM-IN-R2 in
neighbor 172.20.0.2 route-map RM-OUT-R2 out
neighbor 172.20.0.2 maximum-prefix 15 80
neighbor 172.20.0.3 activate
neighbor 172.20.0.3 send-community both
neighbor 172.20.0.3 route-map RM-IN-R3 in
neighbor 172.20.0.3 route-map RM-OUT-R3 out
neighbor 172.20.0.3 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1

```

```

ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65002$
ip as-path access-list 20 permit ^65003$
no ip http server
ip http secure-server
ip route 10.30.1.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.1.100 10.30.1.129 prefix-length 24
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-FAR-R2 seq 40 permit 10.20.2.1/32
ip prefix-list PL-FAR-R2 seq 45 permit 10.30.2.0/24
ip prefix-list PL-FAR-R2 seq 50 permit 192.168.20.0/24
!
ip prefix-list PL-FAR-R3 seq 40 permit 10.20.3.1/32
ip prefix-list PL-FAR-R3 seq 45 permit 10.30.3.0/24
ip prefix-list PL-FAR-R3 seq 50 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.10.0/24
!
ip prefix-list PL-OWN-L0 seq 5 permit 10.20.1.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.1.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.10.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.10.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.10.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.10.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.10.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.10.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.10.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.10.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.10.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.10.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R2 permit 10
  description R1's own L0, tagged 65001:1
  match ip address prefix-list PL-OWN-L0
  set community 65001:1
!
route-map RM-OUT-R2 permit 20
  description R1's own NAT, tagged 65001:2
  match ip address prefix-list PL-OWN-NAT
  set community 65001:2
!
route-map RM-OUT-R2 permit 30
  description R1's own LAN, tagged 65001:3
  match ip address prefix-list PL-OWN-LAN
  set community 65001:3
!
route-map RM-OUT-R2 permit 55
  description re-advertise R3, preserving its communities

```

```

match ip address prefix-list PL-FAR-R3
!
route-map RM-OUT-R3 permit 10
description R1's own LO, tagged 65001:1
match ip address prefix-list PL-OWN-LO
set community 65001:1
!
route-map RM-OUT-R3 permit 20
description R1's own NAT, tagged 65001:2
match ip address prefix-list PL-OWN-NAT
set community 65001:2
!
route-map RM-OUT-R3 permit 30
description R1's own LAN, tagged 65001:3
match ip address prefix-list PL-OWN-LAN
set community 65001:3
!
route-map RM-OUT-R3 permit 55
description re-advertise R2, preserving its communities
match ip address prefix-list PL-FAR-R2
!
route-map NAT-TO-DOMAIN permit 10
match ip address NAT-DOMAIN
!
route-map RM-IN-R2 permit 10
description accept only known communities on an AS path R2 can produce
match as-path 10
match community CL-VALID
!
route-map RM-IN-R3 permit 10
description accept only known communities on an AS path R3 can produce
match as-path 20
match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold

```



```

snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch

```

```

snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0

```


[illegible]

```
80DDCD16 D6BACECA EEB7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB  
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0  
D697DF7F 28  
quit  
!  
!  
!  
!  
!  
!  
!  
!  
license udi pid C8000V sn 99MT10JL006  
license boot level network-premier addon dna-premier  
memory free low-watermark processor 31774  
diagnostic bootstrap level minimal  
!  
!  
spanning-tree extend system-id  
!  
enable secret 9 $9$IzR6xMPRz4hVOU$0/V1R.nWLX98duswhDwnTGs67nYdgYArKX89ewvQ67s  
!  
username lab privilege 15 secret 9 $9$bqcYxwxYLMCe2E$3qzzD/xblonG/KBzgLi14CgQBnGkOqBWmswiHb1oxLU  
!  
redundancy  
!  
!  
!  
crypto ikev2 proposal LAB-PROP  
encryption aes-cbc-256  
integrity sha256  
group 14  
!  
crypto ikev2 policy LAB-POL  
proposal LAB-PROP  
!  
crypto ikev2 keyring LAB-KR  
peer ANY  
address 0.0.0.0 0.0.0.0  
pre-shared-key LabPreSharedKey123  
!  
!  
!  
crypto ikev2 profile LAB-PROF  
match identity remote address 0.0.0.0  
authentication remote pre-share  
authentication local pre-share  
keyring local LAB-KR  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac  
mode transport  
!  
crypto ipsec profile LAB-IPSEC  
set transform-set LAB-TS  
set ikev2-profile LAB-PROF  
!  
!  
!  
!  
!  
!  
!  
!  
interface Loopback0  
description Protected endpoint
```

```

ip address 172.16.2.1 255.255.255.255
!
interface Loopback1
description BGP-advertised prefix
ip address 10.20.2.1 255.255.255.255
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.2 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode gre multipoint
tunnel key 100
tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
description MGMT - QEMU user-mode net (DHCP)
ip address dhcp
negotiation auto
!
interface GigabitEthernet2
description WAN - shared provider segment, DMVPN underlay
ip address 100.64.0.2 255.255.255.0
negotiation auto
!
interface GigabitEthernet3
description LAN toward the local host
ip address 192.168.20.1 255.255.255.0
ip nat inside
negotiation auto
!
interface GigabitEthernet4
description OOB management - NTP, syslog, SNMP
vrf forwarding MGMT
ip address 192.168.99.2 255.255.255.0
negotiation auto
!
router bgp 65002
bgp router-id 172.16.2.1
bgp log-neighbor-changes
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
neighbor 172.20.0.1 timers 10 30
neighbor 172.20.0.1 fall-over bfd
!
address-family ipv4
network 10.20.2.1 mask 255.255.255.255
network 10.30.2.0 mask 255.255.255.0
network 192.168.20.0
neighbor 172.20.0.1 activate
neighbor 172.20.0.1 send-community both
neighbor 172.20.0.1 route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-OUT-R1 out
neighbor 172.20.0.1 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65003$
no ip http server
ip http secure-server
ip route 10.30.2.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.2.100 10.30.2.129 prefix-length 24

```

```

ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.20.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.2.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.2.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.20.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.20.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.20.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.20.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.20.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.20.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.20.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.20.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.20.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.20.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
  description R2's own LO, tagged 65002:1
  match ip address prefix-list PL-OWN-LO
  set community 65002:1
!
route-map RM-OUT-R1 permit 20
  description R2's own NAT, tagged 65002:2
  match ip address prefix-list PL-OWN-NAT
  set community 65002:2
!
route-map RM-OUT-R1 permit 30
  description R2's own LAN, tagged 65002:3
  match ip address prefix-list PL-OWN-LAN
  set community 65002:3
!
route-map NAT-TO-DOMAIN permit 10
  match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
  description accept only known communities on an AS path R1 can produce
  match as-path 10
  match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa

```

```

snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vds12line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete

```



```

snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!

```

```

!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end

```

R3 — running config

Building configuration...

```

Current configuration : 19832 bytes
!
! Last configuration change at 02:21:14 UTC Tue Sep 8 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
hostname c8000v-r3
!
boot-start-marker
boot-end-marker
!
!
vrf definition MGMT
  description out-of-band management
  !
  address-family ipv4
  exit-address-family
!
aaa new-model
!
!
aaa group server tacacs+ LABTACGRP
  server name LAB-TAC
  ip vrf forwarding MGMT

```

[illegible]

```

!
!
!
!
!
!
!
license udi pid C8000V sn 98WSPSI3CQ2
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootup level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$Bw47UP1ye/.yVU$K5CSpZUsnABYFZykCttzeySotiNrBY/frbuYNtoXY2
!
username lab privilege 15 secret 9 $9$nTmPdPNMf5C6Nk$TZUyweyM6Pgquc3KoKrRbcfdLyeESXbTbt75p43gz7g
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
  encryption aes-cbc-256
  integrity sha256
  group 14
!
crypto ikev2 policy LAB-POL
  proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
  peer ANY
    address 0.0.0.0 0.0.0.0
    pre-shared-key LabPreSharedKey123
!
!
!
crypto ikev2 profile LAB-PROF
  match identity remote address 0.0.0.0
  authentication remote pre-share
  authentication local pre-share
  keyring local LAB-KR

```



```

neighbor 172.20.0.1 fall-over bfd
!
address-family ipv4
network 10.20.3.1 mask 255.255.255.255
network 10.30.3.0 mask 255.255.255.0
network 192.168.30.0
neighbor 172.20.0.1 activate
neighbor 172.20.0.1 send-community both
neighbor 172.20.0.1 route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-OUT-R1 out
neighbor 172.20.0.1 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65002$
no ip http server
ip http secure-server
ip route 10.30.3.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.3.100 10.30.3.129 prefix-length 24
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.3.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.3.0/24
!
ip access-list standard VTY-MGMT
10 remark out-of-band management network
10 permit 192.168.99.0 0.0.0.255
20 remark hypervisor user-mode NAT: how the harness reaches these devices
20 permit 10.0.2.0 0.0.0.255
30 deny any log
!
ip access-list extended NAT-DOMAIN
10 permit ip host 192.168.30.10 10.30.0.0 0.0.255.255
20 permit ip host 192.168.30.11 10.30.0.0 0.0.255.255
30 permit ip host 192.168.30.12 10.30.0.0 0.0.255.255
40 permit ip host 192.168.30.13 10.30.0.0 0.0.255.255
50 permit ip host 192.168.30.14 10.30.0.0 0.0.255.255
60 permit ip host 192.168.30.15 10.30.0.0 0.0.255.255
70 permit ip host 192.168.30.16 10.30.0.0 0.0.255.255
80 permit ip host 192.168.30.17 10.30.0.0 0.0.255.255
90 permit ip host 192.168.30.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.30.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
description R3's own LO, tagged 65003:1
match ip address prefix-list PL-OWN-LO
set community 65003:1
!
route-map RM-OUT-R1 permit 20
description R3's own NAT, tagged 65003:2
match ip address prefix-list PL-OWN-NAT

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set community 65003:2
!
route-map RM-OUT-R1 permit 30
description R3's own LAN, tagged 65003:3
match ip address prefix-list PL-OWN-LAN
set community 65003:3
!
route-map NAT-T0-DOMAIN permit 10
match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
description accept only known communities on an AS path R1 can produce
match as-path 10
match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded

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snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational

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snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end

```